LECTURE 0: INTRODUCTION (V1)

1. Introduction

Computational algebraic geometry. That is the title of the course, and, on the one hand, that title is quite apt. There is a large and varied part of mathematics—more than I could reasonably cover, or even survey, in five lectures—devoted to computation inspired or informed by algebraic geometry. It ranges from the use of combinatorics and polyhedral geometry in toric and tropical geometry, through homological methods based on free resolutions, to numerical and homotopy methods for finding and certifying solutions to particular systems over $\mathbb{R}$ or $\mathbb{C}$, and much more.

However, having finished writing 95% of these notes—the remaining 5% being, well, what I am currently writing—the title also feels like a misnomer. These notes are not about a field separate from algebraic geometry or a niche subfield within algebraic geometry. Instead, I wish to suggest that, in many ways, computation is the spiritual heart of algebraic geometry. All these notes attempt to do is take that viewpoint seriously, with the modern tools at our disposal in mind.

The claim that computation is the spiritual heart of algebraic geometry may seem indefensibly bold. After all, over the past seventy-five years, the field has become known to many outsiders for its abstract and formidable machinery: schemes and sheaves, a plethora of cohomology theories, stacks and derived categories, and, more recently, even more exotic things such as shtukas, homotopical algebra, ∞-categories, etc. So let me take a moment to defend the claim with two arguments.

1.1. **Examples in Algebraic Geometry.** Algebraic geometry seeks to study the shapes cut out by systems of polynomial equations. Such shapes lurk in a rather large and dark forest. The central questions that have driven the modern field ask us to chart that forest: Which shapes exist? How do they vary? Which features are typical? Which apparent possibilities never occur? Faced with a forest this large, it may seem absurd that inspecting a single tree could tell us much. Yet, remarkably, it often can.

An enormous number of these questions—including some of the most celebrated ones—come down, in the end, to a single humble task: *compute something about a single concrete example.* We must not merely describe such an example abstractly, but build it and prove something true about it. What makes this possible is a piece of true magic worth stating up front. For many of the properties we care about, the locus of members having the property is *open* in the relevant parameter space: if one member has the property, then so does every sufficiently nearby member. Provided that parameter space is irreducible, any nonempty open locus is dense, so the property holds for what algebraic geometers call a "general" member. Showing that a property is open, while not always immediate, often seems to come down to finding the right definitions. Of course, the empty set is

© Juliette Bruce 2026.
Licensed under Creative Commons BY-NC-SA 4.0.

open! Thus, once openness is known, to prove that a general object has the nice property we care about—and to be certain that we are not proving a vacuous theorem about nothing—it is enough to construct a specific example with that property. In this sense, one well-chosen example becomes a theorem about a dense open subset of the family. In fact, I do not think I am lying when I say that one could win a Fields Medal simply by constructing the right example.

Once you know this pattern, it is impossible to miss. To convince you that I am not lying—and perhaps to argue from authority by appealing to papers that have appeared in so-called fancy journals—let me briefly list a few of my favorite instances: Brill–Noether theory, Green's conjecture for a general curve, the Maximal Rank Conjecture, and interpolation problems. You are not expected to know these results yet; the common pattern is the point.

1.2. **Abstraction in Algebraic Geometry.** The framing of the previous paragraph deserves to be challenged rather than answered. It supposes that the machinery—sheaves, cohomology, resolutions, derived categories—sits at the opposite end of the subject from computation, so that the best one could hope to show is some historical link between the two. However, the historical record suggests this is quite far from the case.

Take the machinery at its source. Hilbert's theorem that rings of invariants are finitely generated produced no generators, and Paul Gordan—who had spent his career producing generators by hand—is supposed to have complained,"*Das ist keine Mathematik. Das ist Theologie!*" The anecdote is too tidy to be true, and, more importantly, it stops far too early. Pressed on precisely the question of effectivity, Hilbert kept going: the Basis Theorem, the Syzygy Theorem, and the Hilbert function all come out of the effort to get his hands on the invariants he had shown to exist. Faced with a computation he could not finish, he invented graded free resolutions. That is not a retreat from calculation into abstraction. It is what the calculation turned into.

Noether then did to Hilbert what Hilbert had done to Gordan. The Basis Theorem, as Hilbert proved it, is a statement about polynomial rings over a field, and its proof is entangled with the elimination-theoretic setting it came from. Noether asked what the argument had actually used. The answer was: very little. Every ideal finitely generated, or equivalently the ascending chain condition, and nothing else – so that the theorem is better stated as the assertion that this single property passes from R to $R[x]$. A hypothesis extracted from one proof became the definition of a class of rings, and Hilbert's theorem became the statement that the class is closed under adjoining a variable. The path here is the one worth naming:

$$\text{calculation} \longrightarrow \text{pattern} \longrightarrow \text{abstraction} \longrightarrow \text{better calculation},$$

and it is the last arrow that matters for us. Noetherianity is not a piece of bookkeeping we tolerate on the way to the interesting theorems. It is the reason Buchberger's algorithm terminates and the reason free resolutions are finite, which is to say it is the reason that essentially every computation performed in these notes returns an answer at all. Gordan, incidentally, later gave a proof of Hilbert's theorem whose monomial arguments read today like an anticipation of Gröbner bases; the two sides of the quarrel converged on the same object.

The Path Through These Notes. In writing these notes, I have tried—and we shall see to what extent I have succeeded—to weave together a number of threads that appear, disappear, and reappear throughout the lectures. At times, an exercise will deliberately be left hanging, a calculation left unexplained, or a theorem left unproved. This is intentional—or I did not catch it in my hurried proofreading. You will see relatively few proofs presented in the traditional theorem–proof format. Sometimes a proof lies beyond the scope of these notes and is best revisited after you have built more machinery; other times, the proof is buried in the exercises. That, too, is deliberate. If you are reading this, I hope your goal is to learn something, and, well, in my experience, no amount of my knowing something can magically make another person know it, too. (The same statement is even more true with the roles reversed.) Learning requires doing, and I hope these notes serve as a guide while you do that work for yourself. That is not to say that I intend to send you into the woods of algebraic geometry unprepared and unaccompanied. Instead, I have tried to write in the spirit of friends setting out on a hike together: I have walked enough of the trail to point toward the next turn and warn you about the loose rocks, but you still have to walk it yourself.

Concretely, the five lectures follow this route:

- **Lecture 1: Equations and Geometry.** We ask what it could possibly mean to *solve* a system of polynomial equations, a question that turns out to have at least six answers! This leads us to study algebraic varieties, ideals, coordinate rings, the operation of taking the radical of an ideal, and Hilbert's Nullstellensatz. Along the way, our four major recurring players enter the scene: the intersection of a parabola and a circle, a plane conic, the twisted cubic, and an elliptic curve. More about them later.

- **Lecture 2: Division and Hidden Equations.** One-variable division suggests a nonlinear analogue of row reduction, so we try to imitate it. Monomial orders and multivariable division show us why an arbitrary list of generators is not good enough; initial ideals and standard monomials tell us what a better list would have to accomplish; and S-polynomials isolate the one cancellation that seems to build such a list every time we try it.

- **Lecture 3: Finding Gröbner Bases.** Was that luck? Dickson's Lemma delivers finite Gröbner bases and, as a bonus, Hilbert's Basis Theorem; Buchberger's criterion turns an infinite condition into a finite test; and Buchberger's algorithm actually constructs a basis. We then pass to projective space, where a Gröbner basis for a degree-compatible monomial order lets us homogenize an affine ideal without the false components at infinity that naively homogenizing an arbitrary generating set can happily invent.

- **Lecture 4: Questions of Rationality & Interpolation.** Here we reverse the direction of travel and start building. The Elimination Theorem recovers equations for the Zariski closure of a parametrized image; standard monomials turn interpolation on a finite set into linear algebra; and rational and birational maps make precise when two apparently different varieties carry the same rational functions—and when, as for a smooth plane cubic, no

birational parametrization by $\mathbb{P}^1$ exists. Both themes named in the title are entry points into broad research programs in which basic questions remain open.

- **Lecture 5: What an Initial Ideal Remembers.** Finally, we see that what we have been doing with Gröbner bases has an entirely different geometric meaning. We read initial ideals as special fibers of degenerations, count standard monomials using Hilbert functions and Hilbert polynomials, and watch a complicated variety move to a combinatorial model. The point set, smoothness, and components may all change along the way; we ask precisely what does not, because that is what a degeneration argument is allowed to conclude.

1.3. **Using These Notes.** Each lecture builds on its predecessors, and I would not encourage you to read them backwards. At the same time, you should not start at the first page and march steadily forward, moving on only after you have finished and digested every part of every exercise. That is not how these notes are meant to be used, and it is not how I find mathematics is best learned. Instead, I encourage you to try many parts of many exercises. If you stall on an exercise, move forward, do something else, and eventually come back. Let things simmer while you move on: as new ideas appear and the bigger picture comes into focus, earlier exercises that seemed daunting may become clearer.

There are also deliberately more exercises, parts, and remarks than we can cover together. Honestly, I have included far more than I expect anyone to cover and digest deeply in five lectures, in the hope that these notes will serve as a friendly introduction to the subject. Still, if you find the notes at all interesting and continue studying algebraic geometry, I like to imagine that you might, from time to time, come back to them and revisit some old friends and familiar examples in a new light.

When it comes to computation, I encourage you both to compute by hand and to compute with *Macaulay2*. Each is a valuable skill in its own right. *Macaulay2* is free, and a browser-based interface is available, so there is nothing to install before trying it. You can, of course, also download it and run it locally, something you will definitely want to do if you continue using it. For working with *Macaulay2*, I have three suggestions:

(1) Instead of copying and pasting the snippets of code in these notes directly into a session, retype them by hand. Yes, this may be old-fashioned, but the process of typing things out often helps us remember commands, syntax, and structure more deeply than copying and pasting does. Pause over each command and ask what mathematical object it creates or computes.

(2) Consult the documentation frequently. The documentation for *Macaulay2* is written largely by research mathematicians, many of whom work in algebraic geometry, and is a treasure trove of examples, ideas, and papers. It is generally the best way to learn the language.

(3) If you enjoy using *Macaulay2*, consider getting involved! The *Macaulay2* is a vibrant and diverse group of researchers from numerous areas of mathematics. There are frequent

workshops dedicated to building and extending the functionality of the language, and they are just a lot of fun.

1.4. **Apologia.** I must end with a fairly serious warning and an apology. These notes are very, very much a first draft and an experiment. This is both the first time I have taught a course like this and the first time these notes have been used. I am certain that there are mathematical errors, typographical mistakes, idiosyncratic explanations that make sense only to me, and exercises that do not work as I had hoped. These issues and the perhaps peculiar choices scattered throughout the notes are entirely mine. The lectures were written at different times, sometimes weeks or months apart. So I apologize if the tone, voice, or tenor of the writing ever shifts in a way that feels jarring.

In the hope of correcting some of these issues and of saving any future souls who find and use these notes, I have created a Google Form through which you can report any issues you find.

https://forms.gle/eWsQYhPhvA3edpfSA

I would be genuinely grateful for any corrections, comments, or other feedback you might have. When submitting feedback, please keep in mind that it is much easier to act on specific, isolated comments. You are, of course, welcome to use the form to tell me that you hate everything about these notes and that they are useless, but I doubt I will know what to do with that. On the other hand, being told that I missed a sign on the third line of the fourth exercise in the second lecture should result in a speedy and satisfying resolution.

LECTURE 1: EQUATIONS AND GEOMETRY (V1)

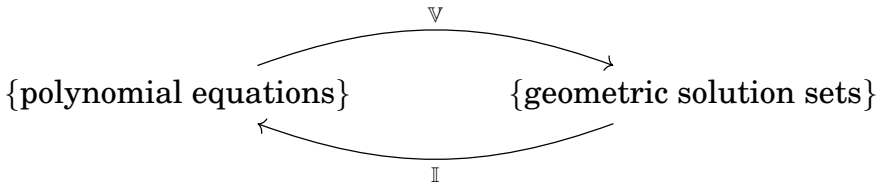
Throughout this lecture, $\mathbb{K}$ denotes a field. The algebra–geometry dictionary we shall build works over an algebraically closed field, but we will often restrict to $\mathbb{R}$ when we wish to visualize things. Further, for reasons we will explain shortly we generally do computations over $\mathbb{Q}$.

1. What Does It Mean to Solve?

The passage between algebra and geometry is familiar to us from high school. Consider the equation

$$y = x^2,$$

the solutions to which determines a subset of the plane $\mathbb{R}^2$, namely its graph. Conversely, if we are shown a graph in $\mathbb{R}^2$, we may try to recover the equation(s) which cut it out. There is something remarkable hidden in this familiar example. Formally the parabola as a subset of $\mathbb{R}^2$ consists of infinitely many points, however, the single expression $y - x^2$ encodes all of them in one piece of finite data. Algebraic geometry takes this familiar exchange seriously:



The two directions lead to different questions. The *forward problem*, which we denote $\mathbb{V}$ for reasons that will become clear shortly, asks us to find the common solution set to a given system of polynomial equations, i.e. we are asked to solve a polynomial system. The *inverse problem*, denoted by $\mathbb{I}$ for reasons that we will see later in this lecture, asks how we can find polynomial equations that cut out a given subset, i.e. we are asked to identify the graph.

The bulk of this lecture is devoted to understanding both directions of this relationship in detail, beginning with the forward problem and then considering the inverse problem. Towards this let us consider a relatively simple system of polynomial equations, which none the less will show that there are several different questions hidden in the instruction *solve the system*.

---

(1) **A Circle & A Parabola.** Consider the following system of two polynomial equations:

$$f_1 = x^2 - y = 0, \qquad f_2 = x^2 + y^2 - 1 = 0.$$

Since f_1 defines a parabola and f_2 the unit circle at the origin we are looking at the intersection of two very familiar shapes.

© Juliette Bruce 2026.
Licensed under Creative Commons BY-NC-SA 4.0.

(a) Substitute the first equation into the second and show that every solution satisfies

$$x^4 + x^2 - 1 = 0.$$

The roots of this new polynomial are the possible x-coordinates of the intersection. What information was eliminated in passing from two equations in x, y to one equation in x? Here the equation $y = x^2$ reconstructs the discarded coordinate. Explain why every root lifts uniquely.

(b) Now work over $\mathbb{R}$. Set $u = x^2$ and solve $u^2 + u - 1 = 0$. Show that exactly one of its roots is nonnegative.

(c) Find the two real solutions of the original system and sketch the picture.

(d) Now work over $\mathbb{C}$. Show that $x^4 + x^2 - 1$ has four distinct roots and conclude that the system has four complex solutions. You need not choose a preferred square root of each complex number: an answer of the form $(\pm\sqrt{u}, u)$ is more informative.

(e) Subtract the first equation from the second to obtain a polynomial involving only y. Explain why this polynomial vanishes at every solution.

(f) Compare what is recorded by the real picture, the complex solution set, and the two one-variable elimination equations obtained in parts (a) and (e). What information does each description retain, and what does it leave out?

(g) A projection can lose information in two different ways. Over $\mathbb{C}$, every nonzero x-coordinate on the curve $y^2 = x$ has two lifts, so the discarded coordinate cannot be recovered uniquely. Now assume $\mathbb{K}$ is infinite and consider the hyperbola

$$H = \left\{(x, y) \in \mathbb{K}^2 \mid xy = 1\right\}.$$

Show that its projection to the x-axis is $\mathbb{K}^\times$. Prove that the only polynomial in $\mathbb{K}[x]$ which vanishes on this image is the zero polynomial. (Recall that a nonzero one-variable polynomial has at most as many roots as its degree.) Thus any common zero set of polynomials in $\mathbb{K}[x]$ which contains the image must be the whole line, even though the origin is not in the image. Here $y = 1/x$ does recover the discarded coordinate on the image; the failure is instead that the image cannot itself be described as a common zero set of polynomial equations. We will recover this distinction by elimination later in the lecture.

The previous exercise suggests that the instruction *solve the system* is a bit ambiguous. For a collection of polynomials $\mathscr{F} \subset \mathbb{K}[x_1, \ldots, x_n]$, there are several different geometric questions one might be asking. The following table captures several more specific versions of what one might hope to do when solving a polynomial system. this one might have when actually trying to solve a system of polynomial equations

Geometric question	**Algebraic form**	***Macaulay2***
Does a solution exist?		
Which equations hold on every solution?		
Which consequences involve only selected coordinates?		
Is the solution set finite or positive-dimensional, and how many points occur when multiplicity is remembered?		
Does the set break into irreducible pieces?		
Which equations vanish on the image of a parametrization or on the closure of that image?		

Throughout this course we will use the word *solve* in each of these senses. Each geometric question has an algebraic counterpart, and our eventual goal is to make these algebraic questions computable. We shall frequently return to this table throughout every lecture, and please keep it in mind.The model is Gaussian elimination, which answers the linear versions of these questions.

---

(2) **Linear Equations - The Gold Standard.** Work over an arbitrary field $\mathbb{K}$ and consider the following system of linear polynomials:

$$\ell_1 = x + y + z - 1, \qquad \ell_2 = 2x - y + z.$$

(a) Row reduce the system $\ell_1 = \ell_2 = 0$ and write its solutions as a one-parameter family.

(b) Show that

$$x - 2y + 1 = \ell_2 - \ell_1, \qquad 3y + z - 2 = 2\ell_1 - \ell_2.$$

Use these equations to recover your parametrization from part (a).

(c) Explain why an elementary row operation replaces a list of linear polynomials by another list generating the same ideal. Conclude, without appealing to the geometry, that row operations preserve the common zero set.

(d) Suppose a linear system in n variables is consistent and its coefficient matrix has rank r. What is the dimension of its solution set? What comparison between the coefficient matrix and the augmented matrix detects inconsistency?

(e) Consider instead the linear equations

$$m_1 = x + y, \qquad m_2 = 2x + 2y - 1.$$

Verify the identity $1 = 2m_1 - m_2$. Explain why this identity is an algebraic certificate that the system $m_1 = m_2 = 0$ has no solutions over *any* field.

(f) Start now with the parametrized line

$$L = \{(2t - 1, t, 2 - 3t) : t \in \mathbb{K}\} \subseteq \mathbb{K}^3.$$

Find two independent affine-linear polynomials which vanish on L. This is the inverse problem for a linear set.

(g) Match parts (a)–(f) with the rows of the opening table which they address. Which row has not yet appeared in a substantial way? Which steps use only ideal-preserving combinations of equations, and which use the special echelon organization of leading variables? What nonlinear replacement is still missing?

---

2. From Equations to Ideals

For systems of linear equation, replacing a subset of the equations by suitable linear combinations, does not change the underlying geometry of their solution set. Further, via these operations we may eventually move our system of equations into echelon form from which the geometry is very visible. For systems of non-linear equations, we do not have an analogous notion of echelon form, but there are natural operations, analogous to linear combinations, which preserve the solution set. Namely, we may replace the care list of equations by the ideal the list of equations generates. In partocular, if $f_1, \ldots, f_s$ vanish at a point $p \in \mathbb{K}^n$, then every expression of the form

$$h_1 f_1 + \cdots + h_s f_s$$

for $h_1, \ldots, h_s \in \mathbb{K}[x_1, \ldots, x_n]$ also vanishes at p. At a moment to ponder why the converse is also true. Thus passing from a list of equations to its ideal records all consequences obtained by these elementary algebraic operations without changing the common zero set.

Definition 1. Let $\mathscr{F} \subseteq \mathbb{K}[x_1, \ldots, x_n]$ be an arbitrary subset. The *vanishing set* of $\mathscr{F}$ over $\mathbb{K}$ is

$$\mathbb{V}_{\mathbb{K}}(\mathscr{F}) = \{p \in \mathbb{K}^n \mid f(p) = 0 \text{ for every } f \in \mathscr{F}\}.$$

When the ground field $\mathbb{K}$ is clear, we simply write $\mathbb{V}(\mathscr{F})$. A subset of $\mathbb{K}^n$ of the form $\mathbb{V}(\mathscr{F})$ is called an *affine algebraic set* or an *affine algebraic variety*.

Out of economy, or perhaps laziness, we often drop adjectives and simply say *algebraic set*, *algebraic variety*, or just *variety* when it seems little confusion will arise. As a slight warning, there are dueling notations throughout the algebraic geometry over the use of the word variety. Some authors reserve the term algebraic variety for an algebraic set, which is also irreducible. (A topological property we shall see later in this exercise.) Others adopt the slightly broader definition we have taken where algebraic set and variety are synonymous and we shall stress the adjective irreducible when appropriate.

Notice the collection $\mathscr{F}$ in the above definition is allowed to be infinite. Nothing, so far, rules out an algebraic set which genuinely requires infinitely many equations. One of the central results of these lectures, Hilbert's Basis Theorem, which we will prove in Lecture 3, says that this cannot occur: every affine algebraic variety can be cut out by finitely many equations.

The notation $\mathbb{V}(\mathscr{F})$ admits two useful mnemonics: $\mathbb{V}$ for the common *vanishing* locus and $\mathbb{V}$ for the *variety* it defines. The historical origin of the this notation is a bit mysterious, however, the mathematical word *variety* comes to use from various translations of Riemann's *Mannigfaltigkeit*. While in modern usage this word translates as manifold, Riemann's use of the word was much more general than the current usage of a smooth or even topological manifold. The German adjective *mannigfaltig*, which is a root of Mannigfaltigkeit, roughly translates as plural, many, or perhaps various, which seems to be in roughly line with Riemann's work. As such Italian mathematicians translated Mannigfaltigkeit as *varietà*, and many French mathematicians translated it as *variété*. Note other French mathematicians translated it as *multiplicité* or *continuum*. When this word began solely referring to algebraic solution sets is unclear.

---

(3) **From Equations to Ideals.** Let $\mathscr{F} \subseteq \mathbb{K}[x_1, \ldots, x_n]$ and let $I, J \subseteq \mathbb{K}[x_1, \ldots, x_n]$ be ideals.

(a) Prove that $\mathbb{V}(\mathscr{F}) = \mathbb{V}(\langle \mathscr{F} \rangle)$. Thus the common zero set remembers the ideal generated by the equations, not merely the displayed list.

(b) Prove that $\mathbb{V}$ reverses inclusions: if $I \subseteq J$, then $\mathbb{V}(J) \subseteq \mathbb{V}(I)$.

(c) Describe the following varieties, and when possible, draw the corresponding picture. Over $\mathbb{C}$, a literal sketch is likely difficult, so you may need to be creative.

$$\mathbb{V}_{\mathbb{R}}(x^2+y^2-1), \quad \mathbb{V}_{\mathbb{R}}(x^2-y^2), \quad \mathbb{V}_{\mathbb{R}}(x^2+y^2),$$
$$\mathbb{V}_{\mathbb{C}}(x^2+y^2-1), \quad \mathbb{V}_{\mathbb{C}}(x^2-y^2), \quad \mathbb{V}_{\mathbb{C}}(x^2+y^2).$$

An *affine plane conic* is the zero set of a polynomial of degree two in two variables, and thus, the six varieties above are all examples of plane conics.

(d) Which of the defining polynomials in the previous part are irreducible over $\mathbb{R}$, but reducible over $\mathbb{C}$? For these examples, how do the varieties over $\mathbb{R}$ and $\mathbb{C}$ differ?

(e) Prove that for any polynomials $f, g \in \mathbb{K}[x_1, \ldots, x_n]$ the following identity holds:

$$\mathbb{V}(fg) = \mathbb{V}(f) \cup \mathbb{V}(g).$$

Thus factorization is an algebraic signal that the corresponding variety can be broken into (simpler) pieces.

(f) A nonempty algebraic variety $X \subset \mathbb{K}^n$ is *reducible* if and only if there exists proper algebraic varieties $X_1, X_2 \subset \mathbb{K}^n$, with $X_i \neq X$ for $i = 1, 2$, such that $X = X_1 \cup X_2$. A variety is

irreducible if it is not reducible. Prove that a nonempty algebraic variety $X \subset \mathbb{K}^n$ is irreducible if and only if given proper algebraic varieties $X_1, X_2 \subset \mathbb{K}^n$ such that $X = X_1 \cup X_2$ then $X_i = X$ for some i. (Note here proper means a proper nonempty subset of $\mathbb{K}^n$.)

(g) Note being reducible or irreducible very much depends on the ground field we are working over. For example, prove the following:

(i) The variety $\mathbb{V}_{\mathbb{K}}(xy)$ is reducible over all fields $\mathbb{K}$.

(ii) The variety $\mathbb{V}_{\mathbb{K}}(x^2 - y^2)$ is reducible if and only if $\mathbb{K}$ does not have characterisitc two. What happens when $\mathbb{K}$ is characterisitc two?

(iii) The variety $\mathbb{V}_{\mathbb{K}}(x^2 + y^2)$ is irreducible when $\mathbb{K} = \mathbb{Q}$ or $\mathbb{R}$, but is reducible when $\mathbb{K} = \mathbb{C}$.

The Nullstellensatz

The same algebraic set can be presented by many different ideals. For example, over every field we have that $\mathbb{V}(x) = \mathbb{V}(x^2) = \mathbb{V}(x^3)$. Geometrically, the equations $x = 0$, $x^2 = 0$, and $x^3 = 0$ all impose the same conditions on a point. Algebraically, however, the ideals $\langle x \rangle$, $\langle x^2 \rangle$, and $\langle x^3 \rangle$ are all different. (Be sure you know why.) More generally, if $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$ is an ideal and $f^k \in I$ for some $k \in \mathbb{Z}_{\geq 1}$, then for any point $p \in \mathbb{V}(I)$ we have

$$(f(p))^k = f^k(p) = 0.$$

Since $\mathbb{K}$ is a field, it has no nonzero nilpotents, so this forces $f(p) = 0$. Thus every equation whose power belongs to I already vanishes on $\mathbb{V}(I)$, even if the equation itself does not belong to I. This leads to the following operation.

Definition 2. Let R be a commutative ring. The *radical* of an ideal $I \subseteq R$ is

$$\sqrt{I} \coloneqq \left\{ f \in R \mid f^k \in I \text{ for some } k \geq 1 \right\}.$$

An ideal is *radical* if $I = \sqrt{I}$.

As argued above, and as you will formalize shortly, $\mathbb{V}(I)$ and $\mathbb{V}(\sqrt{I})$ are equal. Hence, taking radicals is one explanation for why the operation $I \mapsto \mathbb{V}(I)$ loses at least some information: it cannot distinguish an ideal from its radical. Amazingly, when $\mathbb{K}$ is algebraically closed, this is the only information that is lost in the process. This is the first of a series of results broadly know as 'Hilbert's Nullstellensatz", each of which highlights a different perspective of what it might mean to "solve" a system of polynomial equations.

Theorem 3 (Hilbert's Nullstellensatz I (Geometric Form))**.** *Let $\mathbb{K}$ be an algebraically closed field. The map $I \mapsto \mathbb{V}(I)$ gives a bijection:*

$$\left\{ \begin{matrix} \textit{radical ideals} \\ \textit{in } \mathbb{K}[x_1, \ldots, x_n] \end{matrix} \right\} \xrightarrow{\quad \mathbb{V} \quad} \left\{ \begin{matrix} \textit{algebraic varieties} \\ \textit{in } \mathbb{K}^n \end{matrix} \right\}.$$

It is worth pausing to see how this mirrors the relationship between equations and solution sets recalled at the beginning of this lecture. A complete proof of the Nullstellensatz is somewhat beyond the scope of this course, and so we will not give one here. However, we will frequently make use of it and will try to refine our understanding of it.

The theorem underlying this modern geometric formulation was proved, for homogeneous polynomials, in Section 3 of Hilbert's 1893 paper *Ueber die vollen Invariantensysteme*. It appeared three years after his 1890 paper *Ueber die Theorie der algebraischen Formen*, which contained what we now call Hilbert's Basis Theorem and applied it to the studying the question of when the ring of invariants is finitely generated. Famously, Hilbert's 1890 argument was highly non-constructive, leading his contempary Paul Gordan to exclaim, "Das ist keine Mathematik, das ist Theologie", "This is not mathematics, this is theology". While the provienance of this remark is a bit shakey, the first record of I could find it in-print is from Max Noether in 1914, Hilbert's 1893 paper also concered the problem of studying invariants, however from a more constructive perpsective. Interesting, Gordan subsequently gave his own proof of Hilbert's Basis Theorem, and it is his approach we shall roughly follow in Lecture 3.

---

(4) **Radical Ideals.** Let R be a ring and $I \subseteq R$ an ideal.

(a) Show that $I \subseteq \sqrt{I}$ and that $\sqrt{\sqrt{I}} = \sqrt{I}$.

(b) Suppose $f, g \in \sqrt{I}$, with $f^r \in I$ and $g^s \in I$. Use the binomial theorem to show that $(f + g)^{r+s} \in I$. Deduce that $\sqrt{I}$ is closed under addition.

(c) Show that if $f \in \sqrt{I}$ and $h \in R$, then $hf \in \sqrt{I}$. Conclude that $\sqrt{I}$ is an ideal.

(d) Let $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$ be an ideal. Prove that $\mathbb{V}(I) = \mathbb{V}(\sqrt{I})$. In other words, passing to the radical does not change its zero set.

(e) Compute $\sqrt{\langle x^3y^2 \rangle} \subseteq \mathbb{K}[x, y]$. Compare the zero sets of $\langle x^3y^2 \rangle$ and its radical.

(f) Give three different ideals in $\mathbb{K}[x, y]$ which define the x-axis. Which, if any, of your ideals are radical?

The geometric form of Hilbert's Nullstellensatz I asserts that $\mathbb{V}$ has an inverse, but it does not yet tell us what that inverse is. This returns us to the inverse problem with which we began: given a set of points, which polynomial equations hold on all of them? The natural answer is to collect every such equation.

Definition 4. For any subset $X \subseteq \mathbb{K}^n$, its *vanishing ideal* is

$$\mathbb{I}(X) \coloneqq \{f \in \mathbb{K}[x_1, \ldots, x_n] \mid f(p) = 0 \text{ for every } p \in X\}.$$

Thus $\mathbb{I}(X)$ records every polynomial vanishing on all of X, rather than one particular list of equations describing X. Notice this definition makes sense whether or not X is a variety. When X is a variety we often call $\mathbb{I}(X)$ the *defining ideal* of X. As we shall see shortly, it only really makes sense to think of $\mathbb{I}(X)$ as defining X when X is itself an algebraic variety.

In general the operations $\mathbb{V}$ and $\mathbb{I}$ are not literal inverses, even when we restrict our attention to only algebraic varieties. They reverse inclusions and satisfy

$$X \subseteq \mathbb{V}(\mathbb{I}(X)) \qquad \text{and} \qquad I \subseteq \mathbb{I}(\mathbb{V}(I)).$$

The first enlargement replaces an arbitrary set by the smallest algebraic set containing it; the second enlarges an ideal to include every polynomial which vanishes on its zero set. Exercise 5 asks you to make the first statement precise.

Two words of warning are useful at this moment: i) A finite list of equations whose common zero set is X need not generate the full ideal $\mathbb{I}(X)$. ii) A priori the ideal $\mathbb{I}(X)$ need not have a finite generating set. This claim is the content of Hilbert's Basis Theorem.

---

(5) **The Inverse Problem & Zariski Closure.** Let $X, Y \subseteq \mathbb{K}^n$ be arbitrary subsets.

(a) Prove that $\mathbb{I}(X)$ is a radical ideal. Hint: First check it is an ideal and then check that it is equal to its radical.

(b) Prove that if $X \subseteq Y$, then $\mathbb{I}(Y) \subseteq \mathbb{I}(X)$.

(c) Show that $X \subseteq \mathbb{V}(\mathbb{I}(X))$.

(d) Show that $\mathbb{V}(\mathbb{I}(X))$ is the smallest algebraic subset of $\mathbb{K}^n$ containing X: if $X \subseteq Z$ and Z is algebraic, then $\mathbb{V}(\mathbb{I}(X)) \subseteq Z$. We call this set the *Zariski closure* of X.

(e) Conclude that X is an algebraic set if and only if $X = \mathbb{V}(\mathbb{I}(X))$.

(f) Assume now that $\mathbb{K}$ is infinite and let $X = \{(t, t^2) \mid t \in \mathbb{K}^\times\} \subseteq \mathbb{K}^2$. If $f \in \mathbb{I}(X)$, consider the one-variable polynomial $f(t, t^2)$. Use the root bound recalled in Exercise 1(g) to show that it is identically zero, and hence that $f(0,0) = 0$. Conclude that $(0,0) \in \mathbb{V}(\mathbb{I}(X))$ even though $(0,0) \notin X$. In particular, a parabola with one point removed is not an algebraic set.

(g) In the previous part, where does the assumption that $\mathbb{K}$ is infinite enter? Over the finite field $\mathbb{F}_q$, set $X_q = \{(t, t^2) \mid t \in \mathbb{F}_q^\times\}$. Show that $x^{q-1} - 1 \in \mathbb{I}(X_q)$ but does not vanish at $(0,0)$, so $(0,0) \notin \mathbb{V}(\mathbb{I}(X_q))$. Also explain why $t^q - t$ is a nonzero polynomial in $\mathbb{F}_q[t]$ even though it vanishes at every element of $\mathbb{F}_q$. Identify the inference from part (f) which fails over a finite field.

Assume now that $\mathbb{K}$ is algebraically closed. Exercises 4 and 5 allow us to identify the inverse promised by Hilbert's Nullstellensatz I. Given $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$, the ideals $\sqrt{I}$ and $\mathbb{I}(\mathbb{V}(I))$ are both

radical, and those exercises show that they have the same zero set:

$$\mathbb{V}\left(\sqrt{I}\right) = \mathbb{V}(I) = \mathbb{V}(\mathbb{I}(\mathbb{V}(I))).$$

The injectivity in Theorem 3 therefore forces them to be equal. The set $\mathbb{V}(\mathbb{I}(X))$ is the smallest algebraic set containing X; in particular, if X is already algebraic, then $\mathbb{V}(\mathbb{I}(X)) = X$. The other composition, $\mathbb{I}(\mathbb{V}(I))$, contains the deeper content of the Nullstellensatz, and the equality $\mathbb{I}(\mathbb{V}(I)) = \sqrt{I}$ is false over a general field. Under our algebraically closed hypothesis, this brings us to our second version of the Nullstellensatz.

Theorem 5 (Hilbert's Nullstellensatz II (Recovery Form))**.** *Let $\mathbb{K}$ be algebraically closed and let $\mathscr{S} \subseteq \mathbb{K}[x_1, \ldots, x_n]$. If $V = \mathbb{V}(\mathscr{S}) \subseteq \mathbb{K}^n$, then*

$$\mathbb{I}(V) = \sqrt{\langle \mathscr{S} \rangle}.$$

An immediate consequence of Hilbert's Nullstellensatz II is that if I is radical, then $\mathbb{I}(\mathbb{V}(I)) = I$. Thus the vanishing-ideal operation $\mathbb{I}$ is precisely the inverse promised by the bijection $I \mapsto \mathbb{V}(I)$ in Hilbert's Nullstellensatz I. Let us record this conclusion one final time, as this will be the most memorable and used form of the Nullstellensatz.

Theorem 6 (Hilbert's Nullstellensatz III (Affine Algebra–Geometry Dictionary))**.** *Let $\mathbb{K}$ be an algebraically closed field. The maps $\mathbb{V}$ and $\mathbb{I}$ induce inverse, inclusion-reversing bijections:*

$$\left\{\begin{matrix}\textit{radical ideals} \\ \textit{in } \mathbb{K}[x_1, \ldots, x_n]\end{matrix}\right\} \underset{\mathbb{I}}{\overset{\mathbb{V}}{\rightleftarrows}} \left\{\begin{matrix}\textit{algebraic varieties} \\ \textit{in } \mathbb{K}^n\end{matrix}\right\}.$$

In many ways much of algebraic geometry can be view as understanding how various properties can and cannot be passed back and forth thru these correspondences. In this form, the Nullstellensatz solves the inverse problem completely: over an algebraically closed field, an algebraic set determines its radical ideal of equations, and that ideal determines the algebraic set.

A particularly important special case asks which ideal corresponds to the empty set. Since every polynomial vanishes vacuously on the empty set, $\mathbb{I}(\varnothing) = \mathbb{K}[x_1, \ldots, x_n]$. Hilbert's Nullstellensatz II therefore gives the following.

Theorem 7 (Hilbert's Weak Nullstellensatz)**.** *If $\mathbb{K}$ is algebraically closed and $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$ is an ideal, then $\mathbb{V}(I) = \varnothing$ if and only if $I = \langle 1 \rangle$.*

Notice the reverse implication that $1 \in I$ implies $\mathbb{V}(I) = \varnothing$ is immediate since as a polynomial 1 never vanishes. This is true over any field since in a field $1 \neq 0$. Hence the difficult part of the Weak Nullstellensatz is the forward implication that if $\mathbb{V}(I) = \varnothing$ then $1 \in I$. This implication is where the hypothesis that $\mathbb{K}$ be algebraically closed is crucial. (Consider $\mathbb{V}_{\mathbb{R}}(x^2 + 1)$.)

Although we have obtained the weak form here as a consequence of Hilbert's Nullstellensatz II, the usual proof proceeds in the opposite direction. One first proves the weak Nullstellensatz and then uses Rabinowitsch's trick to deduce the full recovery statement $\mathbb{I}(\mathbb{V}(I)) = \sqrt{I}$. While we shall see Rabinowitsch's trick shortly, it will be of interest to us mainly for its computational consequences.

As an interesting historical aside, Rabinowitsch's trick appeared in a one-page note by J. L. Rabinowitsch. The note, *Zum Hilbertschen Nullstellensatz*, was received in 1929 and published on page 520 of volume 102 of *Mathematische Annalen* in 1930. The page identifies its author only by the initials J. L. and the location Moscow. A colorful story, often repeated to identify J. L. Rabinowitsch with George Yuri Rainich, goes as follows. Rainich supposedly used the trick during a lecture and was interrupted by someone who indignantly accused him of presenting the famous Rabinowitsch trick as his own. Without replying, Rainich turned to the blackboard, wrote RABINOWITSCH, erased the letters B, O, W, T, S to leave RAINICH, and then continued his lecture. It is a wonderful story, but almost certainly not history. The current best evidence points towards a different Moscow based mathematician: Yuli Lazarevich Rabinovich, about whom little seems to be know.

---

(6) **Consequences and Certificates.** Assume $\mathbb{K}$ is algebraically closed unless otherwise stated.

(a) Let $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$ be an ideal. Use Theorem 5 to prove that a polynomial h vanishes at every point of $\mathbb{V}(I)$ if and only if $h^k \in I$ for some $k \geq 1$.

(b) Let $I = \langle x, xy - 1 \rangle \subseteq \mathbb{K}[x, y]$. Find an explicit identity expressing 1 as a polynomial combination of the displayed generators. Explain what this proves about $\mathbb{V}(I)$.

(c) Why does the fact that $\mathbb{V}_{\mathbb{R}}(x^2 + 1) = \varnothing$ not contradict Hilbert's Nullstellensatz?

(d) Return to the circle–parabola system in Exercise 1 given by $f_1 = x^2 - y$ and $f_2 = x^2 + y^2 - 1$. Prove that the polynomial $h = y^2 + y - 1$ vanishes on the common zeros of $f_1 = f_2 = 0$ by explicitly demonstrating that $h \in \langle f_1, f_2 \rangle$. Hint: Express h in terms of f_1 and f_2.

(e) *Further: Rabinowitsch's trick.* Let $R = \mathbb{K}[x_1, \ldots, x_n]$ and let $I \subseteq S$ be an ideal. Suppose $g \in S$ vanishes on $\mathbb{V}(I)$. Introducing a new variable z consider the polynomial ring $R[z] \cong \mathbb{K}[x_1, \ldots, x_n, z]$ and set $J = \langle I, 1 - zg \rangle \subset R[z]$. Notice that variety $\mathbb{V}(J)$ is a subset of $\mathbb{K}^{n+1}$.

(i) Prove that $\mathbb{V}(J) = \varnothing$. Use the weak Nullstellensatz to conclude that $1 \in J$.

(ii) More generally, prove that $g \in \sqrt{I}$ then $1 \in \langle I, 1 - zg \rangle$. (Hint: If $g^k \in I$ use the geometric series identity for $1 - (zg)^k$.)

The converse is also true, however, it is much more difficult to prove. Put together we get what is know as Rabinowitsch's trick: $g \in \sqrt{I}$ if and only if $1 \in \langle I, 1 - zg \rangle$. We shall use this statement in Lecture 3 to turn a membership test for the ideal on the right hand side into an algorithm to check membership in the radical ideal on the left hand side.

3. Parametrizations and Kernels

Up to this point, we have mostly focused on describing a solution set (i.e. a variety) implicitly, through its defining equations. A second perspective to describing a variety is by parametrizing it. Roughly, one should think of a parametrization as a convenient recipe for producing points of the solution set without having to solve the system of equations defining it For example, the parabola from the beginning of the lecture can be described implicitly as $\mathbb{V}(y-x^2)$, but it can also be described parametrically as the set $\{(t,t^2) \mid t \in \mathbb{K}\}$. These two description are useful for different purposes: the implicit equation $y - x^2$ makes it exceptionally easy to test whether a point $p \in \mathbb{K}^2$ lies on the parabola, while the parametric description makes it easy to construct points on the curve.

More generally, a collection of polynomials $f_1, \ldots, f_n \in \mathbb{K}[t_1, \ldots, t_m]$ determines a polynomial map

$$F : \mathbb{K}^m \longrightarrow \mathbb{K}^n \qquad \text{given by} \qquad a \longmapsto \big(f_1(a), \ldots, f_n(a)\big).$$

We call this a *polynomial parametrization* of its image. Note the parametrization F need not be injective, and the image of F image need not be algebraic, i.e., the image might not be a variety). The process of finding the vanishing ideal of the image of F from the polynomials $f_1, \ldots, f_n$ defining F is called *implicitization*. Moregenerally, we might define rational parameterizations by replacing the $f_1, \ldots, f_n$ by rational functions. Understanding which varieties can be rationally parameterized, and when such a parametrization is generically one-to-one, has be of great interest in algebraic geometry since the 19th century. These questions lead us to the notions of unirational and rational varieties, and more generally birational geometry, which we shall see in Lecture 4.

For the moment, we wish to see how algebra translates the map F into equations. Notice that we may use the same polynomials $g_1, \ldots, g_n$ defining F to construct a ring homomorphism

$$\begin{aligned} F^* : \mathbb{K}[x_1, \ldots, x_n] &\longrightarrow \mathbb{K}[t_1, \ldots, t_m], \\ f &\longmapsto f(g_1, \ldots, g_n), \end{aligned}$$

which we call the *substitution map* for $g_1, \ldots, g_n$. A polynomial f belongs to $\ker(F^*)$ exactly when $f(g_1, \ldots, g_n)$ is the zero polynomial in $\mathbb{K}[t_1, \ldots, t_m]$. Any such f certainly vanishes at every point of the image, since $f(F(a)) = (F^* f)(a) = 0$ for all $a \in \mathbb{K}^m$. Hence we always have

$$\begin{aligned} \ker(F^*) &\subseteq \mathbb{I}\left(F(\mathbb{K}^m)\right), \\ F(\mathbb{K}^m) &\subseteq \mathbb{V}\left(\ker(F^*)\right). \end{aligned}$$

The relationship among these sets depends both on the field and on whether the image is algebraic. Suppose $\mathbb{K}$ is infinite. A polynomial in $\mathbb{K}[t_1, \ldots, t_m]$ which vanishes at every point of $\mathbb{K}^m$ is the zero polynomial; this follows by induction on m from the corresponding fact for one-variable polynomials. Consequently, $f \in \mathbb{I}(F(\mathbb{K}^m))$ forces $F^*(f) = 0$, and therefore $\ker(F^*) = \mathbb{I}\big(F(\mathbb{K}^m)\big)$. In this case, $\mathbb{V}(\ker F^*) = \mathbb{V}\big(\mathbb{I}(F(\mathbb{K}^m))\big)$ is precisely the Zariski closure of the parametrized image, and it cuts out the image itself when that image is already algebraic. For a general field, $\ker(F^*)$ can be strictly smaller than $\mathbb{I}(F(\mathbb{K}^m))$, as we shall see in the exercise below.

There are several closely related but distinct problems concerning a parameterization F, which are worth keeping separate in our minds:

(i) Deciding whether the image of F is algebraic.

(ii) Computing the kernel of the substitution map F^*.

(iii) Computing the vanishing ideal of $F(\mathbb{K}^m)$, i.e., computing $\mathbb{I}(F(\mathbb{K}^m))$.

Answering (i) means proving there exists an ideal $J \subset \mathbb{K}[x_1, \ldots, x_n]$ such that $\mathbb{V}(J) = F(\mathbb{K}^m)$. If we find such an ideal J it need not be equal to $\mathbb{I}(F(\mathbb{K}^m))$, instead over an arbitrary field all we know is that $J \subset \mathbb{I}(F(\mathbb{K}^m))$. Moreover, although when $\mathbb{K}$ is infinite we have that $\ker(F^*) = \mathbb{I}(F(\mathbb{K}^m))$, this can fail over a finite field. Further, this abstract equality is not yet an explicit answer to the problem of implicitization, as we need a way to find explicit generators for $\ker(F^*)$. At the moment, it is not even clear that such a *finite* generating set exists, yet alone how we might construct one. This will be the focus of Lecture 3 where we will see Gröbner bases and Hilbert's Basis Theorem. The following exercise shows how the three questions above can differ in a relatively simple example.

---

(7) **From a Parametrization to Equations: The Affine Twisted Cubic.** Consider the polynomial map

$$F : \mathbb{K} \longrightarrow \mathbb{K}^3 \qquad \text{given by} \qquad t \longmapsto (t, t^2, t^3),$$

and let C denote the image of F. This curve is called the *affine twisted cubic*; the reason for the name will become clearer when we introduce projective space. Substitution gives the ring homomorphism

$$\begin{aligned} \mathbb{K}[x, y, z] &\xrightarrow{\quad F^* \quad} \mathbb{K}[t] \\ x &\longmapsto t \\ y &\longmapsto t^2 \\ z &\longmapsto t^3 \end{aligned}$$

which is the identity on $\mathbb{K}$.

(a) Verify that $y - x^2$ and $z - xy$ belong to $\ker(F^*)$. Let $J = \langle y - x^2, z - xy \rangle$. Explain why this proves that $J \subseteq \ker(F^*)$ and $C \subseteq \mathbb{V}(J)$.

(b) Conversely, let $(a, b, c) \in \mathbb{V}(J)$. Show that $b = a^2$ and $c = a^3$, and hence that $(a, b, c) = F(a)$.

(c) Conclude that $C = \mathbb{V}(J)$. Hence, in this example, the image of the parametrization is already algebraic. This equality holds over every field.

(d) At this point we know that $J \subseteq \ker(F^*)$ and

$$\mathbb{V}_{\mathbb{K}}(J) = \mathbb{V}_{\mathbb{K}}(\ker(F^*)) = C.$$

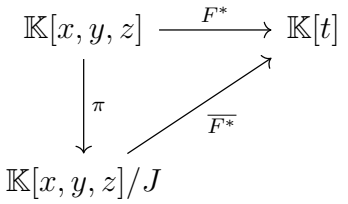
However, as we have seen two ideals defining the same variety does not imply the ideals are equal. Further, over an arbitrary field $\mathbb{V}(I) = \mathbb{V}(J)$ does not imply that $\sqrt{I} = \sqrt{J}$. (What assumption are we missing?) The remainder of the exercise proves $\ker(F^*) = J$. For $\alpha, \beta, \gamma \in \mathbb{Z}_{\geq 0}$, show that

$$x^\alpha y^\beta z^\gamma \equiv x^{\alpha+2\beta+3\gamma} \pmod{J}.$$

Hint: First use the generators of J to show that $y \equiv x^2$ and $z \equiv x^3 \pmod{J}$.

(e) Deduce that every polynomial in $\mathbb{K}[x, y, z]$ is congruent modulo J to a polynomial in x alone.

(f) Since $J \subseteq \ker(F^*)$, the map F^* factors as follows:



where π is the natural quotient map. Show that $\overline{F^*}$ is surjective.

(g) Prove that $\overline{F^*}$ is injective. Then apply the first isomorphism theorem to conclude that $\ker(F^*) = J$. *Hint:* Represent a class in the quotient $\mathbb{K}[x, y, z]/J$ by a polynomial $r(x) \in \mathbb{K}[x, y, z]$ only involving x. Using part (e), then show that $\overline{F^*}(r(x)) = r(t)$.

(h) We always have $\ker(F^*) \subseteq \mathbb{I}(C)$. The reverse inclusion requires more care. Suppose that $\mathbb{K}$ is infinite, and let $h \in \mathbb{I}(C)$. Show that $h(t, t^2, t^3) \in \mathbb{K}[t]$ vanishes at every $t \in \mathbb{K}$, and hence is the zero polynomial. Conclude that $\mathbb{I}(C) = \ker(F^*) = J$.

(i) Explain why the preceding conclusion requires an infinite field. If $\mathbb{K} = \mathbb{F}_q$, show that $x^q - x \in \mathbb{I}(C)$ even though $F^*(x^q - x) = t^q - t \neq 0$ as an element of $\mathbb{F}_q[t]$; the latter is a nonzero polynomial which vanishes as a function on $\mathbb{F}_q$. Thus $\ker(F^*) \subsetneq \mathbb{I}(C)$.

(j) Show that $\langle y - x^2, z - xy \rangle$ is equal to $\langle y - x^2, z - x^3 \rangle$. What does this tell us about the nonuniqueness of a chosen list of equations, even though the ideal of all polynomial relations is uniquely determined?

(k) Verify the further relation

$$xz - y^2 = y(x^2 - y) - x(xy - z) \in J.$$

Record it for now. Lecture 2 will reveal why the two displayed generators do not yet form the kind of organized generating set needed for systematic computation.

As we saw in the previous exercise, the affine twisted cubic admits a polynomial parametrization in a single parameter. Not every curve admits such a parametrization. As a preview of Lecture 4,

consider the affine cubic curve

$$\mathbb{V}_{\mathbb{C}}(y^2 - x^3 + x) \subset \mathbb{K}^2,$$

whose projective closure is the elliptic curve $\mathbb{V}_{\mathbb{P}^2}(y^2z = x^3 - xz^2)$. (Projective space and projective closure will appear in Lecture 3.) We shall see that unlike the twisted cubic, this curve cannot be rationally parameterized.

We can now carry out the same computations in *Macaulay2* (M2), an open-source computer algebra system created by Daniel Grayson and Michael Stillman for algebraic geometry and commutative algebra. It grew out of its predecessor *Macaulay*, written by Dave Bayer and Stillman; David Eisenbud has been a longtime collaborator, and a broad international community now develops the system and its packages. We will use *Macaulay2* through the browser, so nothing needs to be installed. Open either server on the official

"Try it out" page.

In *Macaulay2* one defines something with the = symbol. During a session all defined variables, rings, polynomials, ideals, etc. are remembered. In order to clear these things from memory we use the command `restart`, which starts a fresh session. The *Macaulay2* syntax for multiplication is `*`, powers are written with `^`, and additional and subtraction are `+` and `-` respectively. One can surpress the output of a line by placing a semicolon `;` at the end of the line. Once ran all inputs and outputs `i1`, `o1`, and so on; these labels are not part of the commands. For help with a command, one can enter, for example, `help ideal`.

Since computers cannot store infinite decimal expansions, computations with real or complex numbers generally require floating-point approximations. Although dealing with these numerical issues is itself a beautiful area of mathematics, we will avoid them completely. Instead, we will generally compute over the rational numbers, denoted `QQ` in *Macaulay2*, or over a finite field, where arithmetic is exact. Since our equations have rational coefficients, one may show that any polynomial identity proved over $\mathbb{Q}$ remains true if we view the same identity over $\mathbb{C}$ (The fancy way to say this is "after extending scalars to $\mathbb{C}$.) So for example, a polynomial that factors over $\mathbb{Q}$ also factors over $\mathbb{C}$. However the converse is not true, and identities proven over $\mathbb{C}$ may fail to be true over $\mathbb{Q}$. In general it is good to develop a sense of what precisely a computation done with *Macaulay2* shows, and then think about what theoretical results can be used to generalize it.

One beauty of *Macaulay2* is that much of its syntax mimics mathematical notation; what you type as code will often look like what you would write on the blackboard. For example, `QQ` denotes $\mathbb{Q}$, `ZZ` denotes $\mathbb{Z}$, and, when p is prime, `ZZ/p` denotes $\mathbb{Z}/p\mathbb{Z}$. When q is a prime power, the finite field $\mathbb{F}_q$ is constructed with `GF(q)`. As noted above, there are also `RR` and `CC`, but they use finite-precision approximations rather than exact copies of $\mathbb{R}$ and $\mathbb{C}$. A polynomial ring is specified by listing the coefficient ring followed by variables, separated by commas, in brackets `[]`. For example, the command `R = QQ[x,y,z]` creates the polynomial ring $\mathbb{Q}[x, y, z]$ and gives it the name `R`.

For now, we will treat commands such as `ker` and `eliminate` as black boxes whose outputs we can interpret. Lectures 2 and 3 will develop the algorithms that make these commands possible; both rely on Gröbner basis computations.

The namesake of both *Macaulay2* and the original *Macaulay* is Francis Sowerby Macaulay (1862–1937), an early pioneer in commutative algebra. Macaulay's career was filled with contrasts. He spent most of it as a schoolmaster at St Paul's School in London, yet was elected a fellow of the Royal Society in 1928. His work received little attention from many contemporary algebraic geometers in England but was appreciated by Emmy Noether and those around her who were building the modern language of commutative algebra. His 1916 tract *The Algebraic Theory of Modular Systems* likewise received little attention after his death but became highly influential in the second half of the twentieth century. In Macaulay's terminology, "modular systems" are what we now call ideals and modules. Much of the book is recognizably computational—elimination, resultants, and explicit systems of equations. The name of the software is therefore apt: as Eisenbud and Gray put it, it *"recognizes that much of his work is based on examples created by refined computation."* _

(8) **First *Macaulay2* Session.** Let us return to the circle–parabola equations, which is defined by the ideal $I = \langle x^2 - y, x^2 + y^2 - 1 \rangle \subset \mathbb{K}[x, y]$ Enter the commands one line at a time and compare every output with Exercise 1.

```
restart
R = QQ[x,y];
f1 = x^2-y;
f2 = x^2+y^2-1;
I = ideal(f1,f2); -- encode the intersection as an ideal

R
gens I
f2-f1 -- eliminate x
```

The command `R = QQ[x,y]` creates the polynomial ring $R = \mathbb{Q}[x, y]$. After running this line, we can use the variables x and y to construct polynomials in R. For example `f1 = x^2-y` constructs the polynomial $f_1 = x^2 - y$ and *Macaulay2* knows this is a polynomial in R. The command `I = ideal(f1,f2)` creates the ideal $I = \langle f_1, f_2 \rangle \subset R$. In order to see the generators of I we use the command `gens I`, which displays the chosen generators as a row matrix.

(a) After running the above code explain the three displayed outputs.

(b) Write code that determines whether $y^2 + y - 1$ lies in the ideal I. What does this mean geometrically? *Hint:* One direct test is `isSubset(ideal(y^2+y-1),I)`, but can you find a more direct method?

(9) **A Second *Macaulay2* Session.** Now consider the parametrization of the affine twisted cubic. Before running the block, predict the two zero outputs and at least one generator of the kernel.

```
restart
A = QQ[x,y,z];
T = QQ[t];
phi = map(T,A,{t,t^2,t^3}); -- encode t |-> (t,t^2,t^3)
phi(y-x^2)
phi(z-x*y)
K = ker phi; -- compute the full ideal of relations
trim K
```

Here `ker phi` computes an ideal, while `trim K` asks for a smaller list of generators.

(a) *Macaulay2* lists the target of a ring map first and its source second. Explain why the line `map(T,A,{t,t^2,t^3})` above represents the map $A \to T$ used in Exercise 7.

(b) Explain the two zero outputs. Compare the generators in `trim K` with the generators found by hand. Why do the displayed lists not need to agree term for term?

(c) Use `K == ideal(y-x^2,z-x*y)` to check equality of the two ideals. What mathematical statement is being verified by the output `true`?

(d) Testing one polynomial at a time is already becoming tedious. A *Macaulay2* list is enclosed in braces. Enter

```
candidates = {y-x^2,z-x*y,x*z-y^2,x};
apply(candidates,g -> phi(g))
```

The arrow expression on that line defines a function, in this case, taking something named g to $\phi(g)$, i.e., evaluating the map ϕ defined above at the element g. The command `apply` applies a function to every element of a list or a sequence before returning the outputs as a new list or new sequence. (Note your original list remains unchanged.) Before running the second line, predict the list it should return.

(e) The second useful command to know is `scan`, which also applies a function to every entry of a list or sequence. However, `scan` discards the returned values, as such it useful when the function itself returns an output, such as printing. Run the following code.

```
scan(candidates,g -> print (g,phi(g) == 0))
```

What is the meaning of the output? How does the differ from the output you got when you ran the `apply` in the previous part?

(f) Add polynomial polynomial of your choosing to the list `candidates`, predict whether it belongs to $\ker(\phi)$, and use the same commands to test your prediction.

(g) *Further:* Also verify membership in K using `(x*z-y^2) % gb K`. Explain the mathematical meaning of each test.

In addition to being open source, much of the specialized functionality of *Macaulay2* is developed by researchers in commutative algebra and algebraic geometry through *packages*: collections of commands and documentation devoted to a particular area of mathematics or type of computation. These packages are written and maintained by members of the *Macaulay2* community, allowing the program to grow alongside current research. A list of the packages distributed with *Macaulay2*, organized by subject, is available in the package index.

Before using a package, we load it with the command `needsPackage`. For example, the command `needsPackage "Elimination"` loads the package `Elimination`. Loading a package makes all of the commands in the package available for the current session, however, we must load the package each time we restart if we wish to continue using it. Every *Macaualy2* package comes with extensive documentation, which includes an overview of the package, a list of functions, methods, and objects, and a host of examples. You access the package documentation via the command `viewHelp`, for example, `viewHelp "Elimination"`, which will open the documentation as an HTML file. I would encourage you to frequently consult and use the *Macaulay2* documentation; one rarely memorizes every command, and it often important to carefully understand exactly what a command is computing.

In the following exercise we will use the command `dim I`, which computes the Krull dimension of R/I for an ideal $I \subseteq R$. The precise definition of Krull dimension is beyond this course, but you should informally think of it as the number of locally independent parameters of the corresponding affine variety. For example, an ideal defining a non-empty finite set has Krull dimension zero, an ideal defining a curve has Krull dimension 1, and an ideal defining a surface has Krull dimension 2. (Note we define the dimension relative the the ground field $\mathbb{K}$, and so we think of $\mathbb{C}$ as being one-dimensional while $\mathbb{C}^2$ is two-dimensional. Although we should be careful not to confuse Krull dimension with vector space dimension. We will also use the command `degree I`, which computes the degree of R/I. When $\mathbb{V}(I)$ is zero-dimensional, `degree I` is equal to the $\mathbb{Q}$-vector-space dimension of R/I. As we will see, in this setting, we should roughly think of degree as counting the number of points in $\mathbb{V}(I)$ if we count the points with the correct multiplicity. Exercises 12 and 13 will make this precise by comparing the degree of an ideal defining four distinct points with the degree of an ideal defining a tangential double point.

---

(10) **A Third *Macaulay2* Session.** Let us return to the circle–parabola ideal and start a fresh *Macaulay2* session. Before running the block below, predict the two elimination generators, the dimension, and the degree.

```
restart
needsPackage "Elimination";
R = QQ[x,y];
I = ideal(x^2-y,x^2+y^2-1);

eliminate({y},I) -- algebraic equations involving only x
eliminate({x},I) -- algebraic equations involving only y

dim I == 0  -- check that the intersection is a finite set of points
degree I -- count (with multiplicity) the number of points in the
   intersection
```

(a) Match the two elimination ideals with the hand calculations in Exercise 1. Notice that eliminating y leaves equations in x, and eliminating x leaves equations in y.

(b) Give a geometric interpretation of the outputs of `dim I` and `degree I`. (Note there is something weird going on here; how many points does $x^2 - y$ and $x^2 + y^2 - 1$ intersect in over $\mathbb{Q}$, which is the field we are working over.)

(c) *Further: Closure Rather Than Image.* Set `H = ideal(x*y-1)` and eliminate y with the following command:

```
eliminate({y},H)
```

Compare the output with Exercise 1(g): which point has elimination added, and why?

(d) *Further: Components.* The following computation returns to the factorization examples from Exercise 3. Predict the visible components before asking the computer.

```
restart
needsPackage "MinimalPrimes";
R = QQ[x,y];

axes = ideal(x*y);
decompose axes -- the two irreducible components

twoLines = ideal((x-y)*(x+y));
decompose twoLines -- factorization becomes component decomposition
```

Compare the outputs with the decompositions found by hand. In Lecture 5, we will return to primary decomposition and to nonreduced equations, where they have a geometric job.

---

4. Coordinate Rings and Multiplicity

We now return to a distinction already encountered in the equality $\mathbb{V}(x) = \mathbb{V}(x^2)$. A polynomial $f \in \mathbb{K}[x_1, \ldots, x_n]$ restricts to a function $X \to \mathbb{K}$ on any subset $X \subseteq \mathbb{K}^n$. We say that a function $\phi : X \to \mathbb{K}$ is a *polynomial function on* X if there exists a polynomial $f \in \mathbb{K}[x_1, \ldots, x_n]$ such that $\phi = f|_X$. Note sometimes it is possible for polynomial functions to hide, and not look like polynomials. Consider for example the function $(x, y) \mapsto e^{x^2+y^2}$. This is a well-definined function on all of $\mathbb{R}^2$, but it is certainly not a polynomial function on $\mathbb{R}^2$. However, if we restrict this an consider it as a function on the unit circle, $\mathbb{V}_{\mathbb{R}}(x^2 + y^1 - 1)$, it is a polynomial function. Can you find the polynomial f it agrees with on $\mathbb{V}_{\mathbb{R}}(x^2 + y^2 - 1)$?

Since function addition, subtraction, and multiplication are defined point-wse, the set of all functions $X \to \mathbb{K}$ forms a ring. (You should think about this carefully if it is not obvious.) Further, more, since restriction commutes with these operations the subset of polynomial functions on X forms a subring. This motivates the following definition.

Definition 8. The *coordinate ring* of an affine algebraic variety $X \subseteq \mathbb{K}^n$, which we denote by $\mathbb{K}[X]$, is the ring of polynomial functions on X under point-wise addition and multiplication.

While the above definition makes it seem very abstract, the coordinate ring of a variety X, has a beautiful and simple description. Namely, by definition there is a surjective map of rings:

$$\mathbb{K}[x_1, \ldots, x_m] \xrightarrow{\rho} \mathbb{K}[X] f \longmapsto f|_X$$

where $f|_X$ is the polynomial function on X determined by f, i.e. $p \mapsto f(p)$. A polynomial $f \in \mathbb{K}[x_1, \ldots, x_n]$ restricts to give the zero function on X if and only if $f(p) = 0$ for all $p \in X$. Hence a polynomial f is in the kernel of ρ if and only if f vanishing on all of X, i.e. if and only if $f \in \mathbb{I}(X)$. Hence the kernel of ρ is precisely $\mathbb{I}(X)$. Using everyone's favorite isomorphism theorem we get:

$$\mathbb{K}[X] \coloneqq \mathbb{K}[x_1, \ldots, x_n]/\mathbb{I}(X).$$

Since $\mathbb{I}(X)$ is a radical ideal the coordinate ring of X is always reduced, i.e., it has no nonzero nilpotent elements Further, if $F : X \to Y$ is a map of varieties that is given in coordinates by polynomials and $\phi : Y \to \mathbb{K}$ is a polynomial function then the composition $\phi \circ F : X \to \mathbb{K}$ is a polynomial function. Hence F "pullsback" polynomial functions and induces a ring homomorphism

$$F^* : \mathbb{K}[Y] \longrightarrow \mathbb{K}[X], \qquad \text{given by} \qquad [h] \longmapsto [h \circ F].$$

Notice this ring homomorphism goes in the opposite direction to the map of algebraic varieties.

Now set $S = \mathbb{K}[x_1, \ldots, x_n]$ and suppose that $X = \mathbb{V}(I)$ for a specified ideal $I \subseteq S$. We wish to distinguish the coordinate ring $\mathbb{K}[X]$ from the quotient S/I attached to the ideal with which we began. The inclusion $I \subseteq \mathbb{I}(\mathbb{V}(I))$ induces a canonical surjection

$$S/I \longrightarrow \mathbb{K}[\mathbb{V}(I)] = S/\mathbb{I}(\mathbb{V}(I)).$$

Its kernel is $\mathbb{I}(\mathbb{V}(I))/I$. When $\mathbb{K}$ is algebraically closed, the Nullstellensatz identifies this map with

$$S/I \longrightarrow S/\sqrt{I},$$

whose kernel $\sqrt{I}/I$ consists precisely of the nilpotent elements of S/I. Indeed, the class of f in S/I is nilpotent exactly when $f^m \in I$ for some $m \geq 1$.

This gives a precise meaning to the statement that nilpotents cannot be detected by points. Every $p \in \mathbb{V}(I)$ determines an evaluation homomorphism

$$\mathrm{ev}_p : S/I \longrightarrow \mathbb{K}, \qquad \overline{f} \longmapsto f(p).$$

Every nilpotent is sent to zero by every such homomorphism, since a field has no nonzero nilpotents. Conversely, when $\mathbb{K}$ is algebraically closed, the Nullstellensatz says that an element annihilated by every point evaluation must be nilpotent. Thus nilpotents are not additional points hidden from the picture; they are algebraic information retained by S/I but lost when we pass to the underlying solution set. In the finite intersections below, this information records repeated vanishing and tangency.

The algebraically closed hypothesis in the converse is essential. For example, $\langle x^2 + y^2 \rangle$ is radical in $\mathbb{R}[x, y]$, but its real zero set is only the origin, whose vanishing ideal is $\langle x, y \rangle$. Over a non-algebraically closed field, passing to the set of $\mathbb{K}$-valued solutions may therefore forget more than nilpotents.

---

(11) **Coordinate Rings, Nilpotents, and Points.**

Let $S = \mathbb{K}[x_1, \ldots, x_n]$.

(a) Let $I \subseteq S$. Prove that S/I is an integral domain if and only if I is prime. Show that the nilpotent elements of S/I are precisely the classes $f + I$ with $f \in \sqrt{I}$. Deduce that S/I is reduced if and only if I is radical.

(b) Let X be a nonempty algebraic set. Using $\mathbb{V}(fg) = \mathbb{V}(f) \cup \mathbb{V}(g)$, prove that

$$\begin{aligned} X \text{ is irreducible} &\iff \mathbb{I}(X) \text{ is prime} \\ &\iff \mathbb{K}[X] \text{ is an integral domain.} \end{aligned}$$

Notice that this equivalence does not require $\mathbb{K}$ to be algebraically closed.

Hint: For the reverse implication, if $X = X_1 \cup X_2$ with both $X_i \subsetneq X$, choose $f_i \in \mathbb{I}(X_i) \setminus \mathbb{I}(X)$.

(c) Let $I \subseteq S$ and suppose that $X = \mathbb{V}(I)$ is nonempty. Show that the natural map $q_I : S/I \longrightarrow \mathbb{K}[X]$ has kernel $\mathbb{I}(X)/I$.

(d) Show that every $p \in X$ induces an evaluation homomorphism $\mathrm{ev}_p : S/I \longrightarrow \mathbb{K}$ and that

$$\ker(q_I) = \bigcap_{p \in X} \ker(\mathrm{ev}_p).$$

(e) Assuming that $\mathbb{K}$ is algebraically closed, use the Nullstellensatz to conclude that this intersection is precisely the set of nilpotent elements of S/I.

(12) **A Double Point from Tangency.** Let $A_1 = \mathbb{K}[x]/\langle x \rangle$ and $A_2 = \mathbb{K}[x]/\langle x^2 \rangle$. Notice that both $\langle x \rangle$ and $\langle x^2 \rangle$ define the single point $\{0\} \subseteq \mathbb{K}$.

(a) Prove that $\overline{x} \neq 0$ in A_2 while $\overline{x}^2 = 0$. Identify the kernel of the natural map $A_2 \longrightarrow A_1 \cong \mathbb{K}[\{0\}]$, and explain why evaluation at the unique point cannot detect $\overline{x}$.

(b) The preceding algebra occurs geometrically when curves meet tangentially. The parabola $y = x^2$ is tangent to the line $y = 0$ at the origin. Set

$$I_{\tan} = \langle x^2 - y, y \rangle \subseteq \mathbb{C}[x, y].$$

Show that $\mathbb{C}[x, y]/I_{\tan} \cong \mathbb{C}[x]/\langle x^2 \rangle$.

(c) Compute $\mathbb{V}(I_{\tan})$, $\sqrt{I_{\tan}}$, and $\mathbb{C}[\mathbb{V}(I_{\tan})]$.

(d) Find a basis for $\mathbb{C}[x, y]/I_{\tan}$ as a $\mathbb{C}$-vector space and identify its nonzero nilpotent classes.

(e) Explain why the underlying intersection consists of only the origin, while the quotient ring has dimension 2 as a $\mathbb{C}$-vector space. Since this is the only intersection point, this dimension is its intersection multiplicity: the origin is therefore *counted twice*.

(13) **Four Distinct Intersection Points.** Consider the intersection of the parabola $y = x^2$ with the circle $x^2 + y^2 = 1$, and set

$$I_{\text{circ}} = \langle x^2 - y, x^2 + y^2 - 1 \rangle \subseteq \mathbb{C}[x, y].$$

(a) Show that $\mathbb{C}[x, y]/I_{\text{circ}} \cong \mathbb{C}[x]/\langle x^4 + x^2 - 1 \rangle$.

(b) Deduce that the classes of $1, x, x^2, x^3$ form a $\mathbb{C}$-basis for this quotient, and use $y = x^2$ in the quotient to represent this basis by the classes of $1, x, y, xy$.

(c) Recall from Exercise 1 that $x^4 + x^2 - 1$ has four distinct complex roots $\alpha_1, \ldots, \alpha_4$. Use the Chinese remainder theorem to prove that evaluation induces an isomorphism

$$\mathbb{C}[x]/\langle x^4 + x^2 - 1 \rangle \longrightarrow \mathbb{C}^4, \qquad r(x) \longmapsto (r(\alpha_1), \ldots, r(\alpha_4)).$$

(d) Conclude that $\mathbb{C}[x, y]/I_{\text{circ}}$ is reduced and that the intersection consists of the four distinct points (α_i, α_i^2) for $i = 1, 2, 3, 4$.

(e) Compare this intersection with the tangent intersection in Exercise 12. Explain precisely why the tangent intersection is described as *one point counted twice*, whereas the circle–parabola intersection consists of *four distinct points*. Your answer should compare the underlying point sets, the vector-space dimensions of the quotient rings, and the presence or absence of nonzero nilpotents.

Exercises 12 and 13 are two halves of one classical statement called Bézout's theorem. Roughly, Bézout's theorem states that two plane curves of degrees d and e with no common component intersect in exactly de points. Thus, a conic and a line intersect in exactly $2 \cdots 1 = 2$ points, while two conics meet in $2 \cdot 2 = 4$ points. Note this can be thought of as a generalization of the fact you might have learned in high school: the degree of a polynomial is equal to the number of intersection points of the graph of the polynomial with a generic horizontal line.

As the previous exercises highlight, in order to make bëzout's theorem true three refinements are needed to how we could intersection points: (i) we must work over an algebraically closed field, (ii) we must count points with multiplicity, and (iii) we must work retain points that inersect "at infiniy", which means working in projective space. We will see more about projective space in the later lectures. While this theorem is normally credit to Bézout's workin in 1779, the result is also stated, without proof, in Newton's Principia in 1687

---

(14) **The Affine Dictionary.** Assuming $\mathbb{K}$ is algebraically closed, return to the opening table and complete the algebraic column using $1 \in I$, $\sqrt{I}$, elimination ideals, finite quotient dimension, primality of $\sqrt{I}$, and kernels. Fill in the *Macaulay2* entries learned so far: `eliminate`, `dim`, `degree`, and `ker`; if you completed the *Further: Components* part, also include `decompose`. Mark the rows where an output may describe a Zariski closure rather than the exact image. We will return to the remaining entries in the subsequent lectures.

LECTURE 2: DIVISION AND HIDDEN EQUATIONS (V1)

Throughout this lecture, $\mathbb{K}$ denotes a field. In Lecture 1 we studied the affine twisted cubic $C \subset \mathbb{K}^3$ in detail and saw that it was given by $\mathbb{V}(I)$, where $I = \langle x^2 - y, xy - z \rangle$. While studying this variety we found the following identity

$$xz - y^2 = y(x^2 - y) - x(xy - z),$$

which certifies that $xz - y^2 \in I$. Recall geometrically this means that the polynomial $xz - y^2$ vanishes along the twisted cubic. That said, how could we have found this certificate, i.e., the coefficients y and $-x$, systematically? More importantly, if no such identity exists, how could we ever prove nonmembership?

This is the *ideal-membership problem*: Given an ideal $I = \langle g_1, \ldots, g_s \rangle \subseteq \mathbb{K}[x_1, \ldots, x_n]$ and a polynomial $f \in \mathbb{K}[x_1, \ldots, x_n]$, determine whether $f \in I$. By definition, $f \in I$ if and only if there exist polynomials $q_1, \ldots, q_s \in \mathbb{K}[x_1, \ldots, x_n]$ such that

$$f = q_1 g_1 + \cdots + q_s g_s.$$

Thus an explicit representation of the above form *certifies* or is a *certificate of* membership of f in I. Notice given the q_i checking such an identity is true is easy as all we have to do is multiple and add everything out and see it gives f. Hence the difficulty behind the ideal-membership problem is that the polynomials q_i are unknown, and it seems like the definition of I gives no obvious finite search process. What we need is a some form of constructive algorithm that terminates and produces either a certificate as above (proving $f \in I$) or a conclusive obstruction showing that $f \notin I$.

We have already seen two cases where we have essentially such an algorithm: (i) For systems of linear equations, Gaussian elimination isolates the leading variables, and eventually reduces them to echelon form ii) For polynomials in one-variable the trusty polynomial long division reduces the question of whether the remainder is zero is not. We will solve the ideal-membership problem in general by generalizing the division algorithm, by asking what makes one-variable division work, imitated it as literally as possible, seeing why these imitations fail before making the needed changes.

(1) **What Would an Ideal-Membership Algorithm Decide?** Return to the table near the beginning of Lecture 1.

   (a) Which rows can be approached through membership tests? Identify separately the roles played by $1 \in I$, $f \in I$, and $f \in \sqrt{I}$.

   (b) Explain why $f \in I$ always implies that f vanishes on $\mathbb{V}_{\mathbb{K}}(I)$. When $\mathbb{K}$ is algebraically closed, what does Hilbert's Nullstellensatz say if f vanishes on $\mathbb{V}_{\mathbb{K}}(I)$? Why does its conclusion place f in $\sqrt{I}$, rather than necessarily in I?

© Juliette Bruce 2026.
Licensed under Creative Commons BY-NC-SA 4.0.

(c) Explain why $1 \in I$ always forces $\mathbb{V}_{\mathbb{K}}(I) = \varnothing$. Which hypothesis is required for the converse?

(d) Suppose that you search for polynomials q_i satisfying $f = \sum_i q_i g_i$ and do not find them. Why does this not prove that $f \notin I$? What additional output would turn the failed search into a finite certificate of nonmembership? At this stage, you are not expected to name such an output; describe only what it would need to establish.

The preceding exercise exposes the computational difficulty behind the dictionary. A successful search for $q_1, \ldots, q_s$ produces a checkable certificate of membership, but an unsuccessful search says nothing: we may simply not have searched far enough. We need a systematic way to conduct our search. To begin building such an approach, let us consider a far simpler special case of the problem: principal ideals. Let $\langle g_1 \rangle$ be a principal ideal. Given an element $f \in \mathbb{K}[x_1, \ldots, x_n]$, we know that $f \in \langle g_1 \rangle$ if and only if $f = q_1 g_1$ for some $q_1 \in \mathbb{K}[x_1, \ldots, x_n]$, i.e., if and only if f is divisible by g_1. Hence, for principal ideals, the ideal-membership problem really amounts to finding a systematic way to perform division.

Luckily for us, we are all familiar with at least one special case in which we understand division: polynomials in one variable. Even better, since $\mathbb{K}[x]$ is a principal ideal domain (PID), every ideal is principal. As we will see shortly, division gives a beautiful answer to the ideal-membership question in $\mathbb{K}[x]$. We will then attempt to generalize this to $\mathbb{K}[x_1, \ldots, x_n]$, where ideals need no longer be principal and we do not (yet) have a familiar notion of polynomial division. This leads us naturally to monomial orders, Gröbner bases, normal forms, and beyond.

However, let us not get ahead of ourselves. We begin by considering the one-variable case in depth, adopting the convention that $\deg(0) = -\infty$.

Theorem 1 (Polynomial Division Algorithm)**.** *Let $g \in \mathbb{K}[x]$ be nonzero. For every $f \in \mathbb{K}[x]$ there are unique polynomials $q, r \in \mathbb{K}[x]$ such that*

$$f = qg + r \qquad \textit{and} \qquad \deg(r) < \deg(g).$$

The usual proof already contains an algorithm. For $0 \neq h$, let $\mathrm{LT}(h)$ denote its term of highest degree. Define two new polynomials, the 'working" polynomials p and the quotient polynomial q, which we shall update at each step of our algorithm. Initially define them by $p = f$ and $q = 0$. Now if $\deg(p) \geq \deg(g)$, then define

$$m = \frac{\mathrm{LT}(p)}{\mathrm{LT}(g)}.$$

Subtracting mg cancels the leading term of p, so the new working polynomial is either zero or has smaller degree. We repeat this until either $p = 0$ or $\deg(p) > \deg(g)$. We may write this as an algorithm as the following.

Algorithm: One-Variable Division
Input: $f, g \in \mathbb{K}[x]$ with $g \neq 0$.
Initialize: $q \leftarrow 0$ and $p \leftarrow f$.
Repeat: While $p \neq 0$ and $\deg(p) \geq \deg(g)$, set

$$m \leftarrow \frac{\mathrm{LT}(p)}{\mathrm{LT}(g)}, \qquad q \leftarrow q + m, \qquad p \leftarrow p - mg.$$

Output: q and $r \leftarrow p$.

Notice that at every stage of the algorithm the bookkeeping preserves the identity

$$f = qg + p,$$

and after each update the degree of p strictly decreases. Since the degree of a polynomial is a finite positive number, the well-ordering prinicipal for the natural numbers implies that eventually the algorithm must stop. Exercise 2 asks you to carefully check both of these claims, while Exercise 3(a) isolates what makes the remainder output by the algorithm unique.

The division algorithm also gives us a unique way to represent classes in the quotient ring $\mathbb{K}[x]/\langle g \rangle$. In particular, given $f \in \mathbb{K}[x]$, if we write $f = qg + r$ as in the division algorithm, then the remainder r is the unique representative of the class of f in $\mathbb{K}[x]/\langle g \rangle$ whose degree is smaller than $\deg(g)$.

This gives the remainder a geometric interpretation. Every class in $\mathbb{K}[x]/\langle g \rangle$ determines a polynomial function on $\mathbb{V}(g)$, because every element of $\langle g \rangle$ vanishes there. The phrase "polynomial function" requires some care: distinct classes may determine the same function unless $\langle g \rangle = \mathbb{I}(\mathbb{V}(g))$. The equality $f \equiv r \pmod{\langle g \rangle}$ means that the classes of f and r determine the same function on $\mathbb{V}(g)$. For example, if $g = x^2 - 1$ and $\mathrm{char}(\mathbb{K}) \neq 2$, then $\langle g \rangle = \mathbb{I}(\{-1, 1\})$.

---

(2) **Division and Polynomial Functions on Two Points.** Carefully apply the division algorithm to divide $f = x^5 - 1$ by $g = x^2 - 1$.

(a) Make a table recording the values of p and q at each stage. Verify that at every stage of the algorithm we have the identity $f = qg + p$.

(b) Find the final quotient q and remainder r.

(c) Explain why the degree of p strictly decreases after every pass through the loop.

(d) Assume $\mathrm{char}(\mathbb{K}) \neq 2$. Evaluate both f and the remainder r at $x = -1$ and $x = 1$. Why do both f and r determine the same function on $\mathbb{V}(x^2 - 1)$ even though they are different polynomials?

(e) *Further: Characteristic* 2. Using the fact that $x^2 - 1 = (x - 1)^2$, prove that the class of $x - 1$ in $\mathbb{K}[x]/\langle x^2 - 1 \rangle$ is nonzero and nilpotent. How does this recover the distinction between the

coordinate ring of $\{1\}$ and the quotient ring $\mathbb{K}[x]/\langle x^2-1\rangle$? How does this comapre to the tangent-double point exercise in Lecture 1?

(3) **Ideal Membership in One Variable.** Let $g \in \mathbb{K}[x]$ be nonzero and $f \in \mathbb{K}[x]$.

(a) Prove the uniqueness of the remainder in the division algorithm: if $f = qg + r = q'g + r'$ and each remainder is either zero or has degree strictly less than $\deg(g)$, show that $q = q'$ and $r = r'$.

(b) Carefully prove that $f \in \langle g \rangle$ if and only if the remainder of f divided by g is zero.

(c) Assume that an ideal of $\mathbb{K}[x]$ is presented by a generator g. Describe an algorithm that decides whether a given f lies in $\langle g \rangle$. Why does the fact that $\mathbb{K}[x]$ is a PID mean that this method applies to every ideal once a generator has been found?

(d) *Further: Ideals in* $\mathbb{K}[x]$. Let $I \subseteq \mathbb{K}[x]$ be a nonzero ideal and choose a nonzero element $g \in I$ of least degree. Divide an arbitrary $f \in I$ by g and prove that $I = \langle g \rangle$.

(4) **One-Variable Division in *Macaulay2*.** Return to the browser-based *Macaulay2* session from Lecture 1 and start again with `restart`. In one variable, `//` returns the quotient and `%` returns the remainder. Predict both outputs from Exercise 2 before entering the following block.

```
restart
R = QQ[x];
f = x^5-1;
g = x^2-1;

q = f//g -- quotient in the identity f=qg+r
r = f%g  -- normal form of f modulo the ideal (g)

f == q*g+r -- verify the division identity
```

(a) Compare the outputs with your hand computation. What does the final output `true` verify?

(b) Replace f and g by $x^{1000}-1$ and x^2-1. Predict the remainder before asking the computer. Interpret the result as a statement about the values of $x^{1000}-1$ on $\mathbb{V}_{\mathbb{C}}(x^2-1)$.

(c) Explain why the test `f%g == 0` decides membership in $\langle g \rangle \subseteq \mathbb{K}[x]$. Which part of Theorem 1 is being used?

(d) *Further: The Factor Theorem and a Bound on Roots.* For $a \in \mathbb{K}$, use division by $x-a$ to prove that $f(a)$ is the remainder and deduce that $a \in \mathbb{K}$ is a root of f if and only if f is divisible by $(x-a)$. Then prove that a nonzero polynomial in $\mathbb{K}[x]$ has at most $\deg(f)$ roots in $\mathbb{K}$. This is the one-variable fact used in Lecture 1 to show that a polynomial vanishing at every point of an infinite field is zero; induction on the number of variables gives the corresponding statement for $\mathbb{K}^m$.

Who would have guessed that polynomial division would be so useful? I certainly did not when I first learned about it in high school. We have just seen that the polynomial division algorithm in $\mathbb{K}[x]$ (i) answers the ideal-membership problem for $\mathbb{K}[x]$, (ii) gives us a canonical way to represent classes in quotient rings $\mathbb{K}[x]/\langle g\rangle$, and (iii) proves the factor theorem, among other things. The natural question is whether these ideas extend to several variables.

What prevents us from repeating this calculation in several variables? Let us again return to our running example of the affine twisted cubic, defined by the ideal $I = \langle x^2 - y, xy - z\rangle \subseteq \mathbb{K}[x, y, z]$, and the relation

$$xz - y^2 = y(x^2 - y) - x(xy - z),$$

certifying that $xz - y^2 \in I$. If we wished to obtain this relation from a division algorithm, we would encounter several issues. First, the ideal I is not principal, so we need a notion of division that allows us to divide $xz - y^2$ by two polynomials simultaneously—namely, $x^2 - y$ and $xy - z$—rather than by a single polynomial. Second, if we are to invent such a division algorithm, we need a way to define the leading term of $xz - y^2$. Both monomials in this polynomial have degree two! It is therefore unclear which one the division algorithm should try to cancel first.

We therefore need to make two choices that were invisible in one variable: an order on the monomials and an ordered list of divisors. These choices are not merely bookkeeping. Different ways of declaring a term to be largest can expose different consequences of the same equations, just as different elimination steps can expose different features of a linear system.

Let us turn to the first issue: ordering the monomials in a way that lets us choose between the two terms of $xz - y^2$.

Definition 2. A *monomial order* on $\mathbb{K}[x_1, \ldots, x_n]$ is a relation $<$ on the set of monomials satisfying the following conditions:

(i) $<$ is a strict total order;

(ii) if $m_1 < m_2$ and m is any monomial, then $mm_1 < mm_2$; and

(iii) $<$ is a well-order.

Each condition does a specific job. The first says that any two monomials can be compared: for all monomials m_1 and m_2, exactly one of $m_1 < m_2$, $m_1 = m_2$, $m_2 < m_1$ holds. In particular, every nonzero polynomial has a unique largest term, which is what we set out to obtain. The second says that an inequality survives multiplication by a common monomial; since division proceeds by subtracting monomial multiples of the divisors, this is what keeps a division step from disturbing the order. The third says that every nonempty collection of monomials has a smallest element, or equivalently that there is no infinite strictly decreasing chain $m_1 > m_2 > m_3 > \cdots$. It will replace the fact, used in one-variable division, that a decreasing sequence of nonnegative integer degrees must terminate, and it is what will make our division algorithm stop.

These three conditions are not independent. Given (i) and (ii), condition (iii) is equivalent to requiring that $1 < m$ for every monomial $m \neq 1$. Indeed, if $m < 1$, then multiplying repeatedly by m produces the infinite descending chain $\cdots < m^3 < m^2 < m < 1$; the converse implication rests on Dickson's Lemma, which we shall encounter in the next lecture.

---

(5) **Non-Principal Ideals.**

(a) Prove that the ideal $\langle x, y\rangle \subseteq \mathbb{K}[x, y]$ is not principal. Hint: if $\langle x, y\rangle = \langle g\rangle$, then g divides both x and y. What are their common divisors? Why can the ideal not be the whole ring?

(b) Show directly that over every field $\mathbb{I}(\{(0, 0)\}) = \langle x, y\rangle$. Explain why part (a) says that the ideal of *all* equations vanishing at the origin cannot be generated by one equation.

(c) Why does this not, by itself, rule out the possibility that a single polynomial might have the origin as its set of $\mathbb{K}$-valued zeros? Hint: Describe $\mathbb{V}_{\mathbb{R}}(x^2 + y^2)$.

(6) **Implications of Monomial Orders.**

(a) Deduce from the well-ordering axiom that there is no infinite strictly decreasing sequence of monomials.

(b) Prove that 1 is the smallest monomial in every monomial order. Hint: if $x^\alpha < 1$, use compatibility with multiplication and part (a) to compare $1, x^\alpha, x^{2\alpha}, \ldots$.

(c) Show that ordering monomials in $\mathbb{K}[x, y]$ by total degree is not a total order.

(d) Compare part (a) with the degree drop in one-variable division. What property of a monomial order will replace "the degree strictly decreases" in the proof of termination?

The definition above is appealing because it is intrinsic: it orders the monomials themselves, without first identifying them with anything else. To construct such orders, however, it is far easier to work with exponent vectors. Throughout, we write $\mathscr{M}_n$ for the set of monomials of $\mathbb{K}[x_1, \ldots, x_n]$, which is a commutative monoid under multiplication, just as $\mathbb{Z}^n_{\geq 0}$ is one under addition.

Definition 3. A *monomial order* on $\mathbb{Z}^n_{\geq 0}$ is a relation $\prec$ on $\mathbb{Z}^n_{\geq 0}$ satisfying the analogous conditions:

(i) $\prec$ is a strict total order;

(ii) if $\alpha \prec \beta$ and $\gamma \in \mathbb{Z}^n_{\geq 0}$ then $\alpha + \gamma \prec \beta + \gamma$; and

(iii) $\prec$ is a well-order.

This definition should look very similar to the preceding one. In fact, it is the same definition transported along a dictionary between $\mathscr{M}_n$ and $\mathbb{Z}^n_{\geq 0}$. Let us make that precise. Let $\Phi\colon \mathscr{M}_n \to \mathbb{Z}^n_{\geq 0}$ be a multiplicative bijection, meaning that $\Phi(mm') = \Phi(m) + \Phi(m')$. Given a monomial order $\prec$

on $\mathbb{Z}^n_{\geq 0}$, define a relation $<$ on $\mathscr{M}_n$ by $m_1 < m_2$ if and only if $\Phi(m_1) \prec \Phi(m_2)$. One checks that $<$ is indeed a monomial order.

Thus, specifying a monomial order in this way amounts to specifying two pieces of data: the isomorphism Φ and an order $\prec$ on $\mathbb{Z}^n_{\geq 0}$. The first piece is much simpler than it looks. Since every monomial is a product of variables and Φ is multiplicative, Φ is determined by its values on the variables. Further, the bijection requires Φ to take irreducible monomials to irreducible vectors, i.e., vectors that cannot be written as the sum of two nonzero vectors in $\mathbb{Z}^n_{\geq 0}$. Hence $\Phi(x_i)$ must equal e_j for some j.

Choosing Φ is therefore nothing more than choosing which variable is recorded in which coordinate. This is why a monomial order is customarily specified in two steps: first an ordering of the variables, say $x > y > z$, and then an ordering of $\mathbb{Z}^n_{\geq 0}$. Unless we say otherwise, we use the tautological identification $\Phi(x^\alpha) = \alpha$. We write $\alpha \succ \beta$ for $\beta \prec \alpha$ and $|\alpha| = \alpha_1 + \cdots + \alpha_n$ for the total degree.

There are numerous monomial orders. For example, *Macaulay2* has at least twelve families of them, each of which can highlight a slightly different algebraic or geometric feature of a problem, as we shall see later. In this course we will largely restrict ourselves to the three most commonly used monomial orders. We informally refer to $|\alpha|$ as the (total) degree of α. (Think about why this is.)

(a) *Lexicographic order (Lex).* For distinct $\alpha, \beta \in \mathbb{Z}^n_{\geq 0}$, we declare $\alpha >_{\text{lex}} \beta$ if the leftmost nonzero entry of $\alpha - \beta$ is positive. In other words, in lex we compare α_1 and β_1; if they agree, we compare α_2 and β_2, and so on until the first difference appears.

(b) *Graded lexicographic order (GrLex).* We declare $\alpha >_{\text{grlex}} \beta$ if $|\alpha| > |\beta|$, or if $|\alpha| = |\beta|$ and $\alpha >_{\text{lex}} \beta$. Thus graded lex first compares total degree and uses lex to break ties.

(c) *Graded reverse lexicographic order (GRevLex).* We declare $\alpha >_{\text{grevlex}} \beta$ if $|\alpha| > |\beta|$, or if $|\alpha| = |\beta|$ and the rightmost nonzero entry of $\alpha - \beta$ is negative. Thus, when the total degrees tie, we scan the vectors from the right; at the first place where they differ, the vector with the *smaller* entry is declared larger.

The last two compare total degree first, so both refine the grading; they differ only in how they break a tie. Graded lexicographic order then prefers the monomial with the larger power of the earliest variable, while graded reverse lexicographic order prefers the one with the *smaller* power of the latest variable. Two warnings about the name are in order. Graded reverse lexicographic order is not the reverse of graded lexicographic order, nor is it obtained by reversing the variables; the sign condition and the scanning direction are both flipped, and for $n \leq 2$ the two orders in fact coincide. Notation also varies: grlex means graded lexicographic order in some sources and graded reverse lexicographic order in others, and deglex and $\mathrm{degrevlex}$ are common alternatives.

Each of these is a monomial order on $\mathbb{Z}^n_{\geq 0}$, and all three order the standard basis vectors as

$$e_1 > e_2 > \cdots > e_n.$$

What differs is how that comparison is extended to vectors of higher degree. To obtain monomial orders on $\mathbb{K}[x_1, \ldots, x_n]$, we must specify a multiplicative bijection $\Phi\colon \mathscr{M}_n \to \mathbb{Z}^n_{\geq 0}$ as discussed above. In practice, mathematicians encode Φ by naming a variable order and simply say, for example, lex with $x > y > z$ or lex with $y > x > z$. Let us look at a few examples to make this clearer.

Consider the monomial orders on $\mathbb{K}[x, y, z]$ arising from the variable orders $x > y > z$, $z > y > x$, and $y > z > x$, for lex, graded lex, and graded reverse lexicographic order.

- $x > y > z$: This variable order specifies the familiar bijection $\Phi(x^a y^b z^c) = (a, b, c)$, so that $\Phi(x) = (1, 0, 0)$, $\Phi(y) = (0, 1, 0)$, and $\Phi(z) = (0, 0, 1)$.
- $z > y > x$: Here $\Phi(x^a y^b z^c) = (c, b, a)$, so $\Phi(x) = (0, 0, 1)$, $\Phi(y) = (0, 1, 0)$, and $\Phi(z) = (1, 0, 0)$.
- $y > z > x$: Here $\Phi(x^a y^b z^c) = (b, c, a)$, so $\Phi(x) = (0, 0, 1)$, $\Phi(y) = (1, 0, 0)$, and $\Phi(z) = (0, 1, 0)$.

Now that we understand the three bijections, let us compare the corresponding monomial orders by arranging all monomials of degree at most 2 in the nine possible combinations. Within each cell, the line breaks are purely typographical: every new line continues the same decreasing chain.

	Variable order		
Monomial order	$x > y > z$	$z > y > x$	$y > z > x$
Lex	$x^2 > xy > xz > x$ $y^2 > yz > y$ $z^2 > z > 1$	$z^2 > yz > xz > z$ $y^2 > xy > y$ $x^2 > x > 1$	$y^2 > yz > xy > y$ $z^2 > xz > z$ $x^2 > x > 1$
GrLex	$x^2 > xy > xz$ $y^2 > yz > z^2$ $x > y > z > 1$	$z^2 > yz > xz$ $y^2 > xy > x^2$ $z > y > x > 1$	$y^2 > yz > xy$ $z^2 > xz > x^2$ $y > z > x > 1$
GRevLex	$x^2 > xy > y^2$ $xz > yz > z^2$ $x > y > z > 1$	$z^2 > yz > y^2$ $xz > xy > x^2$ $z > y > x > 1$	$y^2 > yz > z^2$ $xy > xz > x^2$ $y > z > x > 1$

---

(7) **Reading Monomial Orders.** Work in $\mathbb{K}[x, y, z]$ with $x > y > z$.

(a) Arrange the following monomials from largest to smallest in lex, graded lex, and graded reverse lexicographic order:

$$x^2, \quad xy, \quad xz, \quad y^2, \quad y^3, \quad z^2.$$

(b) In each order above, compare x^2 and y^3. Why does lex privilege the first variable while the graded orders first privilege total degree?

(c) Arrange all monomials of total degree at most 3 in $\mathbb{K}[x, y]$ in lex order. Use the definition to explain why $y^k < x$ for every $k \geq 1$. Thus every monomial involving only y is smaller than every monomial involving x. Why might this make lex useful for exposing consequences that eliminate x?

(d) Verify directly from the definitions that lex, graded lex, and graded reverse lexicographic order are compatible with multiplication.

(e) *Further: Verifying That These Are Monomial Orders.* Show that graded lexicographic and graded reverse lexicographic orders are well-orders by first choosing the least total degree occurring in a nonempty set of monomials and using the fact that only finitely many monomials have that degree. For lex, argue by induction on the number of variables: choose the least first coordinate which occurs and then apply the induction hypothesis to the remaining coordinates. Where does the proof use that every exponent is nonnegative?

Our goal in introducing monomial orders is to give each nonzero polynomial a unique distinguished leading term, just as we were able to do in one variable. This is the analogue of choosing a pivot in a row of a matrix. Toward this end, we introduce the following definition.

Definition 4. Let $<$ be a monomial order on $\mathbb{K}[x_1, \ldots, x_n]$. Let $0 \neq f \in \mathbb{K}[x_1, \ldots, x_n]$ and write $f = \sum_{i=1}^{t} c_i x^{\alpha_i}$, where $c_i \in \mathbb{K}^\times$ and the monomials x^{α_i} are distinct. Suppose that x^{α_j} is the largest monomial appearing in f with respect to $<$. The *leading monomial*, *leading coefficient*, and *leading term* of f with respect to $<$ are

$$\mathrm{LM}_<(f) = x^{\alpha_j}, \qquad \mathrm{LC}_<(f) = c_j, \qquad \mathrm{LT}_<(f) = c_j x^{\alpha_j}.$$

When the intended monomial order is clear, we often write $\mathrm{LT}(f)$, $\mathrm{LM}(f)$, and $\mathrm{LC}(f)$ for $\mathrm{LT}_<(f)$, $\mathrm{LM}_<(f)$, and $\mathrm{LC}_<(f)$.

---

(8) **Examples of Leading Terms.** Consider $f = x - y^2$ and $h = xz - y^2$ as elements of $\mathbb{K}[x, y, z]$.

(a) Compute the leading term, leading coefficient, and leading monomial of both f and h with respect to lex, graded lex, and graded reverse lexicographic order, with $x > y > z$.

(b) Run the following in a new *Macaulay2* session. After running the code, replace `Lex` first by `GLex` and then by `GRevLex`, restarting each time. Compare the three outputs with your answers to part (a).

```
restart
R = QQ[x,y,z,MonomialOrder=>Lex];
f = x-y^2;    -- the equation y^2=x in the xy-plane
h = x*z-y^2; -- the hidden twisted cubic relation

{leadMonomial f,leadCoefficient f,leadTerm f}
leadMonomial h
```

(c) In *Macaulay2*, we give a one-input function a name using the arrow notation

```
name = input -> output.
```

A parenthesized block may contain several statements separated by semicolons; the final expression without a semicolon is the value returned by the function. For example, the following function returns the list $\{\mathrm{LM}(p), \mathrm{LC}(p), \mathrm{LT}(p)\}$:

```
leadData = p -> (
    LMp := leadMonomial p;
    LCp := leadCoefficient p;
    LTp := leadTerm p;
    {LMp, LCp, LTp}
)
leadData f
apply({f,h},p -> leadData p)
```

Predict both outputs before running the code, then explain how the final line applies the function to every entry of the list `{f,h}`.

(9) **Working with Leading Terms.** Let $u, v \in \mathbb{K}[x_1, \ldots, x_n]$ be nonzero polynomials and fix a monomial order $<$.

(a) Prove that $\mathrm{LM}(uv) = \mathrm{LM}(u)\,\mathrm{LM}(v)$ and $\mathrm{LT}(uv) = \mathrm{LT}(u)\,\mathrm{LT}(v)$.

(b) Give an example with $u + v \neq 0$ showing that $\mathrm{LM}(u + v)$ need not equal either $\mathrm{LM}(u)$ or $\mathrm{LM}(v)$.

We are finally ready to return to the plan that led us to monomial orders in the first place: imitating long division in several variables. One problem remains. We no longer wish to divide f by a single polynomial g, but instead want to make sense of dividing f by a list of polynomials $G = (g_1, \ldots, g_s)$. We will always require that G be treated as an *ordered* list, and, just as in ordinary division, it is important that none of the g_i be zero. Although we might not yet have much intuition for this goal, let us try to mimic the one-variable algorithm, noting the adjustments we make along the way.

Suppose that we wish to divide a polynomial f by an ordered list of nonzero polynomials $G = (g_1, \ldots, g_s)$. Start by initializing a working polynomial $p = f$ and quotient polynomials $q_1 = \cdots = q_s = 0$, which we shall update as we go. Unlike before, we also keep a separate remainder polynomial $r = 0$, which will be updated at each step. Scanning through the list G, look for the first g_i such that $\mathrm{LM}(g_i)$ divides $\mathrm{LM}(p)$. If one exists, then, just as before, the term

$$\frac{\mathrm{LT}(p)}{\mathrm{LT}(g_i)}$$

is exactly the term by which we must multiply g_i in order to cancel the leading term of p. Here is one new wrinkle. If none of the g_i can cancel $\mathrm{LT}(p)$, we cannot simply stop: one of the smaller terms of p might still be divisible by a leading monomial of one of the g_i. Instead, we move the leading term of p into a separate remainder and continue. Thus, in addition to choosing a monomial order, we must choose an ordering of the divisors. Once both choices have been made, the division procedure is completely determined.

Algorithm: Multivariable Division by $G = (g_1, \ldots, g_s)$
Input: f; nonzero divisors $g_1, \ldots, g_s$ in a fixed order; and a fixed monomial order.
Initialize: $q_1, \ldots, q_s \leftarrow 0$, $r \leftarrow 0$, and $p \leftarrow f$.
Repeat while $p \neq 0$*:*

(1) Scan $g_1, \ldots, g_s$ in order.

(2) If g_i is the first divisor in the list for which $\mathrm{LM}(g_i) \mid \mathrm{LM}(p)$, set

$$m \leftarrow \frac{\mathrm{LT}(p)}{\mathrm{LT}(g_i)}, \qquad q_i \leftarrow q_i + m, \qquad p \leftarrow p - mg_i,$$

and return to the beginning of the list.

(3) If no such g_i exists, set

$$r \leftarrow r + \mathrm{LT}(p), \qquad p \leftarrow p - \mathrm{LT}(p).$$

Output: $q_1, \ldots, q_s, r$.

Notice that just as with one-variable division, we have chosen our bookkeeping so that the identity

$$f = q_1 g_1 + \cdots + q_s g_s + p + r$$

is preserved after every update. This algorithm will terminate since at each stage the lead term of the working polynomial p strictly decreases with respect to the chosen monomial order $<$. Hence termination follows from the well-ordering condition in the definition of a monomial order. When the algorithm does stop, no monomial of the remainder is divisible by any $\mathrm{LM}(g_i)$. Importantly, if the final remainder is zero then at the end the algorithm has found an identity

$$f = q_1 g_1 + \cdots + q_s g_s,$$

which is a certificate that f is in the ideal generated by $g_1, \ldots, g_s$. More generally, the preserved division identity always gives

$$f \equiv r \pmod{\langle g_1, \ldots, g_s \rangle}.$$

This means that f and r represent the same class in the quotient ring $\mathbb{K}[x_1, \ldots, x_n]/\langle g_1, \ldots, g_s \rangle$. As a warning, at the present moment we do not know that r is uniquely determined and so this representation might not be canonical. Even more worryingly while reminder zero proves membership it is not clear the converse – nonzero remainder implies nonmembership – is true. In fact it is not!

---

(10) **Dividing the Circle–Parabola Elimination Equation.** Work in $\mathbb{K}[x, y]$ with lex order $x > y$. Let us return to the circle–parabola intersection from Lecture 1. Let $f = x^4 + x^2 - 1$ and $G = (g_1, g_2) = (x^2 - y, y^2 + y - 1)$. Here g_1 is the parabola equation, g_2 is the y-elimination equation from the circle–parabola problem that opened Lecture 1, and f is the corresponding x-elimination equation. Notice that the divisors have been organized in a special way: Exercise 14 will explain where the second divisor comes from, and Exercise 20 will eventually prove that the resulting list is a Gröbner basis.

(a) Divide f by the ordered list G, recording the values of p, q_1, q_2, r after every update.

(b) Verify after every update that the identity $f = q_1 g_1 + q_2 g_2 + p + r$ remains true.

(c) Show that the final remainder is zero. Record the resulting identity

$$x^4 + x^2 - 1 = q_1(x^2 - y) + q_2(y^2 + y - 1).$$

Why does this identity serve as a certificate for $f \in \langle g_1, g_2 \rangle$?

(d) Viewing $g_1 = g_2 = 0$ as equations, use $y = x^2$ to verify directly that every common zero satisfies $f = 0$.

(e) Why is the ideal-membership certificate from part (c) stronger than this pointwise verification in part (d)? If $\mathbb{K}$ is algebraically closed, what converse follows from the Nullstellensatz, and why does it place f only in the radical of the ideal?

(f) Show that $\langle g_1, g_2 \rangle = \langle x^2 - y, x^2 + y^2 - 1 \rangle$. Conclude that the two displayed systems have the same common zero set, and interpret the certificate from part (c) as an algebraic version of the earlier elimination-by-substitution computation.

(11) **Why the Algorithm Terminates.** Fix a monomial order $<$ on $\mathbb{K}[x_1, \ldots, x_n]$. Let $f, g_1, \ldots, g_s \in \mathbb{K}[x_1, \ldots, x_n]$, with $g_i \neq 0$ for all i. Regard $G = (g_1, \ldots, g_s)$ as an ordered list.

(a) Prove that at every stage of the multivariable division algorithm the identity

$$f = q_1 g_1 + \cdots + q_s g_s + p + r$$

is preserved. Hint: See how the identity changes with both kinds of update in the algorithm.

(b) Suppose some g_i is used to cancel $\mathrm{LT}(p)$. Prove that the new working polynomial p is either zero or has a leading monomial strictly smaller with respect to $<$ than it had at the previous step.

(c) Suppose instead that $\mathrm{LT}(p)$ is moved to the remainder. Prove the same conclusion.

(d) Use Exercise 6(a) to prove that the algorithm terminates.

(e) Prove that on termination no monomial of r is divisible by $\mathrm{LM}(g_i)$ for any i.

(f) Prove that if the remainder is zero, then $f \in \langle g_1, \dots, g_s \rangle$. Hence, a zero remainder certifies ideal membership. Notice that we have not shown the converse!

(g) Prove that the output satisfies the following identities:

$$\mathrm{LM}(q_i g_i) \le \mathrm{LM}(f) \quad \text{for every } i \text{ with } q_i \neq 0, \qquad \mathrm{LM}(r) \le \mathrm{LM}(f) \quad \text{if } r \neq 0.$$

Hint: Every subtraction cancels the current leading monomial, and leading monomials never increase. A representation $f = \sum_i q_i g_i$ satisfying the first set of inequalities is called a *standard representation*.

So far this looks remarkably like one-variable division: the process stops, the remainder contains no further cancellable monomials, and a zero remainder produces a certificate of membership. It is tempting to declare victory. Unfortunately, there are two separate problems. First, the remainder depends on the order in which we list the divisors. Second, and much more seriously, a polynomial lying in the ideal can have a nonzero remainder. Thus the implication

$$\text{the remainder is zero} \quad \Longrightarrow \quad f \in \langle g_1, \dots, g_s \rangle$$

always holds, but the converse can fail. The affine twisted cubic already shows us both failures without requiring any new geometry.

---

(12) **The Remainder Is Not Intrinsic: The Twisted Cubic Again.** Work in $\mathbb{K}[x, y, z]$ with lex order $x > y > z$, and consider the affine twisted cubic. Set $g_1 = x^2 - y$, $g_2 = xy - z$, and $I = \langle g_1, g_2 \rangle$. Assuming $\mathbb{K}$ is infinite, this is the defining ideal of the parametrized curve $t \mapsto (t, t^2, t^3)$ that we called the affine twisted cubic in Lecture 1. (Remember that we showed this, in part, by computing the kernel of the corresponding substitution map.) We have changed the signs of the displayed generators so that their leading coefficients are 1.

(a) Divide $x^2 y$ by (g_1, g_2) and show that the remainder is y^2.

(b) Divide the same polynomial by (g_2, g_1) and show that the remainder is xz.

(c) Verify the identity $y g_1 - x g_2 = xz - y^2$, which appeared at the beginning of the lecture. Notice that it gives a representation of $xz - y^2$ in terms of g_1, g_2, but this is not a standard representation: compare $\mathrm{LM}(y g_1) = \mathrm{LM}(x g_2) = x^2 y$ with $\mathrm{LM}(xz - y^2) = xz$.

(d) Recall that the substitution map sends $x \mapsto t$, $y \mapsto t^2$, and $z \mapsto t^3$. Check that both remainders y^2 and xz map to t^4. Explain why they represent the same polynomial function on the

twisted cubic even though division has produced different representatives. Verify directly that their difference is the relation from part (c).

(e) Compute $\mathrm{LM}(xz-y^2)$. Divide $xz-y^2$ by either displayed list and explain why the algorithm returns the same nonzero polynomial, even though it lies in I.

(f) A zero remainder proves ideal membership. Does a nonzero remainder prove nonmembership? Use part (e) to answer, and explain why this failure is more serious than the mere nonuniqueness in parts (a) and (b).

(g) *Further: Finite Fields.* Over $\mathbb{K} = \mathbb{F}_q$, equality as functions on the finite set is coarser than equality modulo I: the class of $x^q - x$ is nonzero in $\mathbb{K}[x,y,z]/I \cong \mathbb{K}[t]$, but it vanishes at every point of the parametrized curve. Explain why this does not contradict $I = \ker(F^*)$, and compare with the twisted cubic exercise in Lecture 1, where polynomial identities differed from equality as functions over a finite field.

The twisted cubic calculation tells us exactly what the original list failed to see. Its leading monomials were x^2 and xy, but cancellation produced the relation $xz - y^2 \in I$, whose leading monomial is xz. Thus the leading monomials of the displayed generators need not account for the leading monomials of all their polynomial consequences. This suggests collecting the leading terms of every element of the ideal.

Definition 5. Let $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$ be an ideal and fix a monomial order $<$. The *initial ideal* of I with respect to $<$ is

$$\mathrm{in}_<(I) = \langle \mathrm{LT}(f) \mid 0 \neq f \in I \rangle .$$

Because $\mathbb{K}$ is a field, the initial ideal of I is also equal to the ideal generated by the leading monomials $\mathrm{LM}(f)$. The zero ideal requires a bit of careful handling, so we set $\mathrm{in}_<(\langle 0 \rangle) = \langle 0 \rangle$.

If $I = \langle g_1, \ldots, g_s \rangle$, then

$$\langle \mathrm{LT}(g_1), \ldots, \mathrm{LT}(g_s) \rangle \subseteq \mathrm{in}_<(I).$$

The reverse inclusion can fail, so equality need not hold. The twisted cubic has already shown us this: $\langle \mathrm{LT}(g_1), \mathrm{LT}(g_2) \rangle = \langle x^2, xy \rangle$, while $f = xz - y^2 \in \langle g_1, g_2 \rangle$ but $\mathrm{LT}(f) = xz \notin \langle x^2, xy \rangle$.

The issue causing the failure of the other inclusion is cancellation: polynomial combinations of the g_i may have leading monomials that are not divisible by any of the displayed $\mathrm{LM}(g_i)$. This is because the operation of "taking a leading term" does not necessarily commute with taking polynomial combinations. We saw this previously when we found examples in which $\mathrm{LM}(u+v)$ was not determined by $\mathrm{LM}(u)$ and $\mathrm{LM}(v)$.

This suggests a definition designed precisely to prevent the failure.

Definition 6. Let $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$ be an ideal and fix a monomial order $<$. A finite list $G = (g_1, \ldots, g_s)$ of nonzero elements of I is a *Gröbner basis* of I with respect to this order if

$$\mathrm{in}_<(I) = \langle \mathrm{LT}(g_1), \ldots, \mathrm{LT}(g_s) \rangle.$$

For the zero ideal we allow the empty list, whose leading terms generate $\langle 0 \rangle$.

There is something slightly surprising about this definition. We require that each g_i lie in I and that their leading terms generate the initial ideal of I, but we have not assumed that the g_i generate I itself. Additionally, it is not at all clear from the definition that Gröbner bases even exist. In Exercise 18 you will prove that the leading-term condition forces $I = \langle g_1, \ldots, g_s \rangle$ and is exactly what is needed for division to decide ideal membership. Before proving this formally, we will use the circle–parabola and twisted cubic examples to see concretely what the leading-term condition repairs. The proof of the existence of Gröbner bases will be one of the main goals of Lecture 3, where we will seek to prove another celebrated result of Hilbert: Hilbert's Basis Theorem.

A warning: one should not confuse I and $\mathrm{in}_<(I)$; their algebraic and geometric properties can be very different. In general, one should view the passage from an ideal I to its initial ideal $\mathrm{in}_<(I)$ as a noncanonical, lossy process. It is noncanonical because it depends on the choice of monomial order $<$, and lossy because one cannot generally recover I from $\mathrm{in}_<(I)$; much of the geometry and algebra may be lost. For example, if $I = \langle y - x^2 \rangle \subseteq \mathbb{K}[x, y]$ is considered with lex order $x > y$, then $\mathrm{in}_<(I) = \langle x^2 \rangle$. Thus $\mathbb{V}(I)$ is the familiar parabola, while the zero set $\mathbb{V}(\mathrm{in}_<(I))$ is just the y-axis. Lecture 5 will center on the question: which pieces of the algebra and geometry of I does $\mathrm{in}_<(I)$ remember?

One indication of why passing to initial ideals loses information is that initial ideals are a very special type of ideal. In particular, every leading term is a nonzero scalar multiple of a monomial. Since $\mathbb{K}$ is a field, $\mathrm{in}_<(I)$ is generated equally well by the leading monomials, and hence is a monomial ideal.

Definition 7. An ideal $M \subseteq \mathbb{K}[x_1, \ldots, x_n]$ is a *monomial ideal* if it is generated by some collection of monomials.

Notice that we did not require the collection of monomial generators in the definition above to be finite. Lecture 3 will show that a finite collection always suffices.

Monomial ideals are exceptionally nice. For example, a monomial ideal is entirely determined by the set of monomials belonging to it. Since monomials in $\mathbb{K}[x_1, \ldots, x_n]$ are in bijection with $\mathbb{Z}^n_{\geq 0}$, this means that we can—and often do—think of monomial ideals as corresponding to subsets of $\mathbb{Z}^n_{\geq 0}$. Moreover, ideal membership for monomial ideals is particularly simple. We will frequently use the following result.

Here an important question is hiding in plain sight. The definition of $\mathrm{in}_<(I)$ used a leading monomial from every nonzero polynomial in I, so it appears to require infinitely much information. A

monomial ideal, however, looks like an upward-closed staircase. If such a staircase has only finitely many corners, then those corner monomials generate the whole ideal. We will prove in Lecture 3 that this always happens; for now the pictures below should make the claim feel plausible.

Lemma 8. *Let $I = \langle m_1, \ldots, m_s \rangle \subseteq \mathbb{K}[x_1, \ldots, x_n]$ be a monomial ideal, with each m_i a monomial. If $m \in \mathbb{K}[x_1, \ldots, x_n]$ is a monomial, then $m \in I$ if and only if m is divisible by m_i for some i.*

This result motivates the following definition.

Definition 9. If M is a monomial ideal, a monomial not contained in M is called a *standard monomial* of M.

Many of the ideas we have seen thus far were developed in the 1960s. For example, Gröbner bases were introduced by Bruno Buchberger in his 1965 dissertation, *An Algorithm for Finding the Basis Elements of the Residue Class Ring of a Zero-Dimensional Polynomial Ideal*. He named these bases for his advisor, Wolfgang Gröbner. That said, the concept of Gröbner bases had been in the mathematical air. For example, in his celebrated 1964 proof of the existence of resolutions of singularities in characteristic zero—work that would earn him the 1970 Fields Medal—Heisuke Hironaka introduced the closely related notion of *standard bases*.

---

(13) **Monomial Ideals.** Let $M = \langle m_1, \ldots, m_s \rangle \subseteq \mathbb{K}[x_1, \ldots, x_n]$ be an ideal generated by monomials.

(a) Prove that a monomial m lies in M if and only if some m_i divides m. For the less immediate direction, write $m = \sum_i h_i m_i$ and use the linear independence of distinct monomials.

(b) Deduce that a polynomial lies in M if and only if each monomial appearing in it lies in M.

(c) Conclude that $\mathrm{in}_<(M) = M$ for every monomial order.

(d) In two variables, draw $x^a y^b$ at the lattice point (a, b). Explain why every point weakly above and to the right of a point of M also lies in M.

(e) In your picture from the previous part, draw the standard monomials.

(f) Draw pictures of the following monomial ideals and list their standard monomials:

$$\langle x^2, y^2 \rangle \qquad \text{and} \qquad \langle x^2, y \rangle.$$

(14) **A Hidden Consequence of the Circle–Parabola Equations.** Work in $\mathbb{K}[x, y]$ with lex order $x > y$ and set $f_1 = x^2 - y$, $f_2 = x^2 + y^2 - 1$, and $I = \langle f_1, f_2 \rangle$.

(a) Compute $\mathrm{LM}(f_1)$ and $\mathrm{LM}(f_2)$. What is the ideal generated by these two leading monomials?

(b) Consider the difference $h = f_2 - f_1 = y^2 + y - 1$. Show that $h \in I$ and compute $\mathrm{LM}(h)$.

(c) Conclude that $\langle \mathrm{LM}(f_1), \mathrm{LM}(f_2) \rangle$ is a proper subset of $\mathrm{in}_<(I)$. Which leading monomial was invisible in the original list?

(d) Divide h by the ordered list (f_1, f_2). Can either $\mathrm{LM}(f_1)$ or $\mathrm{LM}(f_2)$ be used to cancel the leading term of h at the first step? What remainder does the algorithm return, and why is this apparently uneventful calculation a decisive failure of the membership test?

(e) Geometrically, what does $h(y) = 0$ tell us about the possible y-coordinates of the intersection points? Why is it natural to call h an elimination equation?

The cancellation has exposed the elimination equation which the original leading terms could not see. The natural repair is to replace f_2 by this consequence h. Let us separate two facts about this replacement: it does not change the ideal, and it gives a particularly simple staircase.

---

(15) **The Circle–Parabola Staircase.** Continue with the notation above and set $G' = (f_1, h)$.

(a) Use $h = f_2 - f_1$ to prove that $\langle f_1, f_2 \rangle = \langle f_1, h \rangle$. Thus replacing f_2 by the elimination equation changes the organization of the generators without changing the ideal or the circle–parabola intersection.

(b) Return to Exercise 10 and notice that G' was exactly the ordered list used there.

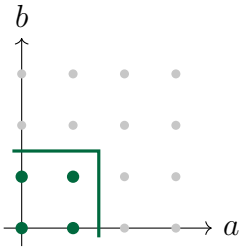
(c) Which monomials are divisible by neither $\mathrm{LM}(f_1) = x^2$ nor $\mathrm{LM}(h) = y^2$? List all of them without imposing a degree bound. Compare the resulting list $1, x, y, xy$ with the basis $1, x, x^2, x^3$ found for the circle–parabola quotient in Lecture 1.

(d) Use $x^2 \equiv y$ and $x^3 \equiv xy$ to explain why the two lists describe the same four-dimensional quotient.

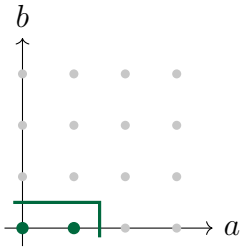
This explains why our first multivariable division worked so well. We had already replaced the original second equation by precisely the consequence whose leading monomial was missing. The ideal and its common zero set did not change; only its presentation became better adapted to division with our chosen monomial order.

The staircase gives us a picture of this improved presentation. In the diagrams below, green lattice points are standard monomials and gray points lie in the monomial ideal. Only the first few exponent vectors are shown.

This picture also returns us to F. S. Macaulay. In studying Hilbert functions—which we have not yet defined—Macaulay organized quotients by counting the monomials that survive modulo a monomial ideal. We are seeing the simplest version of that idea here: the points beneath the staircase will become a basis of the quotient.



$\langle x^2, y^2 \rangle$: $1, x, y, xy$



$\langle x^2, y \rangle$: $1, x$

The monomial quotient $\mathbb{K}[x,y]/\langle x^2, y^2 \rangle$ has $\mathbb{K}$-vector-space dimension 4, while its zero set consists only of the origin. For the finite-dimensional quotient rings considered here, we use *length* to mean this vector-space dimension; thus this quotient has length 4. Its four standard monomials $1, x, y, xy$ also index a basis of the original circle–parabola quotient. Over $\mathbb{C}$, the latter length-4 quotient is instead realized by four reduced intersection points, just as the $2 \cdot 2 = 4$ count from Bézout's theorem predicts. For $\langle x^2, y \rangle$, only 1 and x survive. This quotient is a doubled point of length 2, and the nonzero nilpotent class of x records its tangent direction.

Notice that a staircase is not a drawing of the original algebraic set. It is a picture of monomial divisibility. Its complement will give us representatives in the quotient, and it is the quotient—not merely the set of visible points—that remembers multiplicity.

---

(16) **The Hidden Twisted Cubic Relation.** Return to the ideal $I = \langle x^2 - y, xy - z \rangle \subseteq \mathbb{K}[x,y,z]$ from Exercise 12, which defines the affine twisted cubic. Still working with lex order $x > y > z$, we already found one leading monomial missing from the original list. Let us see what happens after we add the corresponding relation.

(a) Using Exercise 12(c) and (e), conclude that $\langle x^2, xy \rangle \neq \mathrm{in}_<(I)$. Which cancellation produced the invisible monomial?

(b) Set $g_3 = xz - y^2$. The leading monomial of g_3 is xz, and it can cancel against a multiple of $\mathrm{LM}(g_2) = xy$. Form the second cancellation

$$zg_2 - yg_3 = y^3 - z^2.$$

Verify the identity, show that $y^3 - z^2 \in I$, and compute its leading monomial. Why is this leading monomial also absent from $\langle x^2, xy, xz \rangle$?

(c) At this point our candidate for the initial ideal is $M = \langle x^2, xy, xz, y^3 \rangle$. Show that the standard monomials for M are

$$\{x\} \sqcup \{y^b z^c : 0 \leq b \leq 2,\ c \geq 0\}.$$

(d) Under the substitution $x \mapsto t$, $y \mapsto t^2$, $z \mapsto t^3$, determine the power of t represented by each of the standard monomials in the previous part. Show that t^d occurs exactly once for every $d \geq 0$. Hint: Separate d according to its residue modulo 3, treating $d = 1$ using x.

(e) Why is the pattern from the previous part suggested by the ring isomorphism $\mathbb{K}[x,y,z]/I \cong \mathbb{K}[t]$ proved in Lecture 1?

(f) Return for a moment to the three relations g_1, g_2, g_3. Arrange them, up to sign, as the 2×2 minors of

$$\begin{pmatrix} 1 & x & y \\ x & y & z \end{pmatrix}.$$

Thus the cancellation in Exercise 12 did not produce an accidental relation. Which minor had been omitted by the original two-generator list?

(g) *Further: The Same Ideal in a Different Order.* Use graded reverse lexicographic order with $x > y > z$. Show that the three minors have leading monomials x^2, xy, y^2. List the standard monomials of $\langle x^2, xy, y^2 \rangle$ as

$$z^c, \quad xz^c, \quad yz^c \qquad (c \geq 0),$$

and show that they again map bijectively onto the powers of t. Record this calculation; Exercise 20 will turn it into a proof that the three minors already form a Gröbner basis in this order. Which order gives the shorter basis? Which order exposes the equation $y^3 - z^2$ involving only y, z?

(h) The relation $xz - y^2$ has now appeared both as an element of the kernel computed in Lecture 1 and as the polynomial whose leading monomial was missing from the original list. Explain what it tells us in each role. Notice that no affine equation has been lost: $xz - y^2$ is already a polynomial combination of the two displayed generators.

Let us pause over what just happened. In Lecture 1 we chose $x^2 - y$ and $xy - z$ because they make the points of the twisted cubic easy to recognize: once x is known, the other coordinates must be x^2 and x^3. For division in lex order, however, this convenient geometric presentation was badly organized. Both $xz - y^2$ and $y^3 - z^2$ were already in the ideal, but their leading monomials were invisible in the original list. Adding these equations did not change the ideal or the curve. It only made information already present in the ideal visible to the division algorithm. Keep the relation $xz - y^2$ in mind. At the end of Lecture 3 it will become essential when we pass from the affine twisted cubic to its projective closure.

The staircase gives rather convincing evidence that we have now found all the missing leading terms. Its proposed standard monomials map to $1, t, t^2, \ldots$, each exactly once, just as $\mathbb{K}[x,y,z]/I \cong \mathbb{K}[t]$ predicts. Still, evidence is not proof: perhaps some further cancellation produces another leading monomial and shrinks the staircase. We will shortly use the quotient description from Lecture 1 to rule this out without assuming what we are trying to prove.

For this small example we could discover the missing relations by hand. In larger examples, this quickly becomes unreasonable, so let us now ask *Macaulay2* to perform the same reorganization and compare its output with the geometry and algebra we already understand.

---

(17) **Hidden Leading Terms in *Macaulay2*.** Let us return to the circle–parabola ideal via the *Macaualy2* code below, which will compute a Gröbner basis for the ideal $I = \langle x^2 - y, x^2 + y^2 - 1 \rangle$ with respect to the lex monomial order with $x > y > z$. Further, the code snippet below will compute the leading terms of the computed Gröbner basis, and the initial ideal of I.

```
restart
R = QQ[x,y,MonomialOrder=>Lex];
I = ideal(x^2-y, x^2+y^2-1); -- circle--parabola ideal

gens I
shownLeadingIdeal = ideal leadTerm gens I -- only the displayed
   generators

B = gb I;
G = gens B
fullLeadingIdeal = ideal leadTerm G -- leading terms after all
   cancellations
directLeadingIdeal = leadTerm I     -- an ideal input triggers a GB
   computation
directLeadingIdeal == fullLeadingIdeal

h = y^2+y-1;
h % B -- reduce to verify that the elimination equation lies in I
```

In the code above the command `gens I` returns a list of generators for I stored as a row vector. Note this will generally be the generating set you used to initial describe I. The command `gb I` asks *Macaulay2* to compute a Gröbner basis for I. To access the elements in the computed Gröbner basis we use the command `gens gb I`. There is a subtle but important distinction in the code above. Doing `leadTerm I` asks *Macaulay2* to compute the initial ideal of I, which it does by first computing a Gröbner basis for I and then returning the full initial ideal. In contrast, running `ideal leadTerm gens I` computes the ideal generated by the leading terms of a generating set for I. As we have seen these two ideals can be very different.

(a) Explain why `shownLeadingIdeal` uses only the leading terms of the displayed generators, whereas `fullLeadingIdeal` uses the basis computed from the whole ideal. Which missing leading monomial distinguishes the outputs?

(b) The object B stores a special basis computed from I. Compare `G` with `F`. Which new equation makes the missing leading monomial visible?

(c) Compare the zero remainder `h % B` with the hand division in Exercise 14(d). Explain why the two computations use different lists of divisors.

(d) A function with two inputs is defined by placing the inputs in parentheses. Write a function named `hasZeroRemainder` which takes a polynomial p and a Gröbner basis object C and returns `true` precisely when `p % C` is zero. Use it with B and `apply` to test the three polynomials

$$h, \qquad x, \qquad x^4 + x^2 - 1$$

in a single command, predicting the three Boolean outputs before asking the computer. *Hint:* Inside `apply` use a one-input function of the form `p -> hasZeroRemainder(p,B)`. What ideal-membership principle do these outputs suggest? The next exercise will prove that this interpretation is valid.

(e) *Write the code yourself.* Start a fresh session with lex order $y > x$; remember that variables are ordered as they are declared in the ring. Compute a Gröbner basis and verify that its initial ideal is $\langle y, x^4 \rangle$, with standard monomials $1, x, x^2, x^3$. For lex $x > y$, the standard monomials were $1, x, y, xy$. Use $y \equiv x^2$ and $xy \equiv x^3$ to pass between the two lists in the quotient. Why do both orders record quotient dimension 4? Which order reveals an equation in y, and which reveals one in x?

(f) *Further: The Twisted Cubic.* Repeat this exercise for the affine twisted cubic in lex order with $x > y > z$. Locate the hidden monomial xz in the full initial ideal. Verify that the additional leading monomial is y^3, as predicted in Exercise 16. Compare the standard monomials with the powers of t found there.

The computer has not changed our equations in any geometric sense. It has replaced one generating set by another for the same ideal, just as Gaussian elimination replaces a linear system by row-echelon form. The new list may look different, but it has exactly the same polynomial consequences, and its leading terms are organized so that division can see them.

For the circle–parabola ideal, the orders $x > y$ and $y > x$ select the two bases

$$1, x, y, xy \qquad \text{and} \qquad 1, x, x^2, x^3$$

of the same four-dimensional quotient. The remainder chosen by division depends on the monomial order, just as coordinates depend on a choice of basis, but the algebraic intersection itself has not changed.

We have now met the same obstruction by hand and on the computer. The next exercise proves that our definition of a Gröbner basis repairs it. Suppose that $f \in I$ leaves a remainder r. What

does the division identity tell us about r, and what could its leading monomial look like? This is precisely where the equality

$$\mathrm{in}_<(I) = \langle \mathrm{LM}(g_1), \ldots, \mathrm{LM}(g_s) \rangle$$

enters. This is why the definition is the right one: once no leading term can hide, a remainder can no longer falsely claim that an element of the ideal lies outside it.

---

(18) **When Does Division Decide Membership?** Let $S = \mathbb{K}[x_1, \ldots, x_n]$, let $I \subseteq S$ be an ideal, and fix a monomial order. Let $G = (g_1, \ldots, g_s)$ be a finite list of nonzero elements of I.

(a) Suppose the list is a Gröbner basis, meaning that

$$\mathrm{in}_<(I) = \langle \mathrm{LT}(g_1), \ldots, \mathrm{LT}(g_s) \rangle = \langle \mathrm{LM}(g_1), \ldots, \mathrm{LM}(g_s) \rangle.$$

Let $f \in I$ and divide f by $(g_1, \ldots, g_s)$, obtaining remainder r. Show that $r \in I$.

(b) If $r \neq 0$, then $\mathrm{LM}(r) \in \mathrm{in}_<(I)$. Use the monomial-ideal criterion proved above to explain why the displayed equality forces some $\mathrm{LM}(g_i)$ to divide $\mathrm{LM}(r)$.

(c) On the other hand, no monomial of a remainder is divisible by any $\mathrm{LM}(g_i)$. Explain why this contradicts part (b), and conclude that $r = 0$.

(d) For this fixed ordered list G, write $\mathrm{rem}_G(f)$ for the remainder. Conclude that

$$f \in I \quad \Longleftrightarrow \quad \mathrm{rem}_G(f) = 0.$$

For the reverse implication, use $\langle g_1, \ldots, g_s \rangle \subseteq I$.

(e) Since every $f \in I$ has zero remainder, use the division identity to conclude that $I = \langle g_1, \ldots, g_s \rangle$. Thus a Gröbner basis automatically generates its ideal, although this was not required in the definition.

(f) Suppose division by G returns zero for every $f \in I$. Let $0 \neq f \in I$. Explain why, at the first stage of the algorithm, some $\mathrm{LM}(g_i)$ must divide $\mathrm{LM}(f)$: otherwise $\mathrm{LT}(f)$ would be moved permanently into a nonzero remainder. Indeed, the algorithm never removes a term already placed in the remainder, and every later leading monomial is strictly smaller. Deduce that

$$\mathrm{in}_<(I) = \langle \mathrm{LM}(g_1), \ldots, \mathrm{LM}(g_s) \rangle.$$

Be sure to use $g_i \in I$ for the reverse containment.

(g) Suppose $I = \mathbb{I}(X)$ for an algebraic set X. Interpret this membership test geometrically as deciding whether the hypersurface $\mathbb{V}(f)$ contains X. For example, $xz - y^2 \in \mathbb{I}(C)$ says that the corresponding quadric contains the twisted cubic. Why must this interpretation be qualified when I is merely an ideal cutting out X, rather than the full vanishing ideal?

We have now reached the goal stated at the beginning of the lecture. For an arbitrary finite list of divisors, a zero remainder always gives a certificate of membership. While the converse is true, we know that if the list happens to be a Gröbner basis then a nonzero remainder is a finite certificate for nonmembership in the ideal. This begs a number of questions: How does one check if a given finite list of polynomials is a Gröbner basis? Do finite Gröbner bases exist for every ideal? If Gröbner bases do exist how do we find them? Lecture 3 will be dedicated to answering all of these questions.

There is also one other closely related finiteness question floating around, which remains unresolved. We proved that *if* an ideal has a finite Gröbner basis, then that basis generates the ideal. However, we have not yet shown that every ideal is finitely generated, yet along finitely generated by a Gröbner basis. By applying this result to $\mathbb{I}(X)$ for an affine algebraic variety X would imply that every affine variety can be cut out by finitely many equations. This is the classical field case of Hilbert's Basis Theorem, and it will also be one of our goals in Lecture 3.

However, recall that in one-variable, division did more than just answer the ideal membership with a finite certificate. That finite certificate gave us a canonical representative for every class in the corresponding quotient ring. We will now show that our multivariable division algorithm does this as well. A *standard monomial* for an ideal I is a monomial not contained in $\mathrm{in}_<(I($. The standard monomials will play the role that $1, x, \ldots, x^{\deg(g)-1}$ played in the one-variable case, i.e., the will form a basis for $\mathbb{C}[x_1, \ldots, x_n]/I$.

---

(19) **Canonical Normal Forms and Quotient Bases.** Continue with the notation from Exercise 18, and suppose that $G = (g_1, \ldots, g_s)$ is a Gröbner basis for I. Let $f \in \mathbb{K}[x_1, \ldots, x_n]$. We will show that the remainder of f divided by G exists, is unique, and can be expressed in terms of the standard monomials of I.

(a) Suppose that $r, r' \in \mathbb{K}[x_1, \ldots, x_n]$ such that $f - r \in I$ and $f - r' \in I$. Prove that $r - r' \in I$.

(b) Assume that $r - r' \neq 0$. Prove that if every monomial appearing in r or every monomial appearing in r' is a standard monomial for I then $\mathrm{LM}(r - r') \in \mathrm{in}_<(I)$.

(c) Assume that $r - r' \neq 0$. Prove that every monomial in $r - r'$ is a standard monomial. Deduce that $r = r'$.

(d) Divide f by G obtaining the following output division identity:

$$f = q_1 g_1 + \cdots q_s g_s + r$$

where no monomial appearing in r is divisible by any of $\mathrm{LM}(g_1), \ldots, \mathrm{LM}(g_s)$. Prove that every monomial in r is standard. Hint: Since G is a Gröbner basis we know that $\mathrm{in}_<(I) =$

$\langle(\mathrm{LM}(g_1),\ldots,\mathrm{LM}(g_s)\rangle$. Now use the ideal membership criteria for monomial ideals given in Exercise 13(a).

(e) Continuing the previous part, conclude that every class in $\mathbb{K}[x_1,\ldots,x_n]/I$ can be represented uniquely in terms of the classes of standard monomials. (Note you must check that the remainder r is independent of any choices you might have made.)

(f) Deduce that $\mathbb{K}[x_1,\ldots,x_n]/I$ is spanned as a $\mathbb{K}$-vector space by the classes of standard monomials.

(g) Prove that the classes of the standard monomials in $\mathbb{K}[x_1,\ldots,x_n]/I$ are linearly independent over $\mathbb{K}$. Hint: Suppose that $\sum_i c_i m_i \in I$ where $c_i \in \mathbb{K}^\times$ and m_i is standard for all i. Apply the leading monomial argument used above to get a contradiction.

(h) Conclude that the classes of standard monomials form a $\mathbb{K}$-basis for $\mathbb{K}[x_1,\ldots,x_n]/I$.

Notice that the staircase is not merely a helpful picture. Its lattice points index an honest $\mathbb{K}$-basis of the quotient, and division writes every residue class uniquely in that basis. When the staircase is finite, counting its points computes the vector-space dimension of the quotient and hence the lengths we saw in Lecture 1. When it is infinite, the pattern of the surviving monomials can still reveal useful algebraic structure. However, since the underlying vector spaces are infinite-dimensional, we must often move from thinking about vector spaces to thinking about modules.

The unique representative of a class whose monomials are all standard is called its *normal form*. For a fixed monomial order and Gröbner basis G, we denote it by $\mathrm{NF}_G(f)$. Thus we have proved

$$f \in I \qquad \Longleftrightarrow \qquad \mathrm{NF}_G(f) = 0.$$

Although the notation displays G, the normal form depends only on I and the chosen monomial order: every Gröbner basis of I produces the same standard representative.

Accepting for the moment the Gröbner basis computed by *Macaulay2*, the circle–parabola example predicts very concretely that every residue class has a unique representative of the form $a + bx + cy + dxy$. The four green points beneath the staircase therefore do more than suggest the correct dimension: they index another basis of the four-dimensional quotient studied in Lecture 1, and a, b, c, d are the coordinates of a class in this basis.

There is one piece of unfinished business. For the circle–parabola and twisted cubic we have proposed particular initial ideals, guided by hand calculations and by *Macaulay2*. We must be careful not to argue in a circle: we cannot assume our proposed standard monomials form a quotient basis in order to prove that we have found the correct initial ideal. Instead, the next exercise uses quotient descriptions proved independently in Lecture 1 to show directly that every possible remainder of an element of I is zero.

---

(20) **Certifying the Candidate Staircases.** We now replace our computational evidence by proofs that work over an arbitrary field. The strategy is simple: divide an arbitrary $f \in I$ by the candidate basis, and then use a quotient description from Lecture 1 to prove that the resulting remainder must be zero.

(a) Return to the circle–parabola ideal $I = \langle x^2 - y, x^2 + y^2 - 1\rangle \subseteq \mathbb{K}[x, y]$, equipped with lex order $x > y$. Set $G = (x^2 - y, y^2 + y - 1)$. Divide an arbitrary $f \in I$ by G. Explain why its remainder has the form

$$r = a + bx + cy + dxy.$$

(b) Use the division identity and the fact that each element of G lies in I to show that $r \in I$. Under the isomorphism induced by $y \mapsto x^2$,

$$\mathbb{K}[x, y]/I \cong \mathbb{K}[x]/\langle x^4 + x^2 - 1\rangle,$$

which polynomial of degree at most 3 represents the class of r?

(c) Since $r \in I$, this polynomial represents zero in the quotient and is therefore a multiple of the monic polynomial $x^4 + x^2 - 1$. Why must it therefore be zero? Deduce that $r = 0$.

(d) Apply Exercise 18(f) to conclude, over every field, that

$$\mathrm{in}_<(I) = \langle x^2, y^2\rangle.$$

Why does the original list $(x^2 - y, x^2 + y^2 - 1)$ nevertheless fail to be a Gröbner basis?

(e) *Further: The Twisted Cubic in Lex Order.* Return to the affine twisted cubic $I = \langle x^2 - y, xy - z\rangle \subseteq \mathbb{K}[x, y, z]$ with lex order $x > y > z$. Divide an arbitrary $f \in I$ by $G = (x^2 - y, xy - z, xz - y^2, y^3 - z^2)$. Its remainder is a linear combination of the standard monomials listed in Exercise 16(c). As in part (b), the division identity shows that this remainder lies in I. Apply the substitution $x \mapsto t$, $y \mapsto t^2$, $z \mapsto t^3$. Since the standard monomials map to distinct powers of t, prove that the remainder is zero. Deduce

$$\mathrm{in}_<(I) = \langle x^2, xy, xz, y^3\rangle$$

over every field.

(f) *Further: Graded Reverse Lexicographic Order.* Repeat part (e) with graded reverse lexicographic order and the three minors and standard monomials from Exercise 16(f)–(g). Use the monomials z^c, xz^c, yz^c to prove

$$\mathrm{in}_<(I) = \langle x^2, xy, y^2\rangle.$$

Thus lex requires a fourth basis element but exposes an elimination equation, while graded reverse lexicographic order gives the shorter determinantal basis.

The important point is that this proof did not rely on the computer or use Buchberger's criterion, which we have not yet proved. After division, the circle–parabola remainder becomes a polynomial

of degree at most 3 in x. It is simply too small to be a nonzero multiple of $x^4 + x^2 - 1$. If you completed the *Further* parts, the same idea certifies the twisted cubic bases: the candidate standard monomials map to distinct powers of t, so no nonzero linear combination of them can vanish in $\mathbb{K}[t]$.

The two orders for the twisted cubic also give our first clear example of the tradeoff involved in choosing an order. Lex requires one more basis element, but rewards us with the elimination equation $y^3 - z^2$. Graded reverse lexicographic order gives a shorter basis, but does not expose that elimination equation. The quotient remains $\mathbb{K}[t]$ in either order. The computation changes the coordinates in which we see it, not the underlying answer.

The circle–parabola staircase was finite because its quotient was finite-dimensional. The twisted cubic gave an infinite staircase, but its parametrization made the pattern unusually transparent. Let us finish with an equation-defined curve where an infinite staircase has an equally concrete geometric meaning.

---

(21) **An Infinite Staircase: A Cubic Curve.** Assume that $\mathbb{K}$ is algebraically closed of characteristic different from 2, work in $\mathbb{K}[x, y]$ with lex order $y > x$, and set $f = y^2 - x^3 + x$, $I = \langle f \rangle$, and $E = \mathbb{V}(f)$. This is the smooth affine cubic previewed at the end of Lecture 1, which will eventually give an elliptic curve. We are not yet asking whether it can be parametrized. For now we ask a more modest question: what does division tell us about its coordinate ring and about the projection onto the x-axis?

(a) Use Exercise 9(a) to prove, more generally, that the singleton list (f) is a Gröbner basis of the principal ideal generated by every nonzero polynomial f. Deduce that

$$\mathrm{in}_<(I) = \langle y^2 \rangle.$$

Thus division by one polynomial behaves just as well in several variables as it did in one variable. The hidden-leading-term problem only appears when several divisors can cancel against one another.

(b) Draw the staircase of $\langle y^2 \rangle$ and list its standard monomials. Why is the infinite staircase what one should expect for a curve rather than a finite intersection?

(c) Use the normal-form theorem to deduce that every class in $\mathbb{K}[x, y]/I$ has a unique representative $a(x) + b(x)y$, and conclude that this quotient is a free $\mathbb{K}[x]$-module of rank 2 with basis $1, y$.

(d) *Further: What the Staircase Forgets.* Repeat parts (a)–(c) for $y^2 - x^2(x + 1)$, whose curve has a node at the origin. Its initial ideal, staircase, and normal forms are identical. Which geometric difference between the two curves is therefore invisible to the initial ideal?

The description of the quotient $\mathbb{K}[x,y]/I$ found above has a direct geometric meaning. Fixing $x = a$ asks which points of E lie above a; we call this the *fiber* of the projection over a. Let us compare the visible points in a fiber with the algebra that remembers their multiplicities.

---

(22) **Fibers and Multiplicity.** Assume that $\mathbb{K}$ remains algebraically closed of characteristic different from 2. Continue with the cubic curve E and the projection $E \to \mathbb{K}$, $(x,y) \mapsto x$.

(a) Show that the fiber over $a \in \mathbb{K}$ is obtained by solving $y^2 = a^3 - a$. Deduce that it consists of two points when $a \notin \{-1, 0, 1\}$ and one point when $a \in \{-1, 0, 1\}$.

(b) For $a \in \{-1, 0, 1\}$, impose the additional equation $x = a$ and compute the fiber ring

$$\mathbb{K}[x,y]/\langle y^2 - x^3 + x,\ x - a\rangle.$$

Show that it is isomorphic to $\mathbb{K}[y]/\langle y^2\rangle$.

(c) Each exceptional fiber has one visible point but length 2. Explain how the nilpotent class of y records the two points coming together and why this agrees with the rank-2 module in Exercise 21(c).

The infinite staircase has now told us something geometric. The standard monomials x^a and $x^a y$ do much more than say that the quotient is infinite-dimensional: they organize it into two infinite rows. The algebraic way of saying this is that the quotient is a free $\mathbb{K}[x]$-module of rank 2. Correspondingly, the x-projection has fibers of length 2. Usually this length is realized by two distinct points; over each of -1, 0, and 1 it is realized by one double point. The staircase retains exactly the multiplicity needed to make this count constant.

If you completed the *Further* part, the nodal cubic gives an equally important warning. It has the same initial ideal and the same staircase, even though one curve is smooth and the other is singular. A staircase can remember a basis, a module rank, and fiber lengths while forgetting other geometric features. Gröbner methods replace complicated equations by simpler monomial data, so we must always ask which information survived the replacement. With this warning in mind, let us update the affine dictionary and state exactly what we have gained.

---

(23) **Updating the Affine Dictionary.** Consult the table near the beginning of Lecture 1. We can now replace several existence questions by actual finite computations.

(a) For a Gröbner basis G of I, record the test

$$f \in I \quad \Longleftrightarrow \quad \mathrm{NF}_G(f) = 0$$

and the *Macaulay2* commands `gb` and `%`. Why does this decide membership in the specified ideal but not, by itself, decide whether f vanishes on every point of $\mathbb{V}(I)$? When do those questions agree?

(b) Explain why $\mathrm{NF}_G(f)$ is a canonical representative of a class in S/I. Under what hypothesis may it also be interpreted as a canonical representative of a polynomial function on $\mathbb{V}(I)$?

(c) The monomial order changes the standard-monomial basis and hence the chosen representative, but not the truth of $f \in I$. Explain. Why should you now expect the black-box computations from Lecture 1—including `eliminate`, `ker`, and ideal equality—to be implemented using Gröbner bases?

We can finally answer the ideal-membership question with which we began the lecture. Once a Gröbner basis has been found and certified, division terminates with either a membership certificate or a nonzero normal form certifying nonmembership. At the same time, normal forms give concrete coordinates on the quotient S/I.

Before we finish, let us be precise about what this says geometrically. A normal form represents a class in S/I, and therefore restricts to the same polynomial function on $\mathbb{V}(I)$ as the original polynomial. But distinct classes in S/I determine distinct polynomial functions precisely when $I = \mathbb{I}(\mathbb{V}(I))$. Over an algebraically closed field, the Nullstellensatz says that this is equivalent to I being radical. A monomial order can organize an ideal for computation, but no choice of order erases the distinction between an ideal and the vanishing ideal of its visible points.

There remains one central question. The definition of a Gröbner basis refers to the leading terms of every element of an ideal, an infinite collection we cannot inspect one polynomial at a time. How could we ever find such a basis? Our examples suggest an answer: every missing term arose in exactly the same way. We multiplied two equations until their leading terms matched and then subtracted to make those terms cancel. The final exercise isolates this move and turns it into the input for Lecture 3.

---

(24) **The Shape of a Cancellation.** For the final exercise, bring together our two running examples: the circle–parabola ideal and the affine twisted cubic. In $\mathbb{K}[x, y]$, use lex order $x > y$ and set $f_1 = x^2 - y$ and $f_2 = x^2 + y^2 - 1$, the equations defining the parabola and circle, respectively. In $\mathbb{K}[x, y, z]$, use lex order $x > y > z$ and set $g_1 = x^2 - y$, $g_2 = xy - z$, and $g_3 = xz - y^2$. We have previously encountered the three identities

$$f_2 - f_1 = y^2 + y - 1, \qquad yg_1 - xg_2 = xz - y^2, \qquad zg_2 - yg_3 = y^3 - z^2.$$

In each identity, the leading terms of the two polynomial multiples on the left cancel. Our goal is to understand the common pattern behind these cancellations.

(a) Consider, in order, the three pairs (f_2, f_1), (g_1, g_2), and (g_2, g_3). For each pair (u, v), compute $\mathrm{LM}(u)$ and $\mathrm{LM}(v)$, and then compute the least common multiple $L = \mathrm{lcm}(\mathrm{LM}(u), \mathrm{LM}(v))$.

(b) For each pair (u, v) in part (a), check that the monomials multiplying u and v in the corresponding identity above are given by

$$\frac{L}{\mathrm{LM}(u)} \qquad \text{and} \qquad \frac{L}{\mathrm{LM}(v)}.$$

(c) For any two nonzero polynomials u and v, define

$$S(u, v) \coloneqq \frac{\mathrm{lcm}\,(\mathrm{LM}(u), \mathrm{LM}(v))}{\mathrm{LT}(u)} u - \frac{\mathrm{lcm}\,(\mathrm{LM}(u), \mathrm{LM}(v))}{\mathrm{LT}(v)} v.$$

Explain why $S(u, v)$ is itself a polynomial with coefficients in $\mathbb{K}$. Compute the leading terms of the two summands in the definition of $S(u, v)$, and show that these leading terms are equal and therefore cancel.

(d) Compute $S(f_2, f_1)$, $S(g_1, g_2)$, and $S(g_2, g_3)$. Show how this computation recovers the three identities above.

(e) *Further: Uniqueness of the Basic Cancellation.* Let u and v be nonzero polynomials and suppose that h_1 and h_2 are nonzero terms such that the leading terms cancel in $h_1 u - h_2 v$, i.e., suppose $\mathrm{LT}(h_1 u) = \mathrm{LT}(h_2 v)$. Prove that their common leading monomial must be divisible by $L = \mathrm{lcm}(\mathrm{LM}(u), \mathrm{LM}(v))$. Deduce that there is a scalar $c \in \mathbb{K}^\times$ and a monomial m such that

$$h_1 u - h_2 v = c\, m\, S(u, v).$$

Thus every cancellation between two term multiples of u and v is obtained from $S(u, v)$ by multiplying by a nonzero scalar and a monomial. Hint: Write the common leading term as cP. Since P is divisible by both $\mathrm{LM}(u)$ and $\mathrm{LM}(v)$, write $P = mL$. Now use $\mathrm{LT}(h_1 u) = \mathrm{LT}(h_2 v) = cmL$ to solve for h_1 and h_2.

The polynomials defined in part (c) are called *S-polynomials*. We now know what the next algorithm must do: form these basic cancellations between pairs, divide the results by the current list, and adjoin any nonzero remainders. Lecture 3 will make this precise, prove that these pairwise cancellations detect every missing leading term, and show that repeating the process eventually stops.

LECTURE 3: FINDING GRÖBNER BASES (V1)

Throughout this lecture, $\mathbb{K}$ denotes a field, and a monomial order is fixed whenever we use leading terms. At the beginning of Lecture 1, we marveled that the infinitely many points of a parabola could be encoded by one finite expression. We then defined an algebraic set as the common zero set of a collection of polynomials—a collection which we quietly allowed to be infinite—and defined $\mathbb{I}(X)$ by gathering *every* polynomial which vanishes on X. Both definitions left a finiteness question hanging in the air. Must an algebraic set admit a finite list of equations? Better still, can the entire ideal $\mathbb{I}(X)$ of all its equations be generated by finitely many of them?

Lecture 2 introduced a second finiteness question. By the end of Lecture 2 we finally answered the ideal-membership problem. If somebody hands us a Gröbner basis G of an ideal I, then dividing by G gives the test

$$f \in I \qquad \Longleftrightarrow \qquad \mathrm{NF}_G(f) = 0,$$

and the same division gives a canonical representative for every class in S/I, written in the basis of standard monomials. This is exactly the polynomial analogue of row reduction we have been chasing since the opening of Lecture 1. However, this success assumed both that a finite Gröbner basis exists and that we know how to find one.

Let us therefore keep three questions separate:

(1) Why should every ideal in $\mathbb{K}[x_1, \ldots, x_n]$ be generated by finitely many polynomials?

(2) After choosing a monomial order, why should every ideal have a finite Gröbner basis?

(3) Starting from a finite list of generators, how can we actually find such a basis?

The first question is the field case of *Hilbert's Basis Theorem*, which we finally state formally now.

Theorem 1 (Hilbert's Basis Theorem)**.** *Let $\mathbb{K}$ be a field. Every ideal in $\mathbb{K}[x_1, \ldots, x_n]$ is finitely generated.*

We will prove it rather than assume it. Somewhat wonderfully, a positive answer to question (2) will also resolve question (1) and prove Hilbert's Basis Theorem. As we showed in Lecture 2, although generation is not part of the definition, if $(g_1, \ldots, g_s)$ is a finite Gröbner basis of an ideal I, then $I = \langle g_1, \ldots, g_s \rangle$. It turns out that we can reduce the first two questions to an elementary statement about lattice points in $\mathbb{Z}^n_{\geq 0}$. The third is answered by a move we have already made several times by hand without quite recognizing it as an algorithm.

We will first use Dickson's Lemma to settle the two existence questions. We will then return to the move which ended Lecture 2. Whenever a leading monomial went missing from our list, we recovered it in the same way: multiply two polynomials until their leading terms match, subtract,

© Juliette Bruce 2026.
Licensed under Creative Commons BY-NC-SA 4.0.

and see what survives. We called the resulting difference an S-polynomial and then stopped. In every example it exposed precisely the hidden leading monomial. Was that luck? A few friendly examples are not a theorem. Buchberger's criterion will say that pairwise cancellations detect *every* hidden leading term, and Dickson's Lemma will return to guarantee that only finitely many genuinely new ones can appear.

---

(1) **Where Is Finiteness Hiding?**

(a) Let $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$. Suppose that $I = \langle f_1, \ldots, f_s \rangle$. Use Lecture 1 to show that $\mathbb{V}(I) = \mathbb{V}(f_1, \ldots, f_s)$. Thus finite generation of ideals implies that every algebraic set can be described by finitely many equations.

(b) The converse comparison is more delicate. The equations x and x^2 have the same zero set but generate different ideals. Explain why a finite list which merely cuts out X need not generate the full ideal $\mathbb{I}(X)$. Which of these two finite descriptions is stronger?

(c) Lecture 2 proved that every finite Gröbner basis of I automatically generates I. Why does the existence of a finite Gröbner basis for every ideal therefore imply Hilbert's Basis Theorem?

(d) Now start with a finite list $F = (f_1, \ldots, f_r)$. Imagine repeatedly forming S-polynomials, dividing by the current list, and adjoining each nonzero remainder. Why does adjoining a remainder leave the generated ideal unchanged? What information is the enlarged list meant to record more accurately?

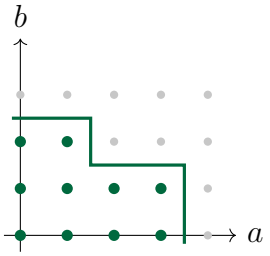
(e) Each new polynomial creates new pairs. Explain why a procedure which makes more work whenever it makes progress has no obvious reason to terminate. State separately the two claims needed to justify the procedure: one about the list when no unchecked pairs remain, and one about whether that moment is ever reached.

Exercise 1 separates finite equations, finite generators, finite Gröbner bases, correctness, and termination. The remarkable point is that all of the unresolved finiteness will come from one statement about monomial ideals. We begin by setting general polynomials aside. A monomial $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ may be drawn as the lattice point $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{Z}^n_{\geq 0}$, exactly as we did when drawing staircases in Lecture 2. Under this bijection divisibility of monomials becomes the componentwise partial order:

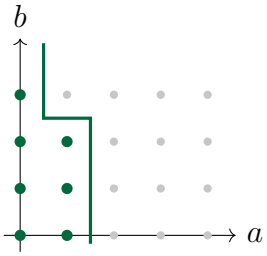
$$x^\alpha \mid x^\beta \qquad \Longleftrightarrow \qquad \alpha_i \leq \beta_i \text{ for every } i.$$

A monomial ideal is therefore an upward-closed region of lattice points: if a point belongs to it, so does everything weakly above and to the right. In two variables we drew the boundary of such a region as a staircase.

Now here is the question. A staircase may perfectly well contain infinitely many lattice points. Can it also have infinitely many *corners*? A corner is a genuinely new piece of information—a monomial in the ideal which is not a multiple of one already there—so this is precisely the question of whether a monomial ideal can need infinitely many generators. Before reading on, try to draw a staircase in the plane with infinitely many corners. The attempt is instructive: every time you place a new corner, the corners already present have fenced off so much of the plane that there is less and less room left for the next one.



$\langle y^3, x^2y^2, x^4\rangle$: three corners



$\langle xy^3, x^2\rangle$: two corners

As before, green lattice points are standard monomials and gray points lie in the monomial ideal. The staircase on the left encloses a finite standard region, the one on the right an infinite one, and the number of corners is not remotely bothered by the difference. This is not an accident of small examples, and it is not special to two variables.

Lemma 2 (Dickson's Lemma)**.** *Every subset $A \subseteq \mathbb{Z}^n_{\geq 0}$ has only finitely many minimal elements under the componentwise partial order. Equivalently, every infinite sequence $\alpha^{(1)}, \alpha^{(2)}, \ldots$ in $\mathbb{Z}^n_{\geq 0}$ contains indices $i < j$ with $\alpha^{(i)} \leq \alpha^{(j)}$ componentwise.*

This seemingly innocuous lemma will do three jobs. First, it will show that every monomial ideal is generated by a finite list of monomials. (Notice that this is a very special case of Hilbert's Basis Theorem.) We will then use the monomial case to show that finite Gröbner bases exist. Finally, Dickson's Lemma will force Buchberger's algorithm, which computes Gröbner bases, to stop.

The naming of Dickson's Lemma comes with a caveat—and a good story. Leonard Dickson stated the lemma in his 1913 paper *Finiteness of the odd perfect and primitive abundant numbers with n distinct prime factors*. His goal was to prove that, for each fixed n, there are only finitely many odd perfect numbers with n distinct prime factors, and the lemma supplied the combinatorial engine. (Whether any odd perfect numbers exist remains open.)

However, the result itself seems to predate Dickson's paper. Paul Gordan used essentially the same statement about monomials in an 1899 note, *Neuer Beweis des Hilbertschen Satzes über homogene Funktionen*, as part of his proof of Hilbert's Basis Theorem. Interestingly, this is the same Gordan to whom Lecture 1 attributed the remark "Das ist keine Mathematik, das ist Theologie." Whatever the truth behind that anecdote, Gordan soon became a developer of Hilbert's ideas, and his 1899 proof is close to the one we are about to reconstruct. The episode is a useful reminder that the constructive and nonconstructive camps were never as sharply divided as the story suggests.

(2) **Understanding Dickson's Lemma.**

(a) Let $A \subseteq \mathbb{Z}_{\geq 0}^n$ and set $I_A \coloneqq \langle x^\alpha \mid \alpha \in A \rangle$. If $A_{\min}$ denotes the set of minimal elements of A, show that every monomial in I_A is divisible by some x^α with $\alpha \in A_{\min}$.

(b) Explain why the statement "every subset of $\mathbb{Z}_{\geq 0}^n$ has finitely many minimal elements with respect to $\leq$" is equivalent to the statement "every monomial ideal in $\mathbb{K}[x_1, \ldots, x_n]$ is finitely generated."

(c) Restate Dickson's Lemma entirely in terms of monomial ideals in $\mathbb{K}[x_1, \ldots, x_n]$.

(d) Prove your monomial ideal version of Dickson's Lemma in the case when $n = 1$.

(e) *Further: A Redundant Axiom.* Lecture 2 postponed the proof that the well-ordering axiom follows from the other assumptions. Suppose an order on monomials is total, is compatible with multiplication, and has 1 as its smallest monomial. Use the sequential form of Dickson's Lemma to prove that there is no infinite strictly decreasing sequence. Thus these assumptions already force the well-ordering property.

(3) **Proving Dickson's Lemma.** We now prove the monomial ideal version of Dickson's Lemma by induction on the number of variables. Your work in Exercise 2d serves as the base case. Assume Dickson's Lemma holds in $n-1$ variables. Let $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$ be a monomial ideal. For $k \geq 0$ let

$$J_k = \left\{ f \in \mathbb{K}[x_1, \ldots, x_{n-1}] \;\middle|\; f x_n^k \in I \right\} \cup \{0\}.$$

(a) Prove that J_k is a monomial ideal in $\mathbb{K}[x_1, \ldots, x_{n-1}]$ for all $k \geq 0$.

(b) Show that the J_k form an ascending chain.

$$J_0 \subseteq J_1 \subseteq J_2 \subseteq \cdots.$$

(Hint: What happens when you multiply something in J_k by x_n?)

(c) Let $L = \bigcup_{d \geq 0} J_d$. Show that L is a monomial ideal in $\mathbb{K}[x_1, \ldots, x_{n-1}]$.

(d) Prove that there exists $N \in \mathbb{Z}_{\geq 0}$ such that $L = J_k$ for all $k \geq N$. (Hint: Use the inductive hypothesis to show that L is finitely generated.)

(e) For each $0 \leq k \leq N$, choose monomial generators $m_{k,1}, \ldots, m_{k,r_k}$ for J_k. Via the inclusion $\mathbb{K}[x_1, \ldots, x_{n-1}] \subset \mathbb{K}[x_1, \ldots, x_n]$ view these as elements in $\mathbb{K}[x_1, \ldots, x_n]$. Prove that I is generated by the finite set

$$\left\{ m_{k,i} x_n^k \;\middle|\; 0 \leq k \leq N,\ 1 \leq i \leq r_k \right\}.$$

(f) Conclude Dickson's Lemma for all n.

(4) **Minimal Monomial Generators.** Let I be a monomial ideal, and let $M(I)$ be the set of monomials in I which are minimal under divisibility.

(a) Use Dickson's Lemma to show that $M(I)$ is finite.

(b) Prove that $I = \langle M(I) \rangle$.

(c) Prove that every monomial generating set of I contains $M(I)$. Conclude that $M(I)$ is the unique minimal monomial generating set of I.

(5) **Existence of Gröbner Bases.** Let $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$ be an ideal and let $<$ be a monomial order.

(a) Show that $\mathrm{in}_<(I)$ is a monomial ideal.

(b) By Dickson's Lemma, choose a minimal monomial generating set $m_1, \ldots, m_t$ such that

$$\mathrm{in}_<(I) = \langle m_1, \ldots, m_t \rangle.$$

Show that there exist elements $g_1, \ldots, g_t \in I$ such that $\mathrm{LT}(g_i)$ divides m_i for all i.

(c) Show that the minimality of $m_1, \ldots, m_t$ implies $\mathrm{LT}(g_i) = c_i m_i$ for some scalar $c_i \in \mathbb{K}^\times$.

(d) Conclude that $\mathscr{G} = \{g_1, \ldots, g_t\}$ is a Gröbner basis of I.

The preceding exercise resolves the question of whether finite Gröbner bases exist; they always do! Although $\mathrm{in}_<(I)$ is defined using the leading monomials of every polynomial in I, it is a monomial ideal. Dickson's Lemma says that finitely many of those monomials generate it. Choosing one polynomial of I with each required leading monomial gives a finite Gröbner basis. We have thus proved the following.

Theorem 3 (Existence of Gröbner Bases)**.** *Let $\mathbb{K}$ be a field and let $<$ be a monomial order on $\mathbb{K}[x_1, \ldots, x_n]$. Every ideal in $\mathbb{K}[x_1, \ldots, x_n]$ has a finite Gröbner basis with respect to $<$.*

Notice the direction of the reasoning. At no point in the lead-up to or the proof of the existence of Gröbner bases did we assume that the ideal I was finitely generated. That is, we did not assume Hilbert's Basis Theorem. Hence this also gives a proof of Hilbert's Basis Theorem, one that uses initial ideals and Gröbner bases to bootstrap from the monomial case, handled by Dickson's Lemma, to the case of a general ideal. Despite this success, the existence proof is not algorithmic: it guarantees that a finite Gröbner basis is sitting inside I without telling us how to write down a single element of it. Once again we are cursed by Gordan's complaint...

Soon we shall address this issue and find an algorithm for producing a Gröbner basis. Before we do, however, let us take a brief aside to mention some beautiful algebra hiding right under our noses—if only so that I can selfishly mention my favorite mathematician, Emmy Noether.

Look back at what the proof of Theorem 3 actually consumed. Dickson's Lemma is a statement about monomials, and it did its work by ruling out an infinite strictly increasing chain of monomial

ideals. Nothing else in the argument really cared that we were in a polynomial ring; what it cared about was a single finiteness phenomenon: *you cannot go on enlarging an ideal forever*. Hilbert proved his basis theorem in 1890 as a theorem about polynomial rings, and for roughly the next thirty years that is what it remained—a fact about polynomials, proved by manipulating polynomials. Then, in her 1921 paper *Idealtheorie in Ringbereichen*, Emmy Noether asked the structural question instead: what is the weakest hypothesis on a ring that makes arguments of this shape run? Her answer was to stop treating the no-infinite-chains property as a conclusion and start treating it as an axiom. The moment you do that, the whole discussion leaves $\mathbb{K}[x_1, \ldots, x_n]$ behind and takes place in an arbitrary commutative ring.

Definition 4. A commutative ring R is *Noetherian* if every ascending chain of ideals

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$$

stabilizes, meaning that there exists $N \in \mathbb{Z}_{\geq 1}$ such that $I_n = I_N$ for all $n \geq N$.

It is common to say that a ring is Noetherian if it satisfies ACC, short for the ascending chain condition. A ring R is Noetherian if and only if every ideal of R is finitely generated. (You will prove this equivalence shortly.) Thus we have proved that $\mathbb{K}[x_1, \ldots, x_n]$ is Noetherian. Since the only ideals in a field $\mathbb{K}$ are the zero and unit ideals, a field is also an example of a Noetherian ring. In this more modern formulation, the version of Hilbert's basis theorem we have seen is a special case of the following more general version.

Theorem 5 (Hilbert's Basis Theorem (General Case))**.** *Let R be a commutative ring. If R is Noetherian, then $R[x]$ is Noetherian.*

Iterating this theorem, starting from the field $\mathbb{K}$, recovers exactly the statement that $\mathbb{K}[x_1, \ldots, x_n]$ is Noetherian. We shall not prove this general version of Hilbert's Basis Theorem, as it would pull us slightly beyond our concrete geometric goals. There is a theory of Gröbner bases over more general coefficient rings, but coefficient divisibility introduces complications that disappear over a field; our division algorithm and Buchberger criterion therefore do not carry over unchanged.

---

(6) **Properties of Noetherian Rings.** For this exercise, let R be any commutative ring, and consider an ascending chain of ideals in R:

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots .$$

(a) Show that $J = \bigcup_{k \geq 1} I_k$ is an ideal of R.

(b) If J is finitely generated, say $J = \langle a_1, \ldots, a_t \rangle$, explain why there exists an integer $N \in \mathbb{Z}_{\geq 1}$ such that $a_1, \ldots, a_t \in I_N$. Conclude that the chain of ideals stabilizes.

(c) Explain why you just showed that if every ideal of R is finitely generated then R is Noetherian.

(d) Let I be an ideal of R and suppose every ascending chain of ideals in R stabilizes. Pick any $a_1 \in I$. If $\langle a_1 \rangle \neq I$, pick $a_2 \in I \setminus \langle a_1 \rangle$, obtaining $\langle a_1 \rangle \subsetneq \langle a_1, a_2 \rangle$. Continue in this way. Use the ascending chain condition to show this process must terminate, and conclude that I is finitely generated.

Emmy Noether was born in Erlangen in 1882, the daughter of the algebraic geometer Max Noether. Women were not permitted to matriculate at Bavarian universities at the time, so she began by sitting in on lectures with the individual permission of each professor. When the rules changed, she enrolled properly, and in 1907 she received her doctorate under Paul Gordan—yes, *that* Gordan, the one who allegedly greeted Hilbert's nonconstructive proof with the remark that it was not mathematics but theology. Her dissertation was a thoroughly Gordan-style piece of work: an explicit computation of a system of invariants of a ternary biquadratic form, complete with hundreds of coefficients written out by hand. She was famously dismissive of it later, calling it *Mist* (roughly: rubbish) and describing that style of invariant theory as a jungle of formulas. It is worth holding on to this detail. The person who did more than anyone to make algebra abstract began by grinding through exactly the sort of computation that abstraction was invented to avoid, and she knew precisely what she was walking away from.

In 1921 Noether published her paper *Idealtheorie in Ringbereichen*, which is widely considered to mark the start of modern commutative algebra. In this and her subsequent works, her method was the one we just used on Hilbert's Basis Theorem: find the property that is actually doing the work, promote it to a definition, and watch a computation turn into a structural fact. Her students, the *Noether-Knaben* or "Noether boys," carried this perspective outward. In April 1933 she was dismissed from Göttingen under the Nazi civil service laws, for being Jewish; by most accounts she simply carried on holding seminars in her apartment. Later that year she emigrated to Bryn Mawr College in Pennsylvania, and also lectured at the Institute for Advanced Study in Princeton. She died on 14 April 1935, aged fifty-three, following surgery. Einstein wrote to the *New York Times* shortly afterwards that she had been, in his judgment, "the most significant creative mathematical genius" produced since women were admitted to higher education.

There is a pleasing symmetry with which to close this aside. Gordan objected to Hilbert's proof because it exhibited nothing; Gordan's own doctoral student then built the framework in which exhibiting nothing became the normal and correct way to do algebra. And the pendulum swings back once more: in 1965 Bruno Buchberger, a student of Wolfgang Gröbner, supplied the algorithm that Hilbert's proof had withheld. That algorithm is where we go next.

Before setting the algorithm in motion, let us put the cancellation from the end of Lecture 2 back on the table. We were not taking arbitrary combinations. Given two nonzero polynomials f and g, we multiplied them just enough to make their leading terms agree and then subtracted.

Definition 6 (S-Polynomial)**.** The *S-polynomial* of two nonzero polynomials f and g is

$$S(f,g) \coloneqq \frac{\operatorname{lcm}(\mathrm{LM}(f),\mathrm{LM}(g))}{\mathrm{LT}(f)} f - \frac{\operatorname{lcm}(\mathrm{LM}(f),\mathrm{LM}(g))}{\mathrm{LT}(g)} g.$$

What survives is the simplest cancellation which the leading terms of f and g can produce. In Lecture 2 we met these cancellations as surprises: a polynomial combination suddenly displayed a leading monomial which the original generators had hidden. The S-polynomial turns that surprise into a deliberate experiment. Let us repeat the last experiment of Lecture 2, but this time knowing exactly what we are looking for.

---

(7) **The Circle–Parabola Cancellation, Now on Purpose.** Work in $\mathbb{K}[x,y]$ with lex order $x > y$. Lecture 2 replaced the original circle–parabola equations by the equivalent list

$$g_1 = x^2 - y, \qquad g_2 = y^2 + y - 1.$$

(a) There is only one pair to examine. Compute

$$S(g_1,g_2) = y^2 g_1 - x^2 g_2$$

and reduce it to zero by (g_1,g_2). Record the complete division identity, since it certifies that the remainder is zero.

(b) We already proved in Lecture 2, by a special argument using the quotient, that (g_1,g_2) is a Gröbner basis. The calculation in part (a) suggests a general test. Before reading on, state the test you hope is true.

(c) Suppose instead that an S-polynomial has nonzero remainder r. Explain why r belongs to the ideal but $\mathrm{LM}(r)$ does not belong to the monomial ideal generated by the current leading monomials. What does this say about the current list?

It is tempting to believe that the phenomenon we saw in the previous example is the whole story: perhaps a generating list is a Gröbner basis exactly when every pairwise cancellation can already be explained by the list. Remarkably, that temptation is correct.

Theorem 7 (Buchberger's Criterion)**.** *Let $G = (g_1,\dots,g_s)$ generate an ideal I. Then G is a Gröbner basis of I if and only if every S-polynomial $S(g_i,g_j)$ has remainder zero upon division by G.*

The definition of a Gröbner basis quantifies over infinitely many polynomials in $\langle g_1,\dots,g_s\rangle$ and allows cancellations among arbitrarily many polynomial multiples of the g_i. Buchberger's criterion replaces all of that by the $\binom{s}{2}$ pairwise experiments. Why should pairs be enough?

The rough idea is familiar from linear algebra, where a dependence among many rows can be untangled by comparing them two at a time. Suppose $f \in I$ is written as $f = \sum_i h_i g_i$ and the

largest monomial among the products $h_i g_i$ cancels. The summands responsible for that monomial can be rewritten as a combination of pairwise S-polynomials (see the final exercise in Lecture 2). If those S-polynomials all reduce to zero, substituting their reductions gives a new representation of f whose largest monomial is strictly smaller. Well-ordering prevents this from continuing forever. Exercise 8 makes this argument precise.

---

(8) **Further: Why Pairwise Cancellations Suffice.** Fix a monomial order $<$ on $\mathbb{K}[x_1, \ldots, x_n]$, let $G = (g_1, \ldots, g_s)$ be a list of monic polynomials, and set $I = \langle g_1, \ldots, g_s \rangle$. Assume that every $S(g_i, g_j)$ reduces to zero modulo G.

(a) Let $0 \neq f \in I$. Explain why the well-ordering property of $<$ implies that there is an expression $f = h_1 g_1 + \cdots + h_s g_s$ for which

$$M = \max\{\mathrm{LM}(h_i g_i) \mid h_i \neq 0\}$$

is as small as possible.

(b) Explain why $\mathrm{LM}(f) \leq M$. If $M = \mathrm{LM}(f)$, show that $\mathrm{LM}(f)$ is divisible by some $\mathrm{LM}(g_i)$, and explain why this is exactly the conclusion we want.

(c) Suppose instead that $M > \mathrm{LM}(f)$, so the top monomial cancels. Reindex so that $1, \ldots, r$ are precisely the summands whose leading monomial is M, and write

$$\mathrm{LT}(h_i) = c_i \frac{M}{\mathrm{LM}(g_i)}.$$

Show that $c_1 + \cdots + c_r = 0$, because the terms carrying the monomial M must cancel.

(d) For $i \geq 2$ set $L_i = \mathrm{lcm}(\mathrm{LM}(g_1), \mathrm{LM}(g_i))$ and verify the identity

$$\sum_{i=1}^{r} c_i \frac{M}{\mathrm{LM}(g_i)} g_i = \sum_{i=2}^{r} c_i \frac{M}{L_i} S(g_i, g_1).$$

Hint: telescope, using $c_1 = -(c_2 + \cdots + c_r)$. This identity is the precise reason pairwise cancellations suffice; everything else in the proof is bookkeeping around it.

(e) Since each $S(g_i, g_1)$ reduces to zero, replace the right-hand side by a combination of elements of G all of whose leading products are strictly smaller than M. Use this to build a representation of f with a smaller maximum, and obtain a contradiction with the minimality in part (a).

(f) Conclude that $M = \mathrm{LM}(f)$ and hence that G is a Gröbner basis. Explain why rescaling removes the assumption that the g_i are monic.

(g) Now the converse, which is much easier. Suppose G is already a Gröbner basis. Since every $S(g_i, g_j)$ lies in I, apply the membership theorem from Lecture 2 to conclude that each has remainder zero.

Buchberger's criterion also gives us a process for "repairing" a list G that fails to be a Gröbner basis. Test every pair. If an S-polynomial leaves a nonzero remainder, adjoin that remainder to the list and continue. This gives us an algorithm for computing a Gröbner basis.

Algorithm: Buchberger's Algorithm
Input: A finite list of nonzero generators $F = (f_1, \ldots, f_s)$ and a monomial order. (Discard any zero generators first.)
Initialize: Set $G \leftarrow F$ and place every unordered pair from G on a list of pairs to be checked.
Repeat: Remove a pair (g_i, g_j) from the list and divide $S(g_i, g_j)$ by the current list G.
(1) If the remainder is zero, continue.
(2) If the remainder r is nonzero, append r to G and add every new pair involving r to the list.
Output: When no unchecked pairs remain, return G.

There is one difficulty hiding in this description of Buchberger's algorithm. Every time our list fails to be a Gröbner basis, we increase its length and continue. Why must this process stop? What prevents a badly chosen input list from causing the algorithm to spiral into an infinite computation?

(9) **Why Buchberger's Algorithm Stops.**

(a) Prove that after every pass through the loop the current list G generates the same ideal as the input list.

(b) Suppose a nonzero remainder r is adjoined. Why is $\mathrm{LM}(r)$ outside the monomial ideal generated by the leading monomials in the old list? Conclude that this monomial ideal strictly increases.

(c) If the current list G stops growing, it has only finitely many pairs, so the algorithm soon exhausts its list. Deduce that an infinite run would require infinitely many additions and hence an infinite strictly ascending chain of monomial ideals.

(d) Apply the sequential form of Dickson's Lemma (Lemma 2) to the sequence of newly adjoined leading monomials and prove that this cannot happen. This is where Dickson's Lemma does its third job.

(e) A pair which earlier had remainder zero still follows the same reduction after new divisors are appended. If its old remainder was a nonzero r, repeat the old reduction to reach r and then reduce by the appended divisor r itself. Thus every processed pair has remainder zero with respect to the final list. Once no unchecked pairs remain, apply Buchberger's criterion. State clearly which part of your argument proves termination and which proves correctness.

(f) Compare this proof with the abstract existence proof in Exercise 5. In what sense does each nonzero remainder add a new corner to the monomial ideal generated by the leading monomials currently known?

At this point it is worth mentioning another slight gremlin I have strategically swept under the rug: run time. I am no computer scientist, so I will shy away from overly precise claims. The actual running time depends on many choices, including the monomial order, the pair-selection strategy, and the implementation of reduction. One useful, very coarse mathematical proxy is the following question: if $\langle g_1, \dots, g_s \rangle \subseteq \mathbb{K}[x_1, \dots, x_n]$ and $\deg(g_i) \leq d$ for every i, can we bound the degrees of the elements in the reduced Gröbner basis, for a fixed monomial order, in terms of n and d? The answer is yes. For $d \geq 2$, a theorem of Dubé gives the bound

$$2\left(\frac{d^2}{2} + d\right)^{2^{n-1}}.$$

Notice that this is a truly horrendous bound: it is doubly exponential in the number of variables. One might hope that the mathematics is merely pessimistic. Sadly, families constructed by Mayr and Meyer, together with homogeneous variants due to Koh, show that doubly exponential behavior in the number of variables genuinely occurs. A more refined way to organize degree growth uses the geometric invariant called Castelnuovo–Mumford regularity, which we will meet later.

Okay, mathematical theory aside, what happens when the rubber meets the road and we see what a computer can do? As part of his thesis, Buchberger implemented his algorithm on a Zuse Z23V at the Innsbruck Computing Center in 1965. One of his examples computed a Gröbner basis for the ideal $\langle g_1, g_2, g_3 \rangle \subseteq \mathbb{Q}[x, y, z]$, where

$$g_1 = z^3 - \frac{1}{2}x^2 - \frac{1}{2}y^2, \qquad g_2 = xz - 2z + xy, \qquad g_3 = x^2 - y$$

He used graded reverse lexicographic order. Buchberger later reported that this computation took approximately 2.5 minutes. On my M5 Pro MacBook Pro, version 1.26.06 of *Macaulay2* is much, much, much faster: it takes only 0.000842 seconds. On the other hand, the fanciest version of public generative AI I tried, ChatGPT 5.6 Sol ProUltra, took more than 1.5 minutes, so perhaps we are not always moving forward.

A computer-algebra joke, as it was told to me, goes like this: in the late 1970s and early 1980s, computer scientists saw the doubly exponential worst case and knew better than to waste their time building a system around Gröbner bases. Algebraic geometers ignored the warnings and built *Macaulay*. The historical record above shows that this is only a joke. Still, it highlights a worthwhile riddle: why are *Macaulay2* and other modern computer algebra systems so useful in research when the worst-case bounds are so severe? Worst-case bounds describe specially constructed families, not every input. Many examples arising in geometry have exploitable structure,

and modern implementations use decades of improvements in pair selection and polynomial reduction. The bounds are a warning rather than a prediction: some computations really do explode, while many important ones finish almost instantly.

With that warning in view, let us run Buchberger's algorithm by hand on the curve which has kept us company since Lecture 1.

---

(10) **The Twisted Cubic, Now Algorithmically.** Work in $\mathbb{K}[x, y, z]$ with graded reverse lexicographic order where $x > y > z$. Consider the ideal $\langle g_1, g_2 \rangle$ defining the affine twisted cubic, where $g_1 = x^2 - y$ and $g_2 = xy - z$. Unlike in Lecture 2, we have changed from lex to a degree-compatible order, meaning that larger total degree implies a larger monomial. This choice will matter later.

(a) Verify that $\mathrm{LM}(g_1) = x^2$ and $\mathrm{LM}(g_2) = xy$, and then show $S(g_1, g_2) = xz - y^2$.

(b) Set $g_3 = y^2 - xz$. Compute $\mathrm{LM}(g_3)$. (Be careful! What monomial order are we using?)

(c) Show that the remainder of $S(g_1, g_3)$ divided by (g_1, g_2, g_3) is zero. Do the same for $S(g_2, g_3)$.

(d) Conclude from Buchberger's criterion that $G = (g_1, g_2, g_3)$ is a Gröbner basis. Why did we not have to check $S(g_1, g_2)$ modulo G?

(e) Before running the following code, predict the initial ideal and the three quadratic relations. Then compare the output with your hand computation.

```
restart
R = QQ[x,y,z,MonomialOrder=>GRevLex];
I = ideal(x^2-y,x*y-z);
B = gb I;

gens B
ideal leadTerm gens B
(x*z-y^2) % B
```

(f) Interpret the zero remainder. Which parts of your hand proof are the computer repeating, and which parts still require mathematical interpretation?

(g) *Further: The Determinantal Pattern.* Arrange these three polynomials, up to sign, as the 2×2 minors of

$$\begin{pmatrix} 1 & x & y \\ x & y & z \end{pmatrix}.$$

The same matrix appeared in Lecture 2. Explain how this presentation places all three relations on the same footing.

(h) *Further: Comparing Orders.* Return to lex order $x > y > z$ and verify that the basis contains $y^3 - z^2$. Interpret this polynomial as an equation for the projection to the yz-plane. Why does the graded reverse lexicographic basis not display this eliminant? This is a first instance of a tradeoff we will meet repeatedly: the useful order depends on the question.

Let us pause for a moment and notice what, at first glance, might feel like an unimportant bookkeeping issue in our application of Buchberger's algorithm to the twisted cubic. We started the algorithm with two equations, $g_1 = x^2 - y$ and $g_2 = xy - z$, which completely determine the affine twisted cubic. Once we know x, the equations force $y = x^2$ and then $z = x^3$. Despite this, Buchberger's algorithm led us to the third equation

$$y^2 - xz.$$

At first glance, this too might feel like bookkeeping whose only purpose is to satisfy Buchberger's criterion. However, this is now the third time we have seen this polynomial. In Lecture 1, it appeared when we found the kernel of the substitution map arising from the parametrization of the affine twisted cubic. We recorded it and moved on. In Lecture 2, the same equation appeared in our search for "hidden" leading monomials. At the time, we might have written it off as an artifact of the somewhat unfamiliar process of multivariable polynomial division. Now it has appeared again.

In all honesty, you might rightly argue that the appearances in Lectures 2 and 3 are the same in disguise. Even so, when something keeps appearing in our examples, it is worth pausing to ask whether something deeper is hiding. In this case, beautiful geometry has been staring us in the face since Lecture 1, and it is time to enjoy the view. To see it, however, we must leave $\mathbb{K}^n$, which algebraic geometers call affine space, and enter $\mathbb{P}^n$, projective space.

We have owed ourselves this move since Lecture 1 anyway. Recall the discussion of Bézout's theorem: a plane curve of degree d and a plane curve of degree e, with no common component, meet in exactly de points, provided we make three corrections to the naive picture. We must work over an algebraically closed field, or the circle and parabola show two points instead of four. We must count with multiplicity, or a tangency shows one point instead of two. And we must work in projective space, so that intersections "at infinity" are not silently discarded. Lectures 1 and 2 dealt with the first two corrections at length. The third has been an IOU sitting in the margin ever since. Let us finally pay it.

Definition 8 (Projective Space)**.** *Projective n-space* over $\mathbb{K}$, denoted $\mathbb{P}^n_{\mathbb{K}}$, is the set of lines through the origin in $\mathbb{K}^{n+1}$.

This definition is terse enough that it is worth slowing down and unpacking what it says. A line through the origin of $\mathbb{K}^{n+1}$, that is, a one-dimensional vector subspace of $\mathbb{K}^{n+1}$, is determined by any nonzero vector on the line. Moreover, two nonzero vectors $v, w \in \mathbb{K}^{n+1}$ determine the same line

exactly when $v = \lambda w$ for some nonzero scalar $\lambda \in \mathbb{K}^\times$. This gives a useful description of $\mathbb{P}^n_{\mathbb{K}}$ as a quotient:

$$\mathbb{P}^n_{\mathbb{K}} \coloneqq \left(\mathbb{K}^{n+1} \setminus \{0\}\right) / \sim, \qquad v \sim w \iff v = \lambda w \text{ for some } \lambda \in \mathbb{K}^\times.$$

Thus, we may think of a point in $\mathbb{P}^n_{\mathbb{K}}$ as an equivalence class of nonzero vectors in $\mathbb{K}^{n+1}$. One should check that $\sim$ is indeed an equivalence relation. It is customary to denote the equivalence class of $(a_0, \ldots, a_n) \in \mathbb{K}^{n+1} \setminus \{0\}$ by

$$[a_0 : \cdots : a_n] \coloneqq \left\{\lambda(a_0, \ldots, a_n) \;\middle|\; \lambda \in \mathbb{K}^\times\right\}.$$

The entries $a_0, \ldots, a_n$ are called *homogeneous coordinates*, but they are defined only up to a common nonzero scalar. The brackets and colons remind us of this fact.

For example, it would not make sense to say that the zeroth homogeneous coordinate of $[1 : 0 : 0]$ equals 1, because $[1 : 0 : 0] = [\lambda : 0 : 0]$ for every $\lambda \in \mathbb{K}^\times$. What is well-defined is whether a particular entry equals zero: multiplying by a nonzero scalar cannot turn a zero entry into a nonzero one, or vice versa. Thus, although it makes no sense to ask for the value of the zeroth homogeneous coordinate of $[a_0 : \cdots : a_n]$, it makes perfect sense to ask whether $a_0 = 0$. This gives a well-defined decomposition of projective space:

$$\mathbb{P}^n_{\mathbb{K}} = \left\{[a_0 : \cdots : a_n] \;\middle|\; a_0 \neq 0\right\} \sqcup \left\{[a_0 : \cdots : a_n] \;\middle|\; a_0 = 0\right\}.$$

Call the first set $U_0 \subset \mathbb{P}^n_{\mathbb{K}}$. If $[a_0 : \cdots : a_n] \in U_0$, then $a_0 \neq 0$, so we may normalize its representative by scaling by $1/a_0$:

$$[a_0 : \cdots : a_n] = \left[1 : \frac{a_1}{a_0} : \cdots : \frac{a_n}{a_0}\right].$$

This normalized representative is unique. Indeed, if $[1 : b_1 : \cdots : b_n] = [1 : c_1 : \cdots : c_n]$, then $(1, b_1, \ldots, b_n) = \lambda(1, c_1, \ldots, c_n)$ for some $\lambda \in \mathbb{K}^\times$. Comparing the first entries gives $\lambda = 1$, so $b_i = c_i$ for every i. Equivalently, there is a bijection

$$\begin{aligned} \mathbb{K}^n &\xrightarrow{\qquad\psi_0\qquad} U_0 \subset \mathbb{P}^n_{\mathbb{K}} \\ (a_1, \ldots, a_n) &\longmapsto [1 : a_1 : \cdots : a_n]. \end{aligned}$$

The inverse of ψ_0 is

$$[a_0 : \cdots : a_n] \longmapsto \left(\frac{a_1}{a_0}, \ldots, \frac{a_n}{a_0}\right).$$

Why is this well-defined? We shall write $\iota \coloneqq \psi_0 : \mathbb{K}^n \hookrightarrow \mathbb{P}^n_{\mathbb{K}}$ for this standard affine embedding.

Thus, although the individual entries of a projective point depend on the representative, their ratios are well-defined whenever the denominator is nonzero. We have found a copy of affine n-space inside projective n-space. The remaining points have the form $[0 : a_1 : \cdots : a_n]$, where at least one of $a_1, \ldots, a_n$ is nonzero. Moreover, $[0 : b_1 : \cdots : b_n] = [0 : c_1 : \cdots : c_n]$ if and only if

$(b_1, \ldots, b_n) = \lambda(c_1, \ldots, c_n)$ for some $\lambda \in \mathbb{K}^\times$. These are precisely the equivalence classes defining $\mathbb{P}^{n-1}_{\mathbb{K}}$. Thus, we have a bijection

$$\begin{aligned} \mathbb{P}^{n-1}_{\mathbb{K}} &\longrightarrow \left\{ [a_0 : \cdots : a_n] \;\middle|\; a_0 = 0 \right\} \subset \mathbb{P}^n_{\mathbb{K}} \\ [a_0 : \cdots : a_{n-1}] &\longmapsto [0 : a_0 : \cdots : a_{n-1}] \end{aligned}$$

Thus, the decomposition above is $\mathbb{P}^n_{\mathbb{K}} \cong \mathbb{K}^n \sqcup \mathbb{P}^{n-1}_{\mathbb{K}}$. We call U_0 a *standard affine chart* and its complement, a copy of $\mathbb{P}^{n-1}_{\mathbb{K}}$, the *hyperplane at infinity*. Continuing inductively gives

$$\mathbb{P}^n_{\mathbb{K}} \cong \mathbb{K}^n \sqcup \mathbb{K}^{n-1} \sqcup \cdots \sqcup \mathbb{K} \sqcup \{\mathrm{pt}\}.$$

Of course, there was nothing special about a_0 in this discussion. For each $i = 0, \ldots, n$, we may define

$$U_i \coloneqq \left\{ [a_0 : \cdots : a_n] \in \mathbb{P}^n_{\mathbb{K}} \mid a_i \neq 0 \right\},$$

and construct a bijection $\psi_i : \mathbb{K}^n \to U_i$ analogous to the one for U_0. At the moment, we have constructed $\mathbb{P}^n_{\mathbb{K}}$ only as a set, but one can also endow it with a topology and, when $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, the structure of a smooth manifold. We shall not need these additional structures. When the field is clear, we shall write $\mathbb{P}^n$ rather than $\mathbb{P}^n_{\mathbb{K}}$.

---

(11) **Finding Infinity.** The phrase *hyperplane at infinity* is evocative, but so far it is only a name. Work over $\mathbb{R}$ throughout. Let us begin with $\mathbb{P}^1_{\mathbb{R}}$, where infinity consists of a single point.

(a) Consider the standard affine chart

$$U_0 = \left\{ [a_0 : a_1] \in \mathbb{P}^1_{\mathbb{R}} \;\middle|\; a_0 \neq 0 \right\}.$$

Prove that every point of U_0 has a unique representative of the form $[1 : t]$. Use this to write down a bijection between $\mathbb{R}$ and U_0.

(b) Show that $\mathbb{P}^1_{\mathbb{R}} \setminus U_0$ consists of exactly one point. We denote this point by ∞. (Hint: If $a_0 = 0$, what must be true of a_1, and how can you scale it?)

(c) In projective coordinates, what point is ∞?

(d) Now look at the other standard affine chart

$$U_1 = \left\{ [a_0 : a_1] \in \mathbb{P}^1_{\mathbb{R}} \;\middle|\; a_1 \neq 0 \right\}.$$

Show that every point of U_1 has a unique representative $[s : 1]$.

(e) Verify the identity $[1 : t] = [\frac{1}{t} : 1]$ assuming $t \neq 0$. What happens as $t \to \pm\infty$?

Now for a genuine problem: which polynomial equations make sense on projective space? The familiar equation $f = x^2 - y$ defines a parabola in affine space $\mathbb{K}^2$, whose projective closure will live

in $\mathbb{P}^2$. The basic obstruction, however, is already visible if we try to use the expression $x^2 - y$ as an equation on $\mathbb{P}^1_{\mathbb{Q}}$. We have

$$f(1,1) = 0, \qquad f(2,2) = 2,$$

even though $[1:1] = [2:2]$ in $\mathbb{P}^1_{\mathbb{Q}}$. Thus, whether f vanishes depends on the representative, so the question "Does f vanish at $[1:1]$?" has no mathematically meaningful answer. Before continuing, consider what changes if we replace $x^2 - y$ by $x^2 - y^2$.

Definition 9 (Homogeneous Polynomials and Ideals)**.** Let $S = \mathbb{K}[x_0, \ldots, x_n]$.

(i) A polynomial $f \in S$ is *homogeneous of degree* d if every monomial in f has total degree d.

(ii) An ideal $I \subset S$ is *homogeneous* if it can be generated by homogeneous polynomials.

The importance of the definition is the following observation: if $f \in S$ is homogeneous of degree d, then

$$f(\lambda x_0, \ldots, \lambda x_n) = \lambda^d f(x_0, \ldots, x_n)$$

for every $\lambda \in \mathbb{K}^\times$. Indeed, each degree-$d$ monomial acquires the factor λ^d. Consequently,

$$f(a_0, \ldots, a_n) = 0 \quad \Longleftrightarrow \quad f(\lambda a_0, \ldots, \lambda a_n) = 0.$$

Thus, whether a homogeneous polynomial vanishes at a projective point is well-defined. Its value rescales by the factor λ^d and therefore does not, in general, define a function on $\mathbb{P}^n$ (degree-zero forms are the basic exception). Homogeneous polynomials do, however, define equations, which have been our north star since the beginning of Lecture 1.

---

(12) **Getting Oriented in Projective Space.** The definition is short, but the first few times one uses it, it is worth being pedantic.

(a) Decide which of the following describe the same point in $\mathbb{P}^2_{\mathbb{Q}}$:

$$[1:2:3], \qquad [2:4:6], \qquad [1:2:4], \qquad [0:2:3].$$

(b) Why is $[0:0:0]$ not a point in $\mathbb{P}^2$?

(c) Let $U_i \subseteq \mathbb{P}^n$ be the set where the ith coordinate is nonzero. Carefully show that U_i is a copy of $\mathbb{K}^n$ by normalizing the ith coordinate to 1. Hint: we did this carefully for $i = 0$ above.

(d) Show that the collection $\{U_i\}$ for $i = 0, 1, \ldots, n$ forms a cover of $\mathbb{P}^n$.

(e) Let $a, b, c, d \in \mathbb{K}$, where a, b are not both zero and $c \neq d$. In $\mathbb{P}^2$ with homogeneous coordinates $[x:y:z]$, verify that $ax + by + cz$ is homogeneous.

(f) Notice that the two affine lines $ax + by = c$ and $ax + by = d$ are parallel and hence never meet in $\mathbb{K}^2$. Consider their projective counterparts, $ax + by = cz$ and $ax + by = dz$. Show that the lines defined by these two homogeneous equations in $\mathbb{P}^2_{\mathbb{K}}$ intersect at $[b:-a:0]$.

(g) The point $[b : -a : 0]$ corresponds to a line through the origin of $\mathbb{K}^3$. Identify this line and explain how it relates to the two planes in $\mathbb{K}^3$ defined by $ax + by = cz$ and $ax + by = dz$.

(h) Which part of the preceding calculation would break down if we tried to use a polynomial with terms of different degrees? Explain why homogeneity is not a technical decoration but the precise condition which makes projective equations possible in the first place.

So $\mathbb{P}^1$ adds one direction at infinity, and $\mathbb{P}^2$ adds a whole line of them. Notice what part (f) is really telling us. The two parallel lines have not been bent, nudged, or otherwise altered; projective space has simply supplied a point for the direction they already shared, and both lines already pass through it. The point $[b : -a : 0]$ depends only on the direction of the lines, not on the constants c and d, which is exactly what "the point where all lines of this direction meet" ought to look like. This is the kind of missing intersection that can make Bézout's count fail in affine space.

We are now in a position to ask the same two questions that organized Lecture 1 and have guided everything since. Given some projective points, how do we gather all of their equations? Given some equations, how do we gather all of their common zeros? Reassuringly, almost nothing in the old definitions needs to change. The only new requirement is that the equations be homogeneous, and we have just seen that this is forced on us rather than chosen.

Definition 10 (Projective Vanishing Sets and Vanishing Ideals)**.** Let $S = \mathbb{K}[x_0, \dots, x_n]$, and write S_d for the $\mathbb{K}$-vector space of homogeneous forms of degree d. For a homogeneous ideal $J \subseteq S$, define

$$\mathbb{V}_{\mathbb{P}^n}(J) = \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid F(a_0, \dots, a_n) = 0 \text{ for every homogeneous } F \in J\}.$$

For a subset $X \subseteq \mathbb{P}^n$, define its *homogeneous vanishing ideal* by

$$\mathbb{I}_{\mathbb{P}^n}(X) = \bigoplus_{d \geq 0} \{F \in S_d \mid F(a_0, \dots, a_n) = 0 \text{ for every } [a_0 : \cdots : a_n] \in X\}.$$

Definition 11 (Projective Variety)**.** A subset $X \subseteq \mathbb{P}^n_{\mathbb{K}}$ is a *projective variety* if

$$X = \mathbb{V}_{\mathbb{P}^n}(J)$$

for some homogeneous ideal $J \subseteq \mathbb{K}[x_0, \dots, x_n]$.

As in Lectures 1 and 2, we are using *variety* in the broad sense of algebraic set. Some books reserve the word for a nonempty irreducible algebraic set, in which case our objects would be called projective algebraic sets. We will say *irreducible projective variety* whenever we intend the extra condition.

Definition 12 (Projective Closure)**.** For $X \subseteq \mathbb{K}^n$, its *projective closure* is the smallest projective variety containing the affine copy $\iota(X)$. Equivalently,

$$\overline{X}^{\mathbb{P}^n} = \mathbb{V}_{\mathbb{P}^n}(\mathbb{I}_{\mathbb{P}^n}(\iota(X))).$$

The word *closure* is carrying real weight here, so let us be clear about what it forbids. We are not permitted to drop the old affine points into a bigger space and then decorate them with whichever points at infinity we find aesthetically pleasing. We must add exactly the projective points forced upon us by every homogeneous equation which already vanishes on the affine set—no more and no fewer. This raises the obvious practical question—the one this whole lecture has been training us to answer: how do we get our hands on those equations without first knowing all of them?

The natural first guess is to take a list of affine equations and homogenize them one at a time.

Definition 13 (Homogenization)**.** Let $0 \neq f = \sum_\alpha c_\alpha x^\alpha \in \mathbb{K}[x_1, \ldots, x_n]$ have total degree d. Its *homogenization with respect to* w is

$$f^h(w, x_1, \ldots, x_n) = \sum_\alpha c_\alpha w^{d-|\alpha|} x^\alpha.$$

Equivalently,

$$f^h(w, x_1, \ldots, x_n) = w^d f\left(\frac{x_1}{w}, \ldots, \frac{x_n}{w}\right).$$

We set $0^h = 0$. The first formula makes clear that f^h really is a polynomial, and setting $w = 1$ recovers f.

So the guess is: homogenize each generator, collect the results, declare victory. Our twisted cubic is about to object. Recall its affine parametrization $t \mapsto (t, t^2, t^3)$ and its two displayed equations $x^2 - y$ and $xy - z$. We have already agreed that once x is known these two equations force $y = x^2$ and $z = x^3$, so they describe every affine point of the curve and nothing else. Let us see what they do at infinity.

---

(13) **The Naive Homogenization Fails.** Let

$$I = \langle x^2 - y,\ xy - z \rangle \subseteq \mathbb{K}[x, y, z].$$

(a) Homogenize the two displayed generators to obtain

$$x^2 - wy, \qquad xy - wz.$$

Show that both vanish on the entire projective line

$$L = \{[w : x : y : z] \in \mathbb{P}^3 \mid w = x = 0\}.$$

Thus two equations which describe exactly the right affine points have quietly picked up a whole line of spurious points at infinity.

(b) Compare this with the parametrization, which ought to know better. On the chart $s \neq 0$ of $\mathbb{P}^1$, write $u = t/s$. We want to recover the affine point $[1 : u : u^2 : u^3]$, but these four coordinates do not yet share a common denominator. Clear denominators so that all four coordinates become homogeneous of the same degree, and obtain

$$\nu_3 : \mathbb{P}^1 \longrightarrow \mathbb{P}^3, \qquad [s : t] \longmapsto [s^3 : s^2 t : s t^2 : t^3].$$

Verify that the four coordinates have no common zero, so the map is well defined. Which single point appears when $s = 0$?

(c) So the parametrization adds exactly one point at infinity while the homogenized equations contribute an entire line. Something has gone missing between the two descriptions. Return to Exercise 10: which relation in the affine ideal is absent from our homogenized list, and how did Buchberger's algorithm discover it?

The missing relation is, of course, $y^2 - xz$. And now we can see exactly what went wrong, which is more interesting than simply noting that it did. Affinely we had the identity

$$x(xy - z) - y(x^2 - y) = y^2 - xz,$$

which is precisely the S-polynomial computation from Exercise 10. Homogenize the two generators and run the identical calculation with the identical multipliers:

$$x(xy - wz) - y(x^2 - wy) = w(y^2 - xz).$$

Look at that stray factor of w. On the chart $w \neq 0$ we may divide by it, and the two homogenized equations do imply $y^2 - xz = 0$, exactly as they did affinely. On the hyperplane at infinity we cannot divide by w, and the identity degenerates into the statement $0 = 0$, which constrains nothing whatsoever. The consequence which held everywhere in affine space survives only where $w \neq 0$, and the false line at infinity slips through the gap.

This tells us that homogenizing an arbitrary generating list is simply not a well-defined operation on the underlying geometry: the answer depends on which generators you happened to display. The repair is to stop homogenizing a list and start homogenizing the whole ideal.

Definition 14 (Homogenization of an Ideal)**.** For an ideal $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$, its *homogenization* is the homogeneous ideal

$$I^h = \langle f^h \mid f \in I \rangle \subseteq \mathbb{K}[w, x_1, \ldots, x_n].$$

When $I = \mathbb{I}(X)$—as it will be for our twisted cubic over an infinite field—the projective zeros of I^h give exactly the projective closure of X. That is the good news. The bad news is that this definition appears to ask us to homogenize infinitely many polynomials, which is not the kind of instruction one can hand to a computer, or indeed to a student.

By now you can probably guess which tool is about to rescue us, and you can probably also guess the catch. A Gröbner basis will let us replace the infinite collection by a finite one—but only if we choose the monomial order carefully. Here is why the choice matters. If $f = \sum_i q_i g_i$, we cannot just homogenize the identity term by term, because the products $q_i g_i$ may have different total degrees, and homogenizing then involves patching in different powers of w on each side. What we need is a division algorithm which never lets the degree run up past $\deg(f)$. And that is precisely what a degree-compatible order buys us.

Theorem 15 (Homogenizing a Gröbner Basis)**.** *Let $G = (g_1, \ldots, g_s)$ be a Gröbner basis of an ideal I with respect to a degree-compatible monomial order, meaning that a monomial of larger total degree is always larger. Then*

$$I^h = \langle g_1^h, \ldots, g_s^h \rangle.$$

So here is the safe move, and it is a satisfying one: compute a Gröbner basis in a graded order first, and only then homogenize. This is why I quietly switched to graded reverse lex back in Exercise 10 and asked you to notice the switch. For the twisted cubic it gives

$$J = \langle x^2 - wy,\ xy - wz,\ y^2 - xz \rangle \subseteq \mathbb{K}[w, x, y, z].$$

What remains is to check that the third equation removes exactly the false points at infinity and none of the genuine ones.

---

(14) **The Twisted Cubic Finds Its Point at Infinity.** Continue with the notation above.

(a) Set $w = 0$ in the three generators of J. Show that the extraneous line from Exercise 13 is cut down to the single point $[0 : 0 : 0 : 1]$. Which equation does the cutting, and why could neither of the first two equations have done it?

(b) Substitute

$$\nu_3([s : t]) = [s^3 : s^2t : st^2 : t^3]$$

into the three generators of J and show that they vanish. Conversely, show that every point of $\mathbb{V}_{\mathbb{P}^3}(J)$ lies in the image of ν_3, treating $w \neq 0$ and $w = 0$ separately. Thus J cuts out precisely the parametrized curve, points at infinity included.

(c) Apply Theorem 15 to the graded reverse lexicographic basis from Exercise 10. Check that its three homogenizations generate J, and conclude that $J = I^h$. Explain in one sentence why the original two generators carried no such guarantee.

(d) *Further: The Projective Determinantal Pattern.* Arrange the three generators of J, up to sign, as the 2×2 minors of

$$\begin{pmatrix} w & x & y \\ x & y & z \end{pmatrix}.$$

Compare this with the affine matrix in Exercise 10. The only change is that the entry 1 has become w, which is homogenization in its most transparent form. Which minor was redundant for describing affine points but indispensable for excluding the false line at infinity?

We can now answer the question we posed at the start of this half of the lecture. The polynomial $y^2 - xz$ was never a new relation: it lay in the affine ideal all along, and Buchberger's algorithm merely dragged it into view. Affinely it repairs a defect of the division algorithm by supplying a

leading term the original list had hidden. Projectively—and this is the same polynomial, doing what looks like a completely unrelated job—it deletes a false line at infinity. That is the geometric content which had been sitting inside a nonzero S-polynomial remainder for two lectures, waiting for us to have the language to read it.

I find this genuinely striking, and it is worth naming the moral. A Gröbner basis is a better presentation of the same ideal, and "better" turned out to mean more than "more convenient for division." A generating set which is adequate for one purpose can be silently inadequate for another, and the extra elements which division demanded were exactly the ones the geometry demanded too.

Two claims are still hiding inside the phrase *projective closure*, and we should not let them pass. We have checked which point set J cuts out, but why is the point at infinity actually *forced* by the affine points, rather than merely compatible with them? And better still: have we found every homogeneous equation of the projective curve, or only enough of them to pin down the points? The next exercise answers both.

---

(15) **Further: Closure and Every Equation.** Assume that $\mathbb{K}$ is infinite.

(a) Let F be homogeneous of degree d and suppose that

$$F(1, t, t^2, t^3) = 0 \qquad \text{for every } t \in \mathbb{K},$$

that is, F vanishes on the affine part of the curve. Set

$$H(s, t) = F(s^3, s^2 t, s t^2, t^3).$$

Show that H is homogeneous of degree $3d$. Since $\mathbb{K}$ is infinite, $H(1, t)$ is the zero polynomial. Use homogeneity to conclude that H is identically zero, and hence that $F(0, 0, 0, 1) = 0$. Why does the definition of projective closure now force $[0 : 0 : 0 : 1]$ into the closure of the affine twisted cubic? Notice that we did not get to choose this point; every equation vanishing on the affine curve insists on it.

(b) We know that $\mathbb{V}_{\mathbb{P}^3}(J)$ is a projective variety containing the affine twisted cubic, so the projective closure is contained in $\mathbb{V}_{\mathbb{P}^3}(J)$. Use part (a) and Exercise 14(b) for the reverse containment, and conclude that

$$\mathbb{V}_{\mathbb{P}^3}(J) = \overline{\{(t, t^2, t^3) \mid t \in \mathbb{K}\}}^{\,\mathbb{P}^3}.$$

(c) Assume, as we established for the affine twisted cubic, that

$$I = \mathbb{I}\big(\{(t, t^2, t^3) \mid t \in \mathbb{K}\}\big).$$

Let F be homogeneous of degree d and vanish on the image of ν_3. Dehomogenize it to $f = F(1, x, y, z)$ and show that $f \in I$. If $F \neq 0$, explain why $f \neq 0$; then, writing $e = \deg(f)$, prove that

$$F = w^{d-e} f^h.$$

Combine this with Exercise 14(c) to conclude that

$$J = \mathbb{I}_{\mathbb{P}^3}\big(\nu_3(\mathbb{P}^1)\big).$$

Thus we have identified not merely the projective points of the curve, but every homogeneous equation which vanishes on them. This is the projective analogue of the inverse problem from Lecture 1, solved completely for this one example.

Having worked one example carefully, let us return to the $\mathbb{V}$–$\mathbb{I}$ dictionary and ask how much of Lecture 1 survives the move to $\mathbb{P}^n$. The definitions looked almost unchanged when we wrote them down, which should make us slightly nervous, because one thing definitely did change: projective space threw away the origin of $\mathbb{K}^{n+1}$. Let us find out exactly what still works and where that missing point causes trouble.

---

(16) **Rebuilding the Projective Dictionary.** Let $S = \mathbb{K}[x_0, \dots, x_n]$, let $J, L \subseteq S$ be homogeneous ideals, and let $X, Y \subseteq \mathbb{P}^n$.

(a) Verify directly that $\mathbb{V}_{\mathbb{P}^n}(J)$ does not depend on the representatives chosen for its projective points. Prove that $\mathbb{I}_{\mathbb{P}^n}(X)$ is a homogeneous ideal. Where exactly does homogeneity get used in each argument?

(b) If $X \subseteq Y$, which way does the inclusion between their vanishing ideals go? If $J \subseteq L$, which way does the inclusion between their vanishing sets go? Prove the familiar containments

$$J \subseteq \mathbb{I}_{\mathbb{P}^n}\big(\mathbb{V}_{\mathbb{P}^n}(J)\big), \qquad X \subseteq \mathbb{V}_{\mathbb{P}^n}\big(\mathbb{I}_{\mathbb{P}^n}(X)\big).$$

Compare each statement with its affine ancestor in Lecture 1. Did any proof have to change?

(c) *Further:* Assume $\mathbb{K}$ is infinite. For $p = [a_0 : \cdots : a_n]$, let $L_p \subseteq \mathbb{K}^{n+1}$ be the line represented by p. Show that the homogeneous forms vanishing at p form the prime ideal of the linear subspace L_p, where $\mathbb{I}(L_p)$ denotes its ordinary affine ideal in S. Deduce that

$$\mathbb{I}_{\mathbb{P}^n}(X) = \bigcap_{p \in X} \mathbb{I}(L_p)$$

is radical.

(d) Now for the place where the missing origin bites. Consider the ideal

$$\mathfrak{m} = \langle x_0, \dots, x_n \rangle.$$

In affine $(n+1)$-space it singles out the origin, and it is about as far from being the unit ideal as an ideal can be. But projective space has thrown the origin away. What is $\mathbb{V}_{\mathbb{P}^n}(\mathfrak{m})$?

Hilbert's Basis Theorem—now ours, rather than borrowed—immediately tells us that every projective variety can be cut out by finitely many homogeneous equations. So have we recovered the entire $\mathbb{V}$–$\mathbb{I}$ dictionary from Lecture 1? Almost, and the gap is exactly the one that part (d) just exposed. Over an algebraically closed field the precise statement is the following, which we record without proof.

Theorem 16 (Projective Nullstellensatz)**.** *Let $\mathbb{K}$ be algebraically closed and let $J \subseteq \mathbb{K}[x_0, \dots, x_n]$ be a homogeneous ideal. If $\mathbb{V}_{\mathbb{P}^n}(J) \neq \varnothing$, then*

$$\mathbb{I}_{\mathbb{P}^n}\big(\mathbb{V}_{\mathbb{P}^n}(J)\big) = \sqrt{J}.$$

That nonemptiness hypothesis is the one new annoyance, and $\mathfrak{m}$ is the reason for it. The ideal $\mathfrak{m} = \langle x_0, \dots, x_n \rangle$ is called the *irrelevant ideal*, a name which for once is completely descriptive: its only affine zero is the origin, and the origin is irrelevant to projective space because it represents no projective point. So $\mathbb{V}_{\mathbb{P}^n}(\mathfrak{m}) = \varnothing$, whence $\mathbb{I}_{\mathbb{P}^n}\big(\mathbb{V}_{\mathbb{P}^n}(\mathfrak{m})\big) = S$, while $\sqrt{\mathfrak{m}} = \mathfrak{m} \neq S$. Projective space discarded a point, and the algebra has not forgotten. In the affine setting the weak Nullstellensatz told us that the only ideal with no zeros is the unit ideal; projectively, a proper homogeneous ideal can have no projective zeros, so the dictionary must explicitly account for the irrelevant ideal.

That completes the main route through the lecture. The next three exercises are marked *Further*. The first proves the homogenization theorem we leaned on; the other two gather up computational certificates and refinements left over from Lectures 1 and 2. If time is short, skip ahead to Exercise 20 and come back to them.

---

(17) **Further: Why Homogenizing a Gröbner Basis Works.** Let $G = (g_1, \dots, g_s)$ be a Gröbner basis of I for a degree-compatible order. The whole proof is an exercise in watching degrees, so watch them closely.

(a) Show that $\deg(\mathrm{LM}(p)) = \deg(p)$ for every nonzero polynomial p. (This is the only thing degree compatibility is for, and everything else follows from it.) Now let $0 \neq f \in I$. Division by G gives

$$f = q_1 g_1 + \cdots + q_s g_s.$$

Explain why each reduction subtracts a multiple whose degree is no larger than the degree of the current polynomial. Deduce that

$$\deg(q_i) + \deg(g_i) \leq \deg(f)$$

whenever $q_i \neq 0$.

(b) Put $d = \deg(f)$ and homogenize the identity to the common degree d:

$$f^h = \sum_{q_i \neq 0} w^{d - \deg(q_i) - \deg(g_i)} q_i^h g_i^h.$$

Explain why part (a) is exactly what makes those exponents of w nonnegative, and deduce that $I^h \subseteq \langle g_1^h, \ldots, g_s^h \rangle$.

(c) Why is the reverse containment immediate? Then point separately at the two places where the Gröbner-basis condition and degree compatibility entered, and describe what goes wrong in the twisted cubic example if you drop either one.

(18) **Further: Two Finite Certificates from Lecture 1.** The identities below are valid over every field. Conclusions about whether a proper ideal has a point require the Weak Nullstellensatz and hence an algebraically closed field; keep the two kinds of statement apart as you work.

(a) In $\mathbb{K}[x, y]$ set

$$h_1 = x^2 + y^2 - 1, \qquad h_2 = xy - 1, \qquad h_3 = x + y - 2.$$

Verify the certificate

$$1 = h_1 + 2h_2 - (x + y + 2)h_3.$$

Why does this one identity rule out a common zero over $\mathbb{K}$ and over every field extension of $\mathbb{K}$, without our ever hunting for a point?

(b) Assume $\mathbb{K}$ is algebraically closed and let $H = \langle h_1, h_2, h_3 \rangle$. Explain why

$$\mathbb{V}(H) = \varnothing \qquad \Longleftrightarrow \qquad 1 \in H \qquad \Longleftrightarrow \qquad \mathrm{NF}_G(1) = 0$$

for a Gröbner basis G of H. Explain why $1 \in H \Longleftrightarrow \mathrm{NF}_G(1) = 0$ and $1 \in H \Longrightarrow \mathbb{V}(H) = \varnothing$ hold over every field, while the reverse implication from an empty zero set is the one that needs algebraic closure.

(c) For $I \subseteq \mathbb{K}[x_1, \ldots, x_n]$ and $f \in \mathbb{K}[x_1, \ldots, x_n]$, Lecture 1 proved Rabinowitsch's equivalence

$$f \in \sqrt{I} \qquad \Longleftrightarrow \qquad 1 \in \langle I, 1 - uf \rangle \subseteq \mathbb{K}[x_1, \ldots, x_n, u].$$

For $I_{\tan} = \langle x^2 - y, y \rangle$, show that $x \notin I_{\tan}$ but $x^2 \in I_{\tan}$. Interpret this in $\mathbb{K}[x, y]/I_{\tan} \cong \mathbb{K}[x]/\langle x^2 \rangle$, and connect it back to the tangential double point from Lecture 1.

(d) Verify

$$1 = (1 + ux)(1 - ux) + u^2(x^2 - y) + u^2 y.$$

Explain why this proves $x \in \sqrt{I_{\tan}}$, and how the same construction converts a question about radical membership—which we had no way to decide—into ordinary ideal membership in one extra variable, which we can now decide.

(e) Check both certificates computationally, after you have checked the identities by hand.

```
restart
R = QQ[x,y,u,MonomialOrder=>Lex];
H = ideal(x^2+y^2-1,x*y-1,x+y-2);
1_R % gb H

I = ideal(x^2-y,y);
Jtan = I + ideal(1-u*x);
x % gb I
1_R % gb Jtan
```

What does each zero or nonzero remainder certify? Be precise about which certificates hold over every field.

(19) **Further: Canonical Bases and Ground Fields.** Buchberger's algorithm makes a great many arbitrary choices along the way—which pair to process next, which divisor to use—so it is reasonable to worry that the answer depends on them. It need not. A Gröbner basis $G = (g_1, \dots, g_s)$ is *reduced* if every g_i is monic and no monomial of g_i is divisible by $\mathrm{LM}(g_j)$ for $j \neq i$. For a fixed monomial order, every ideal has a unique reduced Gröbner basis. So there is a canonical answer sitting at the end of all those choices.

Now let $\mathbb{K} \subseteq L$ be a field extension and set $I_L = I \cdot L[x_1, \dots, x_n]$.

(a) Verify that the circle–parabola basis

$$x^2 - y, \qquad y^2 + y - 1$$

is reduced for lex order $x > y$.

(b) If G is a Gröbner basis of $I \subseteq \mathbb{K}[x_1, \dots, x_n]$, prove using Buchberger's criterion that the same list is a Gröbner basis of I_L. Why do the leading monomials, the S-polynomials, and the reductions all remain unchanged?

(c) Deduce that, for $f \in \mathbb{K}[x_1, \dots, x_n]$,

$$f \in I \qquad \Longleftrightarrow \qquad f \in I_L.$$

Explain why a polynomial identity or membership certificate found over $\mathbb{Q}$ remains valid after passing to $\mathbb{C}$. This is quietly reassuring: it means our habit of computing over $\mathbb{Q}$ and thinking over $\mathbb{C}$ is legitimate.

Be careful not to read more into the preceding exercise than it says, because two nearby statements are false. The ideal $\langle x^2 - 2 \rangle$ is prime over $\mathbb{Q}$ but not after extending scalars to $\mathbb{C}$, so primality need not survive a field extension. Second, reducing modulo a prime is not a field extension: there is no field homomorphism $\mathbb{Q} \to \mathbb{F}_p$. If $2x + y \in \mathbb{Z}[x, y]$ is considered with a monomial order in which $x > y$,

first over $\mathbb{Q}$ and then modulo 2, even its leading monomial changes because its leading coefficient vanishes.

---

(20) **Updating the Computational Dictionary.** Return to the table we began in Lecture 1 and have been filling in for three lectures. This is the point at which several rows stop being promises.

(a) Record how a normal form decides whether $f \in I$. If you completed Exercise 18, add the tests for $f \in \sqrt{I}$ and for inconsistency. Which conclusions hold over every field, and which lean on the Nullstellensatz?

(b) Explain how Buchberger's algorithm differs from the abstract existence proof in Exercise 5. What finite data does the algorithm take as input, and what finite certificate lets somebody else verify that its output really is a Gröbner basis without rerunning it?

(c) Record the roles of `gb`, `gens`, `leadTerm`, and the remainder operator `%`. Which of the commands we treated as black boxes in Lecture 1 can you now describe as Gröbner-basis computations?

(d) Assume $\mathbb{K}$ is infinite and add the projective operations $\mathbb{V}_{\mathbb{P}^n}$ and $\mathbb{I}_{\mathbb{P}^n}$ to the dictionary. For the twisted cubic, identify separately the affine ideal I, the ideal generated by the naive homogenizations of its two displayed generators, and the homogenized ideal I^h. If you completed Exercise 15, also identify the homogeneous vanishing ideal of $\nu_3(\mathbb{P}^1)$. Which equality of point sets did we check directly, and which equality of ideals rested on the graded Gröbner basis?

(e) Finally, write a prose summary. Up to sign, $y^2 - xz$ arose as an S-polynomial remainder; its leading monomial y^2 contributed to the initial ideal; and its homogenization deleted a line at infinity. Identify the single theme connecting these three appearances.

What did the computation prove?

Dickson's Lemma gave us finite Gröbner bases and, as a bonus, Hilbert's Basis Theorem. Buchberger's criterion turned pairwise cancellations into an algorithm, and Dickson's Lemma returned to guarantee that the algorithm stops—though certainly not that it stops quickly. For the twisted cubic, the same missing relation which repairs the initial ideal also removes a false line at infinity, so a degree-compatible Gröbner basis carries us safely from the affine ideal to its projective homogenization.

For three lectures we have mostly traveled from equations to the geometry they cut out. In Lecture 4 we shall reverse the direction of travel. If somebody hands us a parametrization, how can we recover the equations it forces? Standard monomials will turn into interpolation data, and elimination will recover the equations of the image—or, to be honest about it, of its Zariski closure.

As we shall see, those two sets are not always the same, and the difference is where much of the interest lies.

LECTURE 4: INTERPOLATION, IMPLICITIZATION, AND RATIONALITY (V1)

Throughout this lecture, $\mathbb{K}$ denotes a field. We will restate the hypotheses on it whenever they begin to matter, which will be more often than one might like.

Lecture 3 ended with a promise, and it is time to keep it. For three lectures we have traveled almost entirely in one direction. Somebody hands us equations; we ask what geometry they cut out. Today we turn the arrows around. Somebody hands us the geometry—a finite list of points, or a formula which manufactures points out of a parameter—and we ask for the equations.

That one sentence conceals two rather different problems, and it will repay us to keep them apart.

First, suppose we know the values of a polynomial function on a finite set of points. Can we reconstruct the function? This is an interpolation problem, and here is the pleasant surprise: the standard monomials from Lecture 2, which so far have been bookkeeping devices for division, will turn out to be exactly the right coordinates for it.

Second, suppose we are handed a parametrization

$$t \longmapsto \big(g_1(t), \ldots, g_n(t)\big).$$

Can we recover every equation forced by those formulas? This is the problem of *implicitization*. Back in Lecture 1 we typed `ker` into *Macaulay2* and accepted the answer on faith. It is slightly embarrassing that we never asked how the machine produced it. Today we open the box. The kernel computation will turn into an elimination computation, and the Elimination Theorem will explain why the trick works.

There is a third question hiding behind the second, and it is the one I find hardest to put down. Elimination recovers equations once a parametrization has been found. But where do such formulas come from in the first place, and when can they exist at all? Nothing we have proved so far even hints that a given curve should admit one. The circle, the nodal cubic, and the smooth cubic from the earlier lectures will march us from implicitization into birational geometry, and the third of them will refuse to cooperate in a very instructive way.

The two computational problems look quite different, but they share a theme, and it is the theme of this whole course: an infinite question is replaced by finite linear or Gröbner-basis data. There is also a familiar warning waiting for us in both. Interpolation by values sees the quotient by the ideal of the *visible* points, and cannot by itself see a nonreduced thickening. Elimination finds every formal polynomial identity forced by a parametrization, and over an infinite field those identities cut out the *Zariski closure* of the image rather than the image itself. As I warned at the end of Lecture 3, those two sets are not always the same, and much of the interest lives in the gap. As usual, a computation is only as good as our reading of its output.

© Juliette Bruce 2026.
Licensed under Creative Commons BY-NC-SA 4.0.

The unmarked parts develop the main narrative; your instructor may select among them during the lecture. Parts marked *Further* develop refinements and examples which can safely be postponed on a first pass.

We begin with the one-variable model that everything else in the first half of the lecture will imitate.

---

(1) **A Polynomial from Its Values.** Let $a_1, \dots, a_r \in \mathbb{K}$ be distinct, and set

$$g(x) = \prod_{i=1}^{r} (x - a_i).$$

For $1 \le i \le r$, define

$$\ell_i(x) = \prod_{\substack{1 \le j \le r \\ j \neq i}} \frac{x - a_j}{a_i - a_j}.$$

The denominators are nonzero precisely because the a_i are distinct; keep an eye on where that hypothesis is spent.

(a) Show that

$$\ell_i(a_j) = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

(b) Given arbitrary values $b_1, \dots, b_r \in \mathbb{K}$, prove that

$$p(x) = \sum_{i=1}^{r} b_i \ell_i(x)$$

has degree at most $r - 1$ and satisfies $p(a_i) = b_i$ for every i.

(c) Suppose that p and q both have degree at most $r - 1$ and take the same values at $a_1, \dots, a_r$. Use the bound on the number of roots from Lecture 2 to prove that $p = q$. Thus the interpolating polynomial of degree at most $r - 1$ is unique.

(d) Show that the ideals $\langle x - a_i \rangle$ and $\langle x - a_j \rangle$ are comaximal when $i \neq j$. Apply the Chinese remainder theorem to obtain an isomorphism

$$\begin{aligned} \mathbb{K}[x]/\langle g \rangle &\longrightarrow \mathbb{K}^r, \\ [f] &\longmapsto \big(f(a_1), \dots, f(a_r)\big). \end{aligned}$$

(e) Division by g gives every class in the quotient a unique representative of degree at most $r - 1$. Explain why parts (b)–(d) are three translations of a single statement: one into polynomial division, one into interpolation, and one into the Chinese remainder theorem. Which translation would you have found first if you had never seen the other two?

(f) Which basis of $\mathbb{K}[x]/\langle g\rangle$ is selected by the standard monomials? Which basis is selected by the polynomials $\ell_1, \dots, \ell_r$? Describe the change-of-basis matrix between them without computing its inverse.

Part (f) is the one worth lingering over, because it is the whole first half of the lecture in miniature. The monomials

$$1, x, \dots, x^{r-1}$$

give coordinates *by coefficients*, while the Lagrange polynomials $\ell_1, \dots, \ell_r$ give coordinates *by values*. These are two bases of the same r-dimensional vector space, and the matrix carrying the first set of coordinates to the second is the evaluation matrix

$$\begin{pmatrix} 1 & a_1 & a_1^2 & \cdots & a_1^{r-1} \\ 1 & a_2 & a_2^2 & \cdots & a_2^{r-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a_r & a_r^2 & \cdots & a_r^{r-1} \end{pmatrix},$$

the familiar Vandermonde matrix. Notice what Exercise **??** has quietly accomplished: it proves this matrix is invertible without anyone being asked to compute the Vandermonde determinant. Invertibility came instead from the fact that both lists are bases of the same quotient. This is a small mercy, and it is also a template, because in several variables there is no determinant formula waiting for us anyway.

That is precisely where the difficulty lies. In one variable, the list $1, x, \dots, x^{r-1}$ is handed to us for free by the degree. In several variables there is no such canonical list, and it is not even obvious what should play the role of "degree at most $r-1$." The gap is exactly the shape of the tool we built in Lecture 2: fix a monomial order, compute a Gröbner basis of the vanishing ideal of the points, and let the standard monomials supply the replacement. The staircase, which so far has told us about division and multiplicity, is about to tell us about interpolation.

The name "Lagrange interpolation" comes with the same gentle caveat as Dickson's Lemma, and by now you may be developing a healthy suspicion of attributions. Newton had a version by way of divided differences in the 1670s. Edward Waring published this formula in a 1779 paper titled *Problems concerning interpolations*; Euler wrote it down in 1783; and Lagrange published his version in 1795, whereupon the name attached itself and refused to come off. The chronology is more crowded than the name suggests. There is something oddly appropriate in this. The interpolating polynomial is unique, as part (c) proves. The formula used to display it—and the name fastened to that formula—is not.

Let us now watch all of this happen in an example small enough that the geometry and the interpolation basis can both be drawn without solving a single unpleasant equation.

---

(2) **Four Points and Four Monomials.** Assume that $\mathrm{char}(\mathbb{K}) \neq 2$ and let

$$X = \{(1,1),(1,-1),(-1,1),(-1,-1)\} \subseteq \mathbb{K}^2.$$

Work in $S = \mathbb{K}[x,y]$ with lex order $x > y$, and set

$$I = \langle x^2 - 1, y^2 - 1 \rangle.$$

Before starting, guess how many standard monomials you expect to find, and why.

(a) Prove directly that $\mathbb{V}_{\mathbb{K}}(I) = X$.

(b) Use Buchberger's criterion to show that (x^2-1, y^2-1) is a Gröbner basis. Identify the initial ideal and list its standard monomials. (Compare the staircase with the one drawn for $\langle x^2, y^2 \rangle$ in Lecture 2. The staircases agree; the geometry does not.)

(c) Deduce that every class in S/I has a unique representative

$$a + bx + cy + dxy.$$

Why does the number of standard monomials agree with the number of points?

(d) For $\epsilon, \delta \in \{1,-1\}$, define

$$L_{\epsilon,\delta}(x,y) = \frac{(1+\epsilon x)(1+\delta y)}{4}.$$

Show that

$$L_{\epsilon,\delta}(\epsilon',\delta') = \begin{cases} 1 & \text{if } (\epsilon,\delta) = (\epsilon',\delta'), \\ 0 & \text{otherwise.} \end{cases}$$

Where is the characteristic hypothesis used? These are the Lagrange polynomials of Exercise **??** in one variable, multiplied together; explain why that is the natural guess for a product of two two-point sets.

(e) Suppose a polynomial function on X is required to take the value $b_{\epsilon,\delta}$ at (ϵ,δ). Prove that

$$p = \sum_{\epsilon,\delta \in \{1,-1\}} b_{\epsilon,\delta} L_{\epsilon,\delta}$$

has the prescribed values. Explain why p is already in standard normal form, so that no division is required to recognize it.

(f) Show that the four classes $[L_{\epsilon,\delta}]$ are pairwise orthogonal idempotents in S/I:

$$[L_{\epsilon,\delta}]^2 = [L_{\epsilon,\delta}], \qquad [L_{\epsilon,\delta}][L_{\epsilon',\delta'}] = 0 \quad \text{for } (\epsilon,\delta) \neq (\epsilon',\delta').$$

Verify also that their sum is 1. How does this display the product decomposition $S/I \cong \mathbb{K}^4$ inside the quotient ring itself, rather than merely as an abstract isomorphism?

(g) In the order used above, prescribe the four values

$$1, \quad 2, \quad 3, \quad 4.$$

Predict the normal form. Then, and only then, check your answer in *Macaulay2*.

```
restart
R = QQ[x,y];
I = ideal(x^2-1,y^2-1);
B = gb I;

Lpp = (1+x)*(1+y)/4;
Lpm = (1+x)*(1-y)/4;
Lmp = (1-x)*(1+y)/4;
Lmm = (1-x)*(1-y)/4;

p = Lpp+2*Lpm+3*Lmp+4*Lmm;
p % B

apply({{1,1},{1,-1},{-1,1},{-1,-1}},
      q -> sub(p,{x=>q#0,y=>q#1}))
```

Explain the normal form and the list of four outputs. What exactly did the computer verify, and what did your proof establish that the computer did not—namely, over which fields?

Let us name what just happened. The four standard monomials and the four point-evaluation functions are two bases of the same quotient, and the change-of-basis matrix between them is the Vandermonde matrix of the previous section wearing a two-variable disguise. Nothing here was special to a square, or to four points, or to two variables. Let

$$X = \{p_1, \ldots, p_r\} \subseteq \mathbb{K}^n$$

be any finite set of distinct $\mathbb{K}$-rational points. Each point $p = (a_1, \ldots, a_n)$ has maximal ideal

$$\mathfrak{m}_p = \langle x_1 - a_1, \ldots, x_n - a_n \rangle,$$

and

$$\mathbb{I}(X) = \mathfrak{m}_{p_1} \cap \cdots \cap \mathfrak{m}_{p_r}.$$

Distinct point ideals are comaximal, and the reason is almost too simple to write down: if $p_i \neq p_j$, then some coordinate of the two points differs, and subtracting the corresponding two linear generators leaves a nonzero scalar. The Chinese remainder theorem then predicts exactly what we watched happen for the square.

---

(3) **Standard Monomials as Interpolation Data.** Let $X = \{p_1, \dots, p_r\} \subseteq \mathbb{K}^n$ be a finite set of distinct $\mathbb{K}$-rational points, set $S = \mathbb{K}[x_1, \dots, x_n]$, and let $I = \mathbb{I}(X)$. Fix a monomial order and a Gröbner basis G of I. Let $B = \{m_1, \dots, m_N\}$ be the standard monomials of $\mathrm{in}_<(I)$.

(a) Prove that the point ideals $\mathfrak{m}_{p_i}$ are pairwise comaximal and apply the Chinese remainder theorem to show that evaluation induces an isomorphism

$$\begin{aligned} \mathrm{ev} : S/I &\longrightarrow \mathbb{K}^r, \\ [f] &\longmapsto \big(f(p_1), \dots, f(p_r)\big). \end{aligned}$$

Notice that algebraic closure of $\mathbb{K}$ is nowhere required here. We assumed the points were $\mathbb{K}$-rational from the start, which is a much cheaper hypothesis and does all the work.

(b) Lecture 2 tells us that the classes of the monomials in B form a $\mathbb{K}$-basis of S/I. Deduce that $N = r$. Thus the staircase counts the points, and it does so without being told how many there are.

(c) Write down the evaluation matrix

$$E_B = \big(m_j(p_i)\big)_{1 \le i,j \le r}.$$

Prove that E_B is invertible. Resist the urge to compute a determinant: which two isomorphisms is this matrix comparing?

(d) Given values $b_1, \dots, b_r \in \mathbb{K}$, explain how solving the linear system

$$E_B c = (b_1, \dots, b_r)^{\mathsf{T}}$$

constructs the unique normal form

$$c_1 m_1 + \cdots + c_r m_r$$

having those values on X. An infinite-looking reconstruction problem has become an $r \times r$ linear system; say precisely which step did the reducing.

(e) Changing the monomial order can change the list B. Why can it therefore change the interpolation formula without changing the polynomial function on X? Compare this with the two normal-form bases of the circle–parabola quotient in Lecture 2.

(f) Suppose instead that an ideal J merely satisfies $\mathbb{V}_{\mathbb{K}}(J) = X$ but is not radical. Explain why the evaluation map $S/J \to \mathbb{K}^r$ need not be injective. Which familiar algebraic elements must every point evaluation fail to detect? We have met them before, and we are about to meet them again.

(4) **Further: Five Points Remember a Conic.** The homogeneous quadratic forms in $\mathbb{K}[X, Y, Z]$ form the six-dimensional vector space

$$S_2 = \mathrm{span}_{\mathbb{K}}\{X^2, XY, XZ, Y^2, YZ, Z^2\}.$$

A nonzero element of S_2, considered up to multiplication by a nonzero scalar, defines a projective plane conic. Six coefficients, one scaling, five points: the count is suggestive, and this exercise asks when it can be trusted.

(a) Explain why requiring a quadratic form to vanish at one projective point imposes one homogeneous linear condition on its six coefficients. Why is this condition independent of the homogeneous coordinates chosen for the point?

(b) For five points $p_1, \ldots, p_5 \in \mathbb{P}^2$, choose one nonzero homogeneous-coordinate representative $\widetilde{p}_i$ for each point and write the evaluation map

$$\mathrm{ev}_2 : S_2 \longrightarrow \mathbb{K}^5, \qquad Q \longmapsto \big(Q(\widetilde{p}_1), \ldots, Q(\widetilde{p}_5)\big).$$

Explain why replacing a representative by a nonzero scalar multiple only rescales one row of the evaluation matrix, so the rank and kernel do not depend on these choices. If this map has rank 5, show that its kernel is one-dimensional and hence that exactly one conic passes through the five points. This is what we mean here by five points in *general position*.

(c) Assume that $\mathrm{char}(\mathbb{K}) \neq 2, 3$ and consider the five points

$$[1:0:0], \quad [0:0:1], \quad [1:1:1], \quad [1:-1:1], \quad [1:2:4].$$

Write

$$Q = aX^2 + bXY + cXZ + dY^2 + eYZ + fZ^2$$

and impose the five vanishing conditions. Show successively that

$$a = f = 0, \qquad d = -c, \qquad e = -b, \qquad b = 0,$$

and conclude that Q is a scalar multiple of $XZ - Y^2$.

(d) Look again at the five points and recognize them as values of

$$[s:t] \longmapsto [s^2 : st : t^2].$$

How is this conic related, after a linear change of coordinates, to the projective circle we will parametrize later in the lecture?

(e) The word "general" cannot be omitted, and here is the cheapest way to see why. Suppose four of the five points lie on a line L and the fifth point is $p \notin L$. For every line M through p, explain why the reducible conic $L \cup M$ contains all five points. Which linear conditions have failed to be independent?

(f) Compare this problem with interpolation. There we prescribed the values of a polynomial function and solved for a unique quotient class. Here we prescribe the value zero at five projective points and hunt for a one-dimensional kernel inside a degree-bounded space. What job is the chosen degree bound doing, and what plays its role in the interpolation setting?

(5) **Further: A Double Point Needs More Than a Value.** Here is the warning promised in the introduction, in its smallest possible form. Let $A = \mathbb{K}[x]/\langle x^2 \rangle$.

(a) Show that $1, x$ are the standard monomials and that every class has a unique representative $a + bx$.

(b) The underlying zero set is the single point $\{0\}$. Show that evaluation at 0 remembers a but has no way at all to see b. Which nonzero class lies in the kernel?

(c) For $f \in \mathbb{K}[x]$, prove the congruence

$$f(x) \equiv f(0) + f'(0)x \pmod{x^2}.$$

So the class of f is recovered by its value *and* its first derivative at the point. Compare ordinary interpolation at two distinct points with this "value plus tangent value" at one doubled point, and notice that the count of data is the same in both cases.

(d) Return to the tangent intersection $\mathbb{K}[x, y]/\langle x^2 - y, y \rangle$ from Lecture 1. Identify the same two pieces of interpolation data geometrically. Why is multiplicity more than a rule for counting a point twice?

So much for reconstructing a function from its values. Let us turn to the second inverse problem, which is the one we have been quietly dodging since Lecture 1.

Suppose that polynomials $g_1, \ldots, g_n \in \mathbb{K}[t_1, \ldots, t_m]$ give a polynomial map

$$F : \mathbb{K}^m \longrightarrow \mathbb{K}^n, \qquad a \longmapsto \big(g_1(a), \ldots, g_n(a)\big).$$

Lecture 1 translated this map into the substitution homomorphism

$$F^* : \mathbb{K}[x_1, \ldots, x_n] \longrightarrow \mathbb{K}[t_1, \ldots, t_m],$$
$$x_i \longmapsto g_i.$$

This map is usually called the *pullback*, and it is contravariant: points move forward along F, while polynomial functions move backwards along F^*. Its kernel is the ideal of all polynomial identities among the coordinates of the parametrization—exactly the equations we are after. We computed it in Lecture 1 by typing `ker` and reading the answer off the screen. We never asked how the machine found it.

The idea is wonderfully literal, and once seen it is hard to unsee. Write down equations for the *graph* of the map, where the parameters and the coordinates coexist happily. Then forget the parameters.

Definition 1 (Graph Ideal)**.** Set

$$T = \mathbb{K}[t_1, \ldots, t_m], \qquad S = \mathbb{K}[x_1, \ldots, x_n], \qquad R = \mathbb{K}[t_1, \ldots, t_m, x_1, \ldots, x_n].$$

The *graph ideal* of the polynomial map F is

$$J_F = \langle x_1 - g_1, \ldots, x_n - g_n \rangle \subseteq R.$$

The definition is doing something very unmysterious. If we regard $\mathbb{K}^{m+n}$ as having coordinates (t, x), then the equation $x_i = g_i(t)$ says precisely what we want it to say, and $\mathbb{V}(J_F)$ is exactly the graph

$$\{(a, F(a)) : a \in \mathbb{K}^m\}.$$

Projecting the graph onto the x-coordinates gives the image of F. Algebraically, "forget the t-coordinates" has to mean: keep only those elements of J_F which no longer mention the t-variables at all. The following exercise proves that what survives is exactly the kernel of substitution—no more, and, more surprisingly, no less.

---

(6) **The Graph Knows the Kernel.** Continue with the notation above and define

$$\begin{aligned} \Psi : R &\longrightarrow T, \\ t_j &\longmapsto t_j, \\ x_i &\longmapsto g_i(t_1, \ldots, t_m). \end{aligned}$$

(a) Show that Ψ is surjective and that $J_F \subseteq \ker(\Psi)$.

(b) Let $H \in R$. Replace x_1 by g_1, then x_2 by g_2, and so on, one variable at a time. By subtracting consecutive stages and applying the one-variable factor theorem, prove that

$$H(t, x) - H(t, g_1(t), \ldots, g_n(t)) \in J_F.$$

This is nothing but a multivariable version of the schoolroom fact that $h(x) - h(a)$ is divisible by $x - a$, applied n times.

(c) If $H \in \ker(\Psi)$, use part (b) to prove $H \in J_F$. Conclude that

$$\ker(\Psi) = J_F \qquad \text{and} \qquad R/J_F \cong T.$$

Interpret the isomorphism geometrically: the graph is just another copy of the parameter space, which is exactly what a graph ought to be.

(d) The restriction of Ψ to the subring S is the substitution map F^*. Prove the purely algebraic identity

$$\ker(F^*) = J_F \cap S.$$

Notice that this holds over every field, and makes no claim whatsoever about whether the image of F is closed. That caution will earn its keep shortly.

(e) Suppose that $f \in J_F \cap S$. Explain directly, without invoking the isomorphism theorem, why f vanishes at every point of the image of F. Conversely, if $\mathbb{K}$ is infinite and f vanishes

at every point of the image, explain why $f(g_1, \ldots, g_n)$ is the zero polynomial and hence $f \in J_F \cap S$. Where exactly did infinitude enter, and what goes wrong over $\mathbb{F}_q$?

The identity $\ker(F^*) = J_F \cap S$ has traded a kernel problem for an intersection problem. At first glance this looks like no progress at all, and possibly like a step backwards: how are we supposed to extract from an ideal in $m+n$ variables exactly those polynomials which happen to involve only a selected few of them?

But we have already built the tool, and we built it for this very purpose. Recall from Lecture 2 that in lex order with

$$t_1 > \cdots > t_m > x_1 > \cdots > x_n,$$

every monomial involving a t-variable outranks every monomial involving only the x-variables—not by a little, but absolutely, regardless of degree. This is the property that made lex feel so extravagant when we first met it. Here is its reward. If a polynomial has its leading monomial in the x-variables alone, then it cannot be concealing a t-term further down the list, because such a term would have been the leading one.

Definition 2 (Elimination Ideal)**.** Let

$$I \subseteq \mathbb{K}[t_1, \ldots, t_m, x_1, \ldots, x_n].$$

The ideal

$$I \cap \mathbb{K}[x_1, \ldots, x_n]$$

is the *elimination ideal* obtained by eliminating $t_1, \ldots, t_m$.

More generally one can eliminate any selected block of variables. The intersection notation says exactly what the result contains and nothing more: all polynomial consequences of I which no longer mention the forgotten coordinates.

Theorem 3 (Elimination Theorem)**.** *Let*

$$I \subseteq \mathbb{K}[t_1, \ldots, t_m, x_1, \ldots, x_n]$$

and use lex order $t_1 > \cdots > t_m > x_1 > \cdots > x_n$*. If* G *is a Gröbner basis of* I*, then*

$$G_x = G \cap \mathbb{K}[x_1, \ldots, x_n]$$

is a Gröbner basis of the elimination ideal $I \cap \mathbb{K}[x_1, \ldots, x_n]$.

The statement is almost suspiciously efficient. Compute one Gröbner basis in the big ring, then simply *throw away* the elements which still mention the variables you wanted gone, and what remains is not merely a generating set but a Gröbner basis of the elimination ideal. No second computation. No cleverness. If that sounds too good, then good—be suspicious. Then do Exercise **??**, which is four short parts long and consists entirely of the observation about lex order made

in the paragraph above. The theorem is easy because we did the hard work in Lecture 3, when we proved that Gröbner bases exist and that Buchberger's algorithm finds them.

The word *elimination* carries more history than one might expect, and some of it is quite funny. Eliminating variables from systems of polynomial equations was a central occupation of nineteenth-century algebra—Bézout, Sylvester, Cayley, Kronecker, and our own F. S. Macaulay all built machinery for it, mostly by way of resultants. In the twentieth century the fashion turned sharply. In his *Foundations of Algebraic Geometry* (1946), André Weil introduced a device which, he hoped, would finally rid algebraic geometry of the last traces of elimination theory; van der Waerden went so far as to drop the elimination chapter from later editions of his *Modern Algebra*. Then, at the 1970 International Congress, van der Waerden gave a lecture on algebraic geometry in the twentieth century and repeated Weil's verdict. Shreeram Abhyankar, a devoted partisan of nineteenth-century geometry, was in the audience, and responded with a poem whose refrain urged mathematicians to eliminate the eliminators of elimination theory.

The joke, as so often, is that the revival was already underway in a building nobody had thought to check. Buchberger's thesis was five years old by then, and in 1978 Wolfgang Trinks pointed out that Gröbner bases compute elimination ideals—which is to say, he published a version of Theorem **??**. Elimination theory did not die; it acquired an algorithm and a computer, and came back rather stronger than before.

---

(7) **Why Lex Eliminates.** Prove Theorem **??** as follows. The whole argument is the one observation about lex order, applied carefully.

(a) Let $0 \neq f \in I \cap \mathbb{K}[x_1, \ldots, x_n]$. Since G is a Gröbner basis of I, show that some $g \in G$ satisfies

$$\mathrm{LM}(g) \mid \mathrm{LM}(f).$$

(b) Explain why $\mathrm{LM}(g)$ cannot involve any t_j.

(c) Now suppose some *nonleading* term of g involved a t_j. Use the definition of lex order to show that this term would be larger than the purported leading monomial, which involves only the x-variables—a contradiction. Conclude that $g \in G_x$.

(d) Deduce that every leading monomial of an element of the elimination ideal is divisible by the leading monomial of an element of G_x. Explain why this is exactly the Gröbner-basis condition, and notice that we never had to run a second Buchberger computation.

(e) If G_x turns out to be empty, what does the theorem say about the elimination ideal? Before we make it precise, hazard a geometric guess: what should that mean about the projection?

(8) **The Twisted Cubic, Now Backwards.** Our old companion returns, and this time we approach it from the direction we have been avoiding. In Lecture 1 we began with the affine parametrization $t \mapsto (t, t^2, t^3)$ and computed a kernel by hand, with a rather ad hoc argument

about congruences. In Lecture 2 its hidden relation $y^2 - xz$ taught us why Gröbner bases were needed at all. In Lecture 3 the projective parametrization

$$\nu_3 : \mathbb{P}^1 \longrightarrow \mathbb{P}^3, \qquad [s:t] \longmapsto [s^3 : s^2t : st^2 : t^3]$$

showed that the very same quadric was indispensable at infinity. We now recover all of those equations in one stroke, by elimination.

(a) For the affine parametrization, form the graph ideal

$$L_{\text{aff}} = \langle x - t, y - t^2, z - t^3 \rangle \subseteq \mathbb{K}[t, x, y, z].$$

Use Exercise **??** to identify $L_{\text{aff}} \cap \mathbb{K}[x, y, z]$ with the kernel from Lecture 1. Compare the length of this argument with the one we gave there.

(b) Without computing a Gröbner basis, verify that

$$x^2 - y, \qquad xy - z, \qquad y^2 - xz$$

belong to this elimination ideal. The third is a consequence of the first two; why does that not make it useless? (You have now been asked this three times, in three different disguises.)

(c) Now regard the four homogeneous coordinates of ν_3 as a polynomial map $\mathbb{K}^2 \to \mathbb{K}^4$ and form

$$L_{\text{proj}} = \langle w - s^3, x - s^2t, y - st^2, z - t^3 \rangle \subseteq \mathbb{K}[s, t, w, x, y, z].$$

Verify directly that the three quadrics

$$x^2 - wy, \qquad xy - wz, \qquad y^2 - xz$$

lie in $L_{\text{proj}} \cap \mathbb{K}[w, x, y, z]$.

(d) Arrange these quadrics, up to sign, as the 2×2 minors of

$$\begin{pmatrix} w & x & y \\ x & y & z \end{pmatrix}.$$

Which entry was 1 in the affine matrix from Lecture 2, and why did it become w? This is homogenization happening in front of us, in a form which is hard to misread.

(e) Predict the eliminated ideal before running the following block. Then compare the output with the ideal found in Lecture 3.

```
restart
needsPackage "Elimination";
R = QQ[s,t,w,x,y,z,MonomialOrder=>Lex];

L = ideal(w-s^3,x-s^2*t,y-s*t^2,z-t^3);
G = gb L;
K = eliminate({s,t},L);

trim K
J = ideal(x^2-w*y,x*y-w*z,y^2-x*z);
K == J
```

Why are s and t placed first in the ring, and what would go wrong if they were placed last? What does the output `true` certify that a visual comparison of two generating lists would not?

(f) The calculation took place in the affine cone $\mathbb{K}^4$, whereas ν_3 maps into projective space, so we owe ourselves an argument. The kernel, and hence the elimination ideal, is homogeneous: if a relation is decomposed into homogeneous pieces, then each piece is again a relation. Prove this by observing that substituting s^3, s^2t, st^2, t^3 sends a form of degree d to a form of degree $3d$, so pieces of different degrees have nowhere to cancel. Conclude that homogeneous generators of the kernel define genuine projective equations. Assuming $\mathbb{K}$ is infinite, use the work in Lecture 3 to identify J with the full homogeneous vanishing ideal of $\nu_3(\mathbb{P}^1)$.

(g) Finally, compare the two routes. In Lecture 3 we computed a degree-compatible affine Gröbner basis and then homogenized. Here we homogenized the parametrization and then eliminated its parameters. Why should two such different-looking procedures arrive at the same ideal? Which route starts from equations, and which starts from points manufactured by a map?

Let us record what we have. The graph construction and the Elimination Theorem together answer the algebraic implicitization problem:

$$\boxed{\ker(F^*) = J_F \cap \mathbb{K}[x_1, \dots, x_n]}$$

and a lex Gröbner basis computes the ideal on the right. Over an infinite field this kernel is the ideal of every polynomial vanishing on the parametrized image, so that

$$\mathbb{V}\big(\ker(F^*)\big) = \mathbb{V}\big(\mathbb{I}(F(\mathbb{K}^m))\big) = \overline{F(\mathbb{K}^m)}^{\,\mathrm{Zar}}.$$

The last expression is the Zariski closure of the image, and it is extremely tempting to erase that bar. The hyperbola from the very first exercise of Lecture 1 has been waiting since page one to tell us not to.

More generally, assume $\mathbb{K}$ is algebraically closed, let $I \subseteq \mathbb{K}[t_1, \ldots, t_m, x_1, \ldots, x_n]$, and let

$$\pi : \mathbb{K}^{m+n} \longrightarrow \mathbb{K}^n$$

forget the t-coordinates. The *Closure Theorem for Elimination* states that

$$\overline{\pi(\mathbb{V}(I))}^{\text{Zar}} = \mathbb{V}\big(I \cap \mathbb{K}[x_1, \ldots, x_n]\big).$$

Read it as a statement about the limits of the method, and it becomes rather cheerful. Elimination captures every polynomial equation satisfied by the projection. It fails to capture the image itself only because polynomial equations are constitutionally incapable of distinguishing a set from its Zariski closure. The algorithm is not being careless; the language simply cannot express the difference.

---

(9) **The Missing Origin Returns Again.** Assume first that $\mathbb{K}$ is infinite. Let

$$H = \mathbb{V}(tu - 1) \subseteq \mathbb{K}^2$$

and project H to the t-axis. If this feels familiar, it should: it is Exercise 1(g) of Lecture 1, now equipped with an algorithm.

(a) Prove that the image is $\mathbb{K}^\times = \mathbb{K} \setminus \{0\}$.

(b) If $f \in \mathbb{K}[t]$ vanishes on this image, prove that $f = 0$. Conclude that

$$\langle tu - 1 \rangle \cap \mathbb{K}[t] = \langle 0 \rangle.$$

(c) The zero ideal cuts out the whole affine line. Which point has elimination added, and why is that point *forced* upon us by every polynomial equation which holds on the image?

(d) Predict the output, then check it.

```
restart
needsPackage "Elimination";
R = QQ[u,t,MonomialOrder=>Lex];
H = ideal(u*t-1);

eliminate({u},H)
```

Why is the zero output not a failure of the algorithm? What would a correct answer even look like, if the algorithm were permitted to give one?

(e) *Further: A Polynomial Map with Nonclosed Image.* Consider

$$F : \mathbb{K}^2 \longrightarrow \mathbb{K}^2, \qquad (s,t) \longmapsto (s, st).$$

Show that

$$F(\mathbb{K}^2) = \{(x,y) : x \neq 0\} \cup \{(0,0)\}.$$

Prove that no nonzero polynomial vanishes on this image and hence that its Zariski closure is all of $\mathbb{K}^2$. Which points on the y-axis were missing from the actual image? Notice that this image is neither open nor closed, which is a good advertisement for why the Zariski topology needs a vocabulary beyond those two words.

(f) *Further: Finite Fields.* If $\mathbb{K} = \mathbb{F}_q$, the set $\mathbb{K}^\times$ is finite and is therefore itself algebraic—so the whole difficulty above evaporates. Find a nonzero polynomial in $\mathbb{F}_q[t]$ whose zero set is exactly $\mathbb{F}_q^\times$. Explain why this does not contradict the purely algebraic elimination ideal computed from $ut - 1$, and return to the distinction between polynomial identities and polynomial functions from Lecture 1.

There is a pleasant symmetry in what the two halves of the lecture have given us so far. For a finite reduced set, the coordinate ring remembers exactly the values of a function at its points. For a parametrized set over an infinite field, the kernel remembers exactly the polynomial equations holding at its points. The first reconstructs functions from evaluations; the second reconstructs equations from substitutions. And both fail in the same instructive way the moment we ask the points to remember something which is not pointwise data.

We now leave computation for a while and turn to the geometric question underneath it. Finding the equations of a parametrization is one thing. Where does the parametrization come from?

The simplest nonlinear example is the unit circle, and its construction is so elementary that you can perform it with a ruler. Choose one point on the circle, and sweep a line through it.

This also settles an IOU from Lecture 1. There we contrasted the polynomially parametrized twisted cubic with the smooth cubic $y^2 = x^3 - x$, and asserted without proof that the latter admits no rational parametrization at all. We are finally in a position to say what that means, and then to see why it is true.

---

(10) **Drawing the Circle with a Line.** Assume that $\mathrm{char}(\mathbb{K}) \neq 2$ and consider

$$C = \mathbb{V}(x^2 + y^2 - 1) \subseteq \mathbb{K}^2.$$

The point $P = (-1, 0)$ lies on C. A line of slope t through P has equation

$$y = t(x + 1).$$

Draw the picture before doing any algebra; the entire exercise is visible in it.

(a) Substitute the line equation into the circle equation and verify the factorization

$$x^2 + t^2(x+1)^2 - 1 = (x+1)\big((1+t^2)x + t^2 - 1\big).$$

Which factor records the intersection point we already knew about? This is the crucial move: we knew one root in advance, so we may divide it out.

(b) When $1 + t^2 \neq 0$, solve for the second intersection point and obtain

$$\phi(t) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right).$$

Verify directly that $\phi(t) \in C$.

(c) Let $(x, y) \in C$ with $x \neq -1$. The line joining P to (x, y) has slope

$$t = \frac{y}{x+1}.$$

Use $x^2 + y^2 = 1$ to show that

$$1 + t^2 = \frac{2}{x+1}.$$

Deduce that this denominator is nonzero and that substituting this t into ϕ recovers (x, y). Thus the geometric construction has an algebraic inverse, and both directions are given by ratios of polynomials.

(d) Conclude that ϕ gives a bijection

$$\{t \in \mathbb{K} : 1 + t^2 \neq 0\} \longrightarrow C(\mathbb{K}) \setminus \{(-1, 0)\}.$$

Over $\mathbb{R}$, which values of t must be excluded? Over $\mathbb{C}$, which additional values appear, and what geometric feature will projective space eventually assign to them?

(e) The point P was reached by no finite slope at all. Explain geometrically why it ought to correspond to the vertical line through P, or informally to $t = \infty$. We have met this phenomenon before, and we know what to do about it.

(f) Clear denominators in the identity

$$(x, y, 1) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}, 1\right)$$

and compare the resulting quadratics with the classical Pythagorean-triple formula. You have very likely met these three expressions before, in a course which had nothing to do with algebraic geometry.

This calculation produced more than a convenient recipe for manufacturing points. Away from a single point of the circle, the formulas

$$t \longmapsto \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right) \qquad \text{and} \qquad (x, y) \longmapsto \frac{y}{x+1}$$

are honest inverses of one another. The line and the circle are, in a sense we are about to make precise, the same object viewed twice. To say this cleanly we need a little language.

A subset U of an affine variety X is *Zariski open* if its complement is algebraic in X. For example,

$$D(q) = \{p \in X : q(p) \neq 0\}$$

is open. It is *dense* if its Zariski closure is all of X. If X is irreducible, every nonempty open subset is dense—a fact worth pausing on, since it is wildly false in the topology of a metric space, and it is the reason the definitions below are as generous as they are.

Definition 4 (Rational and Birational Maps)**.** Let X and Y be irreducible affine or projective varieties. A *rational map*

$$F : X \dashrightarrow Y$$

is a map given by rational formulas on a nonempty open subset of X, with two descriptions identified when they agree on a dense open subset. On affine charts the coordinates are quotients of polynomials. For a rational map from a projective variety $X \subseteq \mathbb{P}^n$ to $\mathbb{P}^m$, we may instead use homogeneous forms of the same degree which are not all identically zero. The formula is defined at the points where those forms do not vanish simultaneously, and its coordinates are understood up to common scaling.

The varieties X and Y are *birational* if there are rational maps $F : X \dashrightarrow Y$ and $G : Y \dashrightarrow X$ which are inverse on dense open subsets. An irreducible variety of dimension d is *rational* if it is birational to affine d-space, or equivalently in the projective setting to $\mathbb{P}^d$.

The dashed arrow is a warning label, and it should be read as one: a rational map may simply fail to be defined somewhere. Birational does not mean equal, or homeomorphic, or even in bijection on points. It says that after discarding some smaller algebraic subsets—sets of lower dimension, which the Zariski topology regards as negligible—the two varieties are described by mutually inverse rational formulas. Whether that is a reasonable thing to ignore depends entirely on what you want, and a great deal of algebraic geometry consists of finding out.

The circle calculation proves that the affine circle is birational to the affine line. Now let us watch projective space put back the points which the denominators were hiding. The projective equation $X^2 + Y^2 - Z^2$ is smooth when $\mathrm{char}(\mathbb{K}) \neq 2$, and hence irreducible; its affine chart is therefore irreducible, as the definition demands.

---

(11) **The Missing Parameter Goes to Infinity.** Assume again that $\mathrm{char}(\mathbb{K}) \neq 2$. Let

$$\overline{C} = \mathbb{V}_{\mathbb{P}^2}(X^2 + Y^2 - Z^2)$$

be the projective circle. Define

$$\Phi : \mathbb{P}^1 \longrightarrow \mathbb{P}^2, \qquad [s : t] \longmapsto [s^2 - t^2 : 2st : s^2 + t^2].$$

(a) Verify that all three coordinates have the same degree and that they have no common zero in $\mathbb{P}^1$. Where do you use $\mathrm{char}(\mathbb{K}) \neq 2$? Conclude that Φ is defined everywhere—note the missing dashes on that arrow.

(b) Substitute the coordinates into $X^2 + Y^2 - Z^2$ and show that $\Phi(\mathbb{P}^1) \subseteq \overline{C}$.

(c) On the chart $s \neq 0$, write $u = t/s$. Show that the part of the image with $Z \neq 0$ recovers the affine formula from Exercise **??**, so that nothing has been lost in the passage to $\mathbb{P}^1$.

(d) Evaluate $\Phi([0 : 1])$. Which missing affine point does this recover, and does the answer match the geometric prediction you made in Exercise **??**(e)?

(e) If $s^2 + t^2 = 0$, what kind of projective point does $\Phi([s : t])$ produce? Explain what became of the forbidden affine parameters $t = \pm\sqrt{-1}$ when $\mathbb{K}$ happens to contain them. The denominators did not vanish into nothing; they went somewhere.

(f) On the open subset $U_+ \subseteq \overline{C}$ where $Z + X \neq 0$, verify that

$$[X : Y : Z] \longmapsto [Z + X : Y]$$

is inverse to Φ wherever $s \neq 0$.

(g) On the open subset $U_- \subseteq \overline{C}$ where $Z - X \neq 0$, verify that

$$[X : Y : Z] \longmapsto [Y : Z - X]$$

is inverse to Φ wherever $t \neq 0$. Show that U_+ and U_- cover $\overline{C}$. On their overlap, use

$$(Z + X)(Z - X) = Y^2$$

to prove that $[Z + X : Y] = [Y : Z - X]$. The two partial inverses therefore glue into a single map $\overline{C} \to \mathbb{P}^1$ defined at every point, with no exceptions to apologize for.

(h) Conclude that Φ is not merely a birational parametrization: it is an isomorphism

$$\mathbb{P}^1 \cong \overline{C}.$$

Compare this global statement with the affine parametrization, where one point and a couple of denominator values had to be excluded by hand. Very little was true of the circle that was not already true affinely; what changed is that we can now say it without exceptions.

(i) *Further: Every Conic with a Point.* Let $Q \subseteq \mathbb{P}^2$ be a smooth projective conic with a $\mathbb{K}$-rational point P. Explain why the same construction—send a line through P to its second intersection with Q—should give a global parametrization $\mathbb{P}^1 \cong Q$, with the tangent line accounting for the value at P itself. What can go wrong over a field where the conic has no rational point at all? (Consider $\mathbb{V}_{\mathbb{P}^2}(X^2 + Y^2 + Z^2)$ over $\mathbb{R}$, and notice that the obstruction is arithmetic rather than geometric.)

(12) **Further: Eliminating a Denominator.** The affine formula in Exercise **??** is rational rather than polynomial, so our graph-ideal machinery does not literally apply. The repair is a trick we have used twice already, and it is worth noticing that it is the same trick. We encode the condition $1 + t^2 \neq 0$ algebraically by introducing a variable u which is forced to be its inverse. Work over a field which is infinite and has characteristic different from 2, and set

$$L = \left\langle (1+t^2)x - (1-t^2),\, (1+t^2)y - 2t,\, u(1+t^2) - 1 \right\rangle \subseteq \mathbb{K}[t, u, x, y].$$

(a) Explain why a point of $\mathbb{V}(L)$ records a parameter t at which the rational formula is actually defined, together with its image (x, y). Which of the three generators is doing the excluding?

(b) Show directly from the displayed equations that $x^2 + y^2 - 1 \in L$. Write $d = 1 + t^2$ and first prove

$$d^2(x^2 + y^2 - 1) = (dx - (1-t^2))(dx + 1 - t^2) + (dy - 2t)(dy + 2t).$$

Then use $ud - 1 \in L$ to strip the factor d^2 inside the quotient.

(c) Predict the elimination ideal $L \cap \mathbb{K}[x, y]$ and then compute it.

```
restart
needsPackage "Elimination";
R = QQ[t,u,x,y,MonomialOrder=>Lex];

L = ideal((1+t^2)*x-(1-t^2),
          (1+t^2)*y-2*t,
          u*(1+t^2)-1);

C = eliminate({t,u},L);
trim C
```

(d) The actual affine image omits $(-1, 0)$, while the eliminated equation cheerfully cuts out the whole circle. Explain why the output is nevertheless the correct answer to the question elimination was asked.

(e) Compare the auxiliary equation $u(1 + t^2) - 1$ with the equation $1 - uf$ in Rabinowitsch's trick from Lectures 1 and 3. What single algebraic maneuver is being used in both places, and what is the general principle? (Roughly: an ideal cannot express "$f \neq 0$," but it can express "f has an inverse," which is the same thing said in a language ideals happen to speak.)

Before moving on, notice that the projective formula

$$[s : t] \longmapsto [s^2 - t^2 : 2st : s^2 + t^2]$$

has quietly explained the classical Pythagorean triples. Choosing $s, t \in \mathbb{Z}$ gives

$$(s^2 - t^2)^2 + (2st)^2 = (s^2 + t^2)^2.$$

From our point of view this is not a lucky algebraic identity to be verified and admired. It is the homogeneous equation of the conic, evaluated on its own parametrization—it *had* to be true, and the parametrization is where it came from.

That observation is very old, which is part of what makes it satisfying. Sweeping a line through a known rational point to produce new ones is the method Diophantus used in the *Arithmetica*, though he described it algebraically and left no account of the geometry. Attribution here is a little contested in the usual way: Weil credited the chord construction to Newton, who reconstructed and systematized these arguments in the 1670s, while Bashmakova argued that Diophantus was already doing exactly this. Either way, the pattern the ancients found by hand is the one we just found by algebra, and the reason it works is worth stating plainly.

A line meets a conic in two points, counted with multiplicity. Once one intersection point is nailed down in advance, exactly one remains, and a single linear equation recovers it rationally. Everything depends on that arithmetic: $2 - 1 = 1$.

So what happens for a cubic? A line meets it in three points, and fixing one leaves two—which a linear equation cannot separate. Unless, that is, the fixed point already counts twice. A singular point of multiplicity two absorbs two of the three intersections and leaves a single one behind, and suddenly the count works again. This is why the two cubic curves from Lecture 2 behave so differently, and it is time to look at both.

---

(13) **A Node Turns a Cubic into a Parametrized Curve.** Assume that $\operatorname{char}(\mathbb{K}) \neq 2$ and consider

$$N = \mathbb{V}\big(y^2 - x^2(x+1)\big) \subseteq \mathbb{K}^2.$$

Lecture 2 set this nodal cubic beside the smooth cubic $y^2 = x^3 - x$ and observed that the two have identical staircases. Here they part company. The origin is the node of N, and the node is not a blemish—it is the entire trick.

(a) Intersect N with the line $y = tx$ through the origin and verify

$$t^2x^2 - x^2(x+1) = x^2(t^2 - x - 1).$$

How does the factor x^2 record that the line has already met the cubic twice at the node? What changes for the tangent slopes $t = \pm 1$, when the remaining factor also vanishes at $x = 0$?

(b) Solve for the remaining intersection point and obtain the polynomial parametrization

$$\psi : \mathbb{K} \longrightarrow N, \qquad t \longmapsto (t^2 - 1, t(t^2 - 1)).$$

Verify directly that it lands on N. Note that this one is polynomial, not merely rational—the denominators of the circle case have vanished. Why?

(c) Away from the node, show that $t = y/x$. Deduce that ψ is a birational parametrization.

(d) Compute $\psi(1)$ and $\psi(-1)$. Why does this show that a birational parametrization need not be one-to-one everywhere? What geometric information distinguishes the two parameter values, given that they produce the same point?

(e) Form the graph ideal of ψ, predict its elimination ideal, and then check.

```
restart
needsPackage "Elimination";
R = QQ[t,x,y,MonomialOrder=>Lex];

L = ideal(x-(t^2-1),y-(t^3-t));
K = eliminate({t},L);
trim K
```

How does the output certify the implicit equation of the image? Is the affine image already closed in this example, and how would you know without being told?

(f) *Further: Projective Completion.* Homogenize the nodal cubic to

$$Y^2Z = X^2(X + Z).$$

Verify that

$$[s : t] \longmapsto [s(t^2 - s^2) : t(t^2 - s^2) : s^3]$$

lands on its projective closure. Which two points of $\mathbb{P}^1$ map to the node, and which point maps to the point at infinity? Compare with the circle, where the analogous map was injective and we got a genuine isomorphism.

(14) **Why the Smooth Cubic Resists the Same Trick.** Assume now that $\mathbb{K}$ is algebraically closed of characteristic different from 2 and 3. For a projective plane curve $\mathbb{V}_{\mathbb{P}^2}(F)$ over this field, a point $p \in \mathbb{V}_{\mathbb{P}^2}(F)$ is singular when all three first-order partial derivatives of F vanish at p, and the curve is smooth when it has no such point. This is the hypersurface case of the Jacobian criterion.

Consider

$$E = \mathbb{V}(y^2 - x^3 + x).$$

Its projective closure is the smooth cubic

$$\overline{E} = \mathbb{V}_{\mathbb{P}^2}(Y^2Z - X^3 + XZ^2).$$

Here there is no node to absorb an intersection, and we should expect the construction to break. Watch precisely where.

(a) Check that $(0,0) \in E$ and that it is a smooth point by computing one nonzero partial derivative. Further, verify that the three partial derivatives of $Y^2Z - X^3 + XZ^2$ have no common projective zero on $\overline{E}$; this establishes the global smoothness used below.

(b) Intersect E with the line $y = tx$ through $(0,0)$ and obtain

$$x(x^2 - t^2x - 1) = 0.$$

After removing the known intersection $x = 0$, what degree equation remains? Compare with the linear residual equations for the conic and the nodal cubic. This single degree is the whole story.

(c) The residual equation gives

$$x = \frac{t^2 \pm \sqrt{t^4 + 4}}{2}.$$

Show that $t^4 + 4$ is not a square in $\mathbb{K}(t)$. Conclude that this pencil of lines does not produce a rational parametrization: the second intersection point is there, but it cannot be written as a rational function of t.

(d) Be careful about the logic, because it is exactly the sort of place where one is tempted to overclaim. Why does the failure of one attempted construction not prove that no rational parametrization exists?

(e) To get the stronger conclusion we must borrow a theorem from the geometry of curves: a smooth projective cubic has genus 1, whereas $\mathbb{P}^1$ has genus 0, and genus is preserved by birational equivalence of smooth projective curves. Accepting this, conclude that $\overline{E}$ is not rational. Then identify which hypothesis of the theorem fails for the nodal cubic, and thereby permits it to be rational after all.

(f) *Further: Asking Macaulay2 About Birationality.* The package `RationalMaps` can test whether a specified rational map is birational. Predict the two Boolean outputs from the geometry above before running anything.

```
restart
needsPackage "RationalMaps";

P1 = QQ[s,t];

-- The rational inverse from the nodal cubic to P^1
N = QQ[X,Y,Z]/(Y^2*Z-X^2*(X+Z));
rhoN = map(N,P1,{X,Y});
isBirationalMap(rhoN,Verbosity=>0)

-- The x-coordinate map from the smooth cubic to P^1
E = QQ[u,v,w]/(v^2*w-u^3+u*w^2);
rhoE = map(E,P1,{w,u});
isBirationalMap(rhoE,Verbosity=>0)
```

Remember that a map of coordinate rings points in the direction opposite to the corresponding map of varieties. Thus `rhoN` represents $[X : Y : Z] \dashrightarrow [X : Y]$, while `rhoE` represents $[u : v : w] \dashrightarrow [w : u]$. Identify the point where each displayed coordinate pair vanishes simultaneously. For `rhoN` the map cannot extend across the node. For `rhoE` the rational map does extend across $[0 : 1 : 0]$, sending it to the point at infinity $[0 : 1] \in \mathbb{P}^1$; only this particular coordinate pair has a base point there. Why is the first map the rational inverse to the nodal parametrization, while the second is generically two-to-one by the fiber calculation from Lecture 2?

The expected outputs are `true` and `false`. Now be careful again: why does the second output show only that *this particular* map is not birational, rather than prove by itself that no birational map $\overline{E} \dashrightarrow \mathbb{P}^1$ exists? Which argument above supplies the stronger conclusion, and notice that it is not one a computer gave us.

(g) Gröbner bases can recover the implicit equations of a parametrization once one is handed to us. What part of this example shows that *finding* a parametrization is a genuinely different problem from implicitizing one?

The contrast is worth stating bluntly, because it is easy to come away from three lectures of algorithms with the wrong impression. Elimination is a general algorithm for passing from polynomial formulas, or from suitably encoded rational ones, to equations. It always works. There is no corresponding theorem saying that every curve admits a rational parametrization, and there cannot be, because the statement is false. A smooth conic with a rational point has one; a nodal cubic has one; a smooth cubic does not. The Gröbner basis does not settle this by magic. It faithfully organizes the algebraic consequences of whatever data we hand it, and if we hand it nothing, it organizes nothing.

I want to point at one more thing before we move on, because the failure in Exercise **??** is more interesting than a failure has any right to be. The chord construction on a cubic does not simply break down. Given two rational points on a smooth cubic, the line through them meets the curve in a third, which is again rational. So instead of a parametrization we obtain a way of *combining* points to make new ones—and that operation turns out to be a group law. Diophantus and Fermat used it to generate solutions; Newton systematized it; Poincaré, in a 1901 paper, asked the modern structural question about the resulting group of rational points; and Mordell proved in 1922 that this group is finitely generated. The curve which refused to be a line was hiding an abelian group instead, which is a considerably better consolation prize. This is the beginning of the theory of elliptic curves, and it starts precisely where our attempt to parametrize ran out of road.

The word *birational* deliberately ignores smaller closed subsets, and by now we have seen what that permission buys. It is what allowed the two parameter values 1 and -1 to collide at the node without spoiling the rational inverse $t = y/x$ away from it. It is what allowed the affine circle to omit one point and still count as a copy of the line. Projective completion and Zariski closure put missing limit points back; birational geometry asks instead which rational-function information survives once such lower-dimensional exceptions have been discarded. These are two different kinds of forgiveness, and it is worth being clear about which one is being extended in any given argument.

We finish, as always, by returning to the computational dictionary which has followed us since the opening pages of Lecture 1.

---

(15) **Updating the Two-Way Dictionary.** Return to the table near the beginning of Lecture 1.

(a) Add the interpolation problem: given values on a finite reduced set X of $\mathbb{K}$-rational points, the evaluation isomorphism

$$\mathbb{K}[X] \cong \mathbb{K}^X$$

and a standard-monomial basis turn interpolation into a finite linear system. Here $\mathbb{K}^X$ denotes the ring of all functions $X \to \mathbb{K}$. Which hypothesis fails for a double point?

(b) For a polynomial parametrization F, record

$$\ker(F^*) = J_F \cap \mathbb{K}[x_1, \ldots, x_n].$$

Explain separately the role played by the graph ideal, by lex order, and by the Elimination Theorem. Each does exactly one job.

(c) Which statement is valid over every field: the graph-kernel identity, or the claim that the kernel equals the vanishing ideal of the set-theoretic image? What hypothesis did we need for the second, and where did we spend it?

(d) State carefully what $\mathbb{V}(\ker(F^*))$ describes. When may the word "closure" be safely omitted? Use the hyperbola and the nodal cubic as contrasting examples.

(e) Record the *Macaulay2* commands `ker`, `eliminate`, `gb`, and `%`. Which of them can now be understood as interfaces to a single Gröbner-basis computation? What question does each one answer, and why should a human still inspect the variables, the monomial order, and the field before interpreting any output? If you completed the *Further* computation with `RationalMaps`, also record `isBirationalMap`: what does it answer about one specified map, and what does a `false` output conspicuously fail to prove?

(f) *Further: Unirational versus Rational.* Explain the difference between the following three claims about a variety X:

(i) X is the Zariski closure of a polynomial image;

(ii) there is a rational map from affine or projective space to X whose image is Zariski dense (such a map is called *dominant*, and X is then called *unirational*);

(iii) X is birational to affine or projective space.

Birationality implies unirationality, and in dimension one they agree, but in higher dimensions the converse genuinely fails—which was an open problem of some notoriety before it was settled. Which claims were illustrated by the circle and by the nodal cubic? For the smooth cubic, record only what we actually proved: claim (iii) fails.

(g) Finally, bring the three movements of the lecture together. In interpolation, what finite matrix converts standard-monomial coefficients into values? In implicitization, what finite Gröbner-basis computation converts a graph into equations? What extra evidence is required before a rational formula certifies birationality? And in each of the three cases, where exactly can pointwise information lose something algebraic?

What did the computation prove?

For a finite reduced set, the standard monomials form an interpolation basis: prescribed values determine a unique normal form, and the evaluation matrix is invertible for reasons no determinant needs to know. For a parametrization, the graph ideal plus the Elimination Theorem compute the kernel of substitution, and over an infinite field that kernel cuts out the Zariski closure of the image—closure, not image. The circle and the nodal cubic are rational because a line through a chosen point leaves exactly one new intersection to recover; a smooth cubic leaves two, and no amount of computing will fix that. Gröbner bases can implicitize a parametrization, but geometry decides whether one exists.

We have now put an initial ideal to work in two quite different ways. Its standard monomials gave us coordinates for interpolation, and its leading terms made elimination possible. Yet the pictures from Lecture 2 have left a question sitting unanswered all this time, and it has only grown more pointed. The circle–parabola intersection and the monomial ideal $\langle x^2, y^2 \rangle$ have entirely different points but the same four standard monomials. A smooth cubic and a nodal cubic can have the same staircase, and we have just spent several pages establishing that these two curves are not remotely the same. So what exactly survives when an ideal is replaced by its initial ideal?

Lecture 5 answers this degree by degree. Counting standard monomials will produce the Hilbert function and the Hilbert polynomial, and a one-parameter family will let us actually watch a variety flatten into the geometry of its initial ideal. The point set, the smoothness, and the irreducible components may all change in the process. Dimension and degree will not: they were visible in the staircase the whole time.